
HSC Mathematics Extension 2
Last Resorts Mastery
Vu Hung Nguyen
Problem 16



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*"Mathematics is the door and key to the sciences."*

Roger Bacon

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# 1 Introduction

## 1.1 Project Overview

This booklet compiles the most challenging problems from HSC Mathematics Extension 2 examinations—the notorious Problem 16, as we call them "Last Resort." These problems represent the pinnacle of high school mathematics, typically combining multiple advanced topics and requiring sophisticated problem-solving techniques. Every problem showcases the depth and complexity expected at the highest level of the HSC Mathematics curriculum.

The collection draws from authentic examination problems that have challenged Extension 2 students over many years. Each problem has been carefully selected to represent the diverse range of mathematical concepts and proof techniques that appear in Problem 16, from advanced inequalities and complex number theory to vector optimization and differential geometry applications.

## 1.2 What Makes Problem 16 Special

Problem 16 is uniquely challenging because it:

- Combines multiple mathematical topics within a single problem
- Requires advanced proof techniques and mathematical maturity
- Often involves multi-part questions that build complexity progressively
- Tests both computational skills and conceptual understanding
- Demands clear mathematical communication and rigorous reasoning

## 1.3 Target Audience

This booklet is designed for Extension 2 students who want to:

- Master the most challenging problems in the HSC curriculum
- Develop advanced problem-solving strategies
- Build confidence in tackling complex multi-step problems
- Understand how to approach unfamiliar mathematical scenarios
- Prepare thoroughly for the demands of Problem 16 in examinations

## 1.4 How to Use This Booklet

- **Study the Fundamentals:** Review key theorems and techniques
- **Attempt Problems Independently:** Try each problem without looking at solutions first
- **Part 1 Strategy:** Compare your work against solutions to understand model reasoning
- **Part 2 Approach:** Use upside-down hints sparingly, then review concise solutions
- **Practice Multiple Times:** Rework problems from memory to build technique mastery
- **Focus on Communication:** Pay attention to how mathematical arguments are structured and presented

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# 2 Fundamentals Review

## 2.1 Limit Tools for Sequences

### 2.1.1 The Squeeze (Pinching) Theorem

**Theorem 0.1: Squeeze Theorem for Sequences**

Let $(a_n)$, $(b_n)$ and $(c_n)$ be real sequences such that

$$a_n \leq b_n \leq c_n$$

for all sufficiently large $n$. If

$$\lim_{n\to\infty} a_n = L \quad \text{and} \quad \lim_{n\to\infty} c_n = L,$$

then

$$\lim_{n\to\infty} b_n = L.$$

**Remark 0.1: Extension 2 application**

In the problem proving $\pi^2$ is irrational, we introduced the integral sequence $(I_n)$ and showed

$$0 < I_n < (\text{a quantity that tends to } 0 \text{ as } n \to \infty).$$

The Squeeze Theorem then forces $I_n \to 0$, even though $I_n$ itself is not evaluated in closed form.

### 2.1.2 The Monotone Convergence Theorem

**Theorem 0.2: Monotone Convergence for Sequences**

Let $(x_n)$ be a real sequence.

- If $(x_n)$ is increasing (i.e. $x_{n+1} \geq x_n$ for all $n$) and bounded above (there exists $M$ with $x_n \leq M$ for all $n$), then $(x_n)$ converges to a finite limit $L \leq M$.
- If $(x_n)$ is decreasing and bounded below, then $(x_n)$ converges to a finite limit $L \geq m$.

**Remark 0.2: Extension 2 application**

This theorem underpins many recursive–sequence problems. In the Mandelbrot escape problem, to show that a sequence diverges instead of converging, we proved it was strictly increasing and *unbounded above*, so no finite limit can exist. In other problems (especially those involving iteration to a root), one typically shows a sequence is monotone and bounded to deduce convergence, then identifies the limit by solving the fixed–point equation.

## 2.2 Key Inequalities

Problem 16 frequently employs fundamental inequalities as building blocks for more complex arguments.

### 2.2.1 Arithmetic-Geometric Mean (AM-GM) Inequality

**Theorem 0.3: AM-GM Inequality**

For positive real numbers $a_1, a_2, \ldots, a_n$:

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}$$

Equality occurs if and only if $a_1 = a_2 = \cdots = a_n$.

**Theorem 0.4: Weighted AM-GM Inequality**

For positive real numbers $a_1, a_2, \ldots, a_n$ and positive weights $w_1, w_2, \ldots, w_n$ with $w_1 + w_2 + \cdots + w_n = 1$:

$$w_1 a_1 + w_2 a_2 + \cdots + w_n a_n \geq a_1^{w_1} a_2^{w_2} \cdots a_n^{w_n}$$

### 2.2.2 Cauchy-Schwarz Inequality

**Theorem 0.5: Cauchy-Schwarz Inequality**

For vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$:

$$|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$$

Equivalently, for real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$:

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)$$

## 2.3 Complex Number Theory

### 2.3.1 De Moivre's Theorem

**Theorem 0.6: De Moivre's Theorem**

For complex numbers in polar form $z = r(\cos\theta + i\sin\theta)$:

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

### 2.3.2 Roots of Unity

**Theorem 0.7: Roots of Unity**

The $n$th roots of unity are given by:

$$\omega_k = e^{2\pi ik/n} = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$$

for $k = 0, 1, 2, \ldots, n-1$.

## 2.4 Vector Algebra

### 2.4.1 Dot Product

For vectors $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$:

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = |\mathbf{u}||\mathbf{v}|\cos\theta$$

### 2.4.2 Cross Product (Optional Enrichment, Not in the NESA Syllabus)

This idea is included only as enrichment. The cross product is *not* prescribed in the current NESA Mathematics Extension 1 or Extension 2 syllabus, so it should not be treated as core HSC content. It is mainly useful as background for later university study in areas such as mathematics, physics, and engineering.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Its magnitude satisfies $|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta$, which equals the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

## 2.5 Polynomial Theory

### 2.5.1 Newton's Identities

**Theorem 0.8: Newton's Identities**

For a polynomial with roots $\alpha_1, \alpha_2, \ldots, \alpha_n$, if $S_k = \sum_{i=1}^{n} \alpha_i^k$ and $e_k$ are the elementary symmetric polynomials, then:

$$S_k - e_1S_{k-1} + e_2S_{k-2} - \cdots + (-1)^{k-1}e_{k-1}S_1 + (-1)^k k e_k = 0$$

**Remark 0.3: Quadratic and cubic cases (Viète's formulas)**

For low degrees, Newton's identities reduce to the familiar Viète relations:

- **Degree 2:** For $P(x) = x^2 - sx + p$ with roots $\alpha, \beta$, we have $e_1 = \alpha + \beta = s$ and $e_2 = \alpha\beta = p$. Newton's identities give $S_1 = \alpha + \beta = e_1$ and $S_2 = \alpha^2 + \beta^2 = e_1^2 - 2e_2 = s^2 - 2p$.
- **Degree 3:** For $P(x) = x^3 - sx^2 + px - q$ with roots $\alpha, \beta, \gamma$, we have $e_1 = \alpha + \beta + \gamma = s$, $e_2 = \alpha\beta + \beta\gamma + \gamma\alpha = p$, and $e_3 = \alpha\beta\gamma = q$. Newton's identities yield

  $$S_1 = e_1 = s, \qquad S_2 = e_1S_1 - 2e_2 = s^2 - 2p, \qquad S_3 = e_1S_2 - e_2S_1 + 3e_3 = s^3 - 3sp + 3q,$$

  which express power sums like $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha^3 + \beta^3 + \gamma^3$ purely in terms of the coefficients—exactly the spirit of Viète's formulas.

## 2.6 Chebyshev Polynomials (First Kind)

The Chebyshev polynomials of the first kind $T_n(x)$ are a family of orthogonal polynomials on $[-1, 1]$ with the defining trigonometric identity:

$$T_n(x) = \cos\big(n \arccos x\big).$$

### 2.6.1 Recurrence Formula

They satisfy the three-term recurrence for $n \geq 1$:

$$T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x), \quad T_0(x) = 1,\ T_1(x) = x.$$

### 2.6.2 Explicit Low-Degree Polynomials

| $n$ | $T_n(x)$ | $\cos(n\theta)$ as a polynomial in $\cos\theta$ |
|---|---|---|
| 0 | $1$ | $1$ |
| 1 | $x$ | $\cos\theta$ |
| 2 | $2x^2 - 1$ | $2\cos^2\theta - 1$ |
| 3 | $4x^3 - 3x$ | $4\cos^3\theta - 3\cos\theta$ |
| 4 | $8x^4 - 8x^2 + 1$ | $8\cos^4\theta - 8\cos^2\theta + 1$ |
| 5 | $16x^5 - 20x^3 + 5x$ | $16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$ |

Notice that the leading coefficient doubles each time, so $\deg T_n = n$ and the coefficient of $x^n$ is $2^{n-1}$ for $n \geq 1$.

### 2.6.3 Zeros and Extremal Points

From $T_n(\cos\theta) = \cos(n\theta)$ we see:

- **Zeros:** $T_n(x) = 0$ at the $n$ points

  $$x_k = \cos\frac{(2k-1)\pi}{2n}, \quad k = 1, 2, \ldots, n,$$

  all lying in $(-1, 1)$.

- **Extremal points when $n = 5$:** Since $\cos(5\theta)$ takes values $\pm 1$ at $\theta = k\pi/5$, the polynomial $T_5(x)$ attains its maxima and minima

$$T_5(\cos\tfrac{k\pi}{5}) = \cos(5 \cdot \tfrac{k\pi}{5}) = \cos(k\pi) \in \{1, -1\}, \quad k = 0, 1, 2, 3, 4, 5.$$

  Thus $T_5(x)$ oscillates between $-1$ and $1$ at the six points $x = \cos(k\pi/5)$ on $[-1, 1]$.

**Example 0.1: Solve $\cos(5\theta) = 0$ and deduce an exact value**

Solve

$$\cos(5\theta) = 0.$$

Since $\cos\phi = 0$ when $\phi = \frac{\pi}{2} + k\pi$ $(k \in \mathbb{Z})$, we get

$$5\theta = \frac{\pi}{2} + k\pi \quad \Longrightarrow \quad \theta = \frac{\pi}{10} + \frac{k\pi}{5}, \quad k \in \mathbb{Z}.$$

Now let $x = \cos\theta$. Using $T_5(x) = \cos(5\theta)$,

$$0 = T_5(x) = 16x^5 - 20x^3 + 5x = x(16x^4 - 20x^2 + 5).$$

For nonzero roots, set $y = x^2$:

$$16y^2 - 20y + 5 = 0 \quad \Longrightarrow \quad y = \frac{20 \pm \sqrt{400 - 320}}{32} = \frac{5 \pm \sqrt{5}}{8}.$$

Hence

$$\cos^2\theta \in \left\{ \frac{5 + \sqrt{5}}{8}, \frac{5 - \sqrt{5}}{8} \right\}$$

for solutions of $\cos(5\theta) = 0$ with $\cos\theta \neq 0$.
Take $\theta = \frac{3\pi}{10}$ (since $5 \cdot \frac{3\pi}{10} = \frac{3\pi}{2}$, so it is a solution). Also $\frac{3\pi}{10} \in \left(0, \frac{\pi}{2}\right)$, so

$$\cos\frac{3\pi}{10} > 0, \qquad \cos\frac{3\pi}{10} < \cos\frac{\pi}{10}.$$

Thus $\cos^2\left(\frac{3\pi}{10}\right)$ is the smaller of the two values, giving

$$\cos^2\left(\frac{3\pi}{10}\right) = \frac{5 - \sqrt{5}}{8}.$$

**Example 0.2: Evaluate $\tan^2\frac{\pi}{14}+\tan^2\frac{3\pi}{14}+\tan^2\frac{5\pi}{14}$**

Expand $(\cos\theta+i\sin\theta)^7$ (De Moivre) to obtain

$$\cos(7\theta)=64\cos^7\theta-112\cos^5\theta+56\cos^3\theta-7\cos\theta.$$

Let $x=\cos\theta$. Then

$$\cos(7\theta)=0\Longrightarrow x\left(64x^6-112x^4+56x^2-7\right)=0.$$

For $0<\theta<\pi$, the solutions of $\cos(7\theta)=0$ are

$$\theta=\frac{(2k+1)\pi}{14},\quad k=0,1,\dots,6.$$

The nonzero $x=\cos\theta$ values therefore satisfy

$$64x^6-112x^4+56x^2-7=0.$$

Set $u=x^2$. Then

$$64u^3-112u^2+56u-7=0,$$

whose three roots are

$$u_1=\cos^2\frac{\pi}{14},\quad u_2=\cos^2\frac{3\pi}{14},\quad u_3=\cos^2\frac{5\pi}{14}.$$

Now put $y=\frac{1}{u}$. Substituting $u=\frac{1}{y}$ gives

$$64-112y+56y^2-7y^3=0\Longleftrightarrow 7y^3-56y^2+112y-64=0.$$

So for $y_i=\frac{1}{u_i}$, Viète gives

$$y_1+y_2+y_3=\frac{56}{7}=8.$$

Hence

$$\tan^2\alpha=\frac{1-\cos^2\alpha}{\cos^2\alpha}=\frac{1}{u}-1=y-1,$$

and therefore

$$\tan^2\frac{\pi}{14}+\tan^2\frac{3\pi}{14}+\tan^2\frac{5\pi}{14}=(y_1+y_2+y_3)-3=8-3=5.$$

## 2.7 Chebyshev Polynomials (Second Kind)

The Chebyshev polynomials of the second kind $U_n(x)$ form another family of orthogonal polynomials on $[-1,1]$, closely related to the first kind. They are defined by the trigonometric identity:

$$U_n(x)=\frac{\sin\left((n+1)\arccos x\right)}{\sin(\arccos x)}.$$

Equivalently, if $x=\cos\theta$, then:

$$U_n(\cos\theta)=\frac{\sin((n+1)\theta)}{\sin\theta}.$$

### 2.7.1 Recurrence Formula

The second kind Chebyshev polynomials satisfy the same three-term recurrence as the first kind, but with different initial conditions:

$$U_{n+1}(x) = 2x\,U_n(x) - U_{n-1}(x), \quad U_0(x) = 1,\ U_1(x) = 2x.$$

### 2.7.2 Explicit Low-Degree Polynomials

| $n$ | $U_n(x)$ | $\dfrac{\sin((n+1)\theta)}{\sin\theta}$ as a polynomial in $\cos\theta$ |
|---|---|---|
| 0 | $1$ | $1$ |
| 1 | $2x$ | $2\cos\theta$ |
| 2 | $4x^2 - 1$ | $4\cos^2\theta - 1$ |
| 3 | $8x^3 - 4x$ | $8\cos^3\theta - 4\cos\theta$ |
| 4 | $16x^4 - 12x^2 + 1$ | $16\cos^4\theta - 12\cos^2\theta + 1$ |
| 5 | $32x^5 - 32x^3 + 6x$ | $32\cos^5\theta - 32\cos^3\theta + 6\cos\theta$ |

### 2.7.3 Zeros

Using $U_n(\cos\theta) = \dfrac{\sin((n+1)\theta)}{\sin\theta}$, the zeros of $U_n$ occur when the numerator vanishes but $\sin\theta \neq 0$, i.e.

$$U_n(x_k) = 0 \quad \text{for} \quad x_k = \cos\frac{k\pi}{n+1}, \quad k = 1, 2, \ldots, n.$$

All these zeros lie strictly inside the interval $(-1, 1)$.

### 2.7.4 Relationship to First Kind

The Chebyshev polynomials of the first and second kind are related by:

$$U_n(x) = \frac{T'_{n+1}(x)}{n+1}, \quad T_n(x) = U_n(x) - x\,U_{n-1}(x).$$

Additionally, they satisfy:

$$T_n(x) = \cos(n \arccos x) \quad \text{and} \quad U_n(x) = \frac{\sin((n+1)\arccos x)}{\sqrt{1-x^2}}.$$

### 2.7.5 Key Properties

- $T_n(x)$ has the minimal leading coefficient among all monic polynomials of degree $n$ bounded by 1 on $[-1, 1]$.
- The zeros of $U_n(x)$ are simple and lie in the interior of $(-1, 1)$.
- Both families play a crucial role in approximation theory and the study of polynomial bounds.

These polynomials appear frequently in Problem 16 when dealing with bounded polynomial coefficients and optimal approximations.

## 2.8 Optimization Techniques

### 2.8.1 Lagrange Multipliers (Conceptual)

**Definition 0.1: Method of Lagrange Multipliers**

To optimize a function $f(x, y, z)$ subject to a constraint $g(x, y, z) = 0$, look for points where:

$$\nabla f = \lambda \nabla g$$

Here, $\nabla f$ and $\nabla g$ are the gradients of $f$ and $g$, respectively, and $\lambda$ is a scalar known as the Lagrange multiplier.
The gradient of a function $f$ is the vector of its partial derivatives:

$$\nabla f = \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

In HSC problems, this often reduces to applying AM-GM or Cauchy-Schwarz inequalities.

**Note:** This technique is not covered at HSC level but is useful attacking hard problems (and for university studies as well).

Here is how you could use it to attack the problems:

- Guess when the max/min occurs: Use the condition $\nabla f = \lambda \nabla g$ to solve for the variables (often resulting in $x = y = z$ for symmetric expressions).
- Identify the "Target Inequality": Once you know the values that achieve the optimum (e.g., $x = 2, y = 2$), you know exactly what the right-hand side of your inequality should look like.
- Reverse-engineer the formal proof: Work backward from that optimum using standard HSC tools—most commonly by setting up AM-GM terms or choosing the vectors for Cauchy-Schwarz that align with your discovered ratios.

**Example 0.3: Constrained Optimization on an Ellipse**

Find the maximum value of $f(x, y) = x + y$ subject to the constraint $2x^2 + 3y^2 = 1$.
**Solution:** Let $f(x, y) = x + y$ and $g(x, y) = 2x^2 + 3y^2 - 1 = 0$. The Lagrange multiplier condition $\nabla f = \lambda \nabla g$ gives:

$$1 = 4\lambda x \implies x = \frac{1}{4\lambda}$$
$$1 = 6\lambda y \implies y = \frac{1}{6\lambda}$$

From these, $y = \frac{2}{3}x$. Substituting into the constraint:

$$2x^2 + 3\left(\frac{2}{3}x\right)^2 = 1 \implies \frac{10}{3}x^2 = 1 \implies x = \sqrt{\frac{3}{10}}$$

Then $y = \frac{2}{3}\sqrt{\frac{3}{10}} = \sqrt{\frac{4}{30}} = \sqrt{\frac{2}{15}}$. The maximum value is $f_{\max} = \sqrt{\frac{3}{10}} + \sqrt{\frac{2}{15}} = \sqrt{\frac{5}{6}}$.
*Connection to HSC:* This identifies the correct vectors for Cauchy-Schwarz: $(\sqrt{2}x, \sqrt{3}y) \cdot (1/\sqrt{2}, 1/\sqrt{3})$.

**Example 0.4: Proving AM-GM Type Inequalities**

Maximize the product $xyz$ for $x, y, z > 0$ subject to $x+y+z = k$ (where $k$ is a constant).
**Solution:** Let $f(x, y, z) = xyz$ and $g(x, y, z) = x + y + z - k = 0$. The condition $\nabla f = \lambda \nabla g$ gives:

$$(yz, xz, xy) = \lambda(1, 1, 1)$$

This implies $yz = xz = xy$, so $x = y = z$ (since all are positive). Substituting into the constraint: $3x = k \implies x = k/3$. The maximum value of the product is $(k/3)^3 = k^3/27$. This confirms that for any fixed sum, the product is maximized when the terms are equal, which is the geometric intuition behind AM-GM: $\frac{x+y+z}{3} \geq \sqrt[3]{xyz}$.

## 2.9 The Big O Notation

### 2.9.1 What is Big O?

The Big O notation is a mathematical tool used to describe the behavior of functions as their input approaches a particular value (often infinity or zero). It provides a way to express approximations and compare the growth rates of functions.

**Definition 0.2: Big O Notation**

For functions $f$ and $g$, we write $f(x) = O(g(x))$ as $x \to a$ if there exist constants $C > 0$ and $\delta > 0$ such that

$$|f(x)| \leq C|g(x)|$$

for all $x$ near $a$.

**Remark 0.4**

Informally, this means $f(x)$ grows no faster than $g(x)$ near $a$.

**Remark 0.5**

The same notation is used for *sequences* $(u_n)$ as $n \to \infty$, where we write $u_n = O(v_n)$ if $|u_n| \leq C|v_n|$ for all sufficiently large integers $n$.

### 2.9.2 Important Note for HSC Students

- Big O notation is **NOT** covered in the HSC/NESA syllabus
- It is **NOT required** for HSC examinations
- **NOT recommended** to use or mention in HSC assessments or exams
- However, keep it as your **"secret weapon"** for Extension 2! It can help you:
  - Find limits quickly by identifying dominant terms
  - Approximate expressions efficiently
  - Understand the behavior of complex expressions
- Excellent preparation for university mathematics and further studies

### 2.9.3 Applications

Big O notation is particularly useful for:

- **Approximating functions:** Understanding which terms dominate as $x$ becomes large or small
- **Finding limits:** Identifying the leading behavior to evaluate challenging limits
- **Sequences of numbers:** Describing how sequences grow or decay
- **Functions:** Comparing growth rates of different functions

### 2.9.4 Common Examples

Here are some fundamental approximations using Big O notation:

**Exponential Function:**

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + O(x^4) \quad \text{as } x \to 0$$

**Natural Logarithm:**

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + O(x^5) \quad \text{as } x \to 0$$

**Sine Function:**

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + O(x^7) \quad \text{as } x \to 0$$

**Cosine Function:**

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + O(x^6) \quad \text{as } x \to 0$$

### 2.9.5 Practical Examples: The "Secret Weapon" in Action

**Example 0.5: Avoiding the L'Hopital Marathon**

Consider the classic limit:

$$L = \lim_{x \to 0} \frac{x - \sin(x)}{x^3}$$

*The Hard Way:* Applying L'Hopital's Rule three separate times until the $x^3$ vanishes.
*The Big O Way:* Substitute $\sin(x) = x - \frac{x^3}{6} + O(x^5)$.

$$\begin{aligned} \frac{x - \sin(x)}{x^3} &= \frac{x - \left(x - \frac{x^3}{6} + O(x^5)\right)}{x^3} \\ &= \frac{\frac{x^3}{6} - O(x^5)}{x^3} \\ &= \frac{1}{6} - O(x^2) \end{aligned}$$

As $x \to 0$, the term $O(x^2) \to 0$. Thus, $L = \frac{1}{6}$.

**Example 0.6: Mixed Function Simplification**

Evaluate the limit involving both Logarithms and Trigonometry:

$$L = \lim_{x \to 0} \frac{\ln(1+x) - \sin(x) + \frac{1}{2}x^2}{x^3}$$

*The Hard Way:* Differentiating this repeatedly is messy due to the quotient and chain rules. *The Big O Way:* Use the known expansions:

- $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + O(x^4)$
- $\sin(x) = x - \frac{x^3}{6} + O(x^5)$

Substitute these into the numerator:

$$\begin{aligned} \text{Num} &= \left(x - \frac{x^2}{2} + \frac{x^3}{3} + \dots\right) - \left(x - \frac{x^3}{6} + \dots\right) + \frac{x^2}{2} \\ &= (x - x) + \left(-\frac{x^2}{2} + \frac{x^2}{2}\right) + \left(\frac{x^3}{3} - (-\frac{x^3}{6})\right) + O(x^4) \\ &= \frac{3x^3}{6} + O(x^4) = \frac{1}{2}x^3 + O(x^4) \end{aligned}$$

Now, divide by the denominator $x^3$:

$$L = \lim_{x \to 0} \left(\frac{1}{2} + O(x)\right) = \frac{1}{2}$$

### 2.9.6 Connection with Taylor Series

The Big O notation naturally connects with Taylor series expansions. The Taylor series represents a function as an infinite sum of terms calculated from the function's derivatives at a single point. The $O(x^n)$ term represents all the remaining terms of order $n$ and higher, giving us a precise way to express truncation error.

For example, when we write $e^x = 1 + x + \frac{x^2}{2} + O(x^3)$, we're saying that the error in approximating $e^x$ by $1 + x + \frac{x^2}{2}$ behaves like $x^3$ for small $x$.

## 2.10 Notation and Conventions

- Complex numbers are denoted by $z, w, \alpha, \beta, \dots$
- Vectors are written in bold: $\mathbf{u}, \mathbf{v}, \mathbf{a}, \mathbf{b}$
- Real numbers are assumed unless stated otherwise
- $\mathbb{R}^n$ denotes $n$-dimensional real space
- $|z|$ denotes the modulus of complex number $z$
- $|\mathbf{v}|$ denotes the magnitude of vector $\mathbf{v}$

# 3 Part 1: Problems and Solutions (Detailed)

Part 1 contains 16 carefully selected problems representing the core techniques and challenge levels found in Problem 16. Each solution includes a strategy explanation, complete step-by-step working, and takeaway insights to build your problem-solving toolkit.

## 3.1 Medium Last Resort Problems

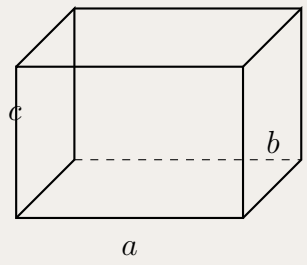
Problems that combine 2–3 mathematical topics with sophisticated reasoning.

**Problem 3.1: AM-GM Surface Area Optimization**

(a) Prove that for positive real numbers $x, y, z$:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{3}{\sqrt[3]{xyz}}$$

(b) A rectangular prism has dimensions $a, b, c$ and a fixed, constant Volume $V$. Using part (a), show that the Total Surface Area $S$ is minimized when the prism is a cube.




**Hint:** For part (a), apply AM-GM inequality directly to the reciprocals $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$. For part (b), express the surface area $S = 2(ab+bc+ca)$ in terms of the volume constraint $V = abc$, then use part (a) with strategic substitution.

## Solution 3.1

**(a) Proof:** Since $x, y, z > 0$, then $\frac{1}{x}, \frac{1}{y}, \frac{1}{z} > 0$. Applying the AM-GM inequality to these three terms:

$$\frac{\frac{1}{x}+\frac{1}{y}+\frac{1}{z}}{3} \geq \sqrt[3]{\frac{1}{x}\cdot\frac{1}{y}\cdot\frac{1}{z}}$$
$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z} \geq 3\sqrt[3]{\frac{1}{xyz}}$$
$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z} \geq \frac{3}{\sqrt[3]{xyz}}$$

**(b) Application:** Let the dimensions be $a, b, c$.

- Fixed Volume: $V = abc$ (constant).
- Surface Area: $S = 2(ab + bc + ca)$.

We want to relate $S$ to the reciprocals in part (a).

$$S = 2abc\left(\frac{1}{c}+\frac{1}{a}+\frac{1}{b}\right)$$
$$S = 2V\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$$

From part (a), we know $\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq \frac{3}{\sqrt[3]{abc}}$. Substituting this into the expression for $S$:

$$S \geq 2V\left(\frac{3}{\sqrt[3]{V}}\right)$$
$$S \geq 6V^1V^{-1/3}$$
$$S \geq 6V^{2/3}$$

Since $V$ is constant, $6V^{2/3}$ is a constant minimum value. Equality holds (minimum $S$ occurs) when the terms in the AM-GM are equal:

$$\frac{1}{a} = \frac{1}{b} = \frac{1}{c} \implies a = b = c$$

Therefore, the prism is a cube. □

## Takeaways 3.1

1. **Factoring Strategy:** Recognizing that factoring out the volume product $(abc)$ from the surface area expression $(ab + bc + ca)$ perfectly isolates the sum of the reciprocals.

2. **The Equality Condition:** Emphasizing that optimization using AM-GM relies entirely on the equality condition—the minimum or maximum is strictly achieved when all terms are equal (i.e., $a = b = c$).

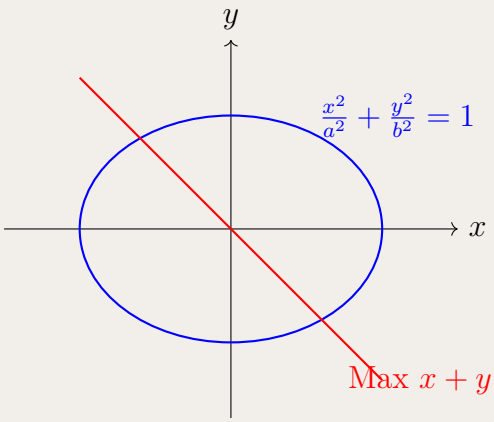
## Problem 3.2: Cauchy-Schwarz on Ellipsoid

The point $P(x, y, z)$ lies on the surface of the ellipsoid defined by:

$$\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{25} = 1$$

(i) By choosing appropriate vectors $\mathbf{u}$ and $\mathbf{v}$ and applying the Cauchy-Schwarz inequality ($\mathbf{u} \cdot \mathbf{v} \leq |\mathbf{u}||\mathbf{v}|$), find the maximum value of $x + y + z$.

(ii) Find the coordinates of $P$ in the first octant where this maximum occurs.




**Hint:** The standard Cauchy-Schwarz inequality is $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}||\mathbf{b}|$. You cannot define $\mathbf{u} = (x, y, z)$ directly because the sum of squares is not 1. Instead, define $\mathbf{u}$ such that $|\mathbf{u}|^2$ exactly matches the left-hand side of the ellipsoid equation. Then, choose a constant vector $\mathbf{v}$ such that the dot product $\mathbf{u} \cdot \mathbf{v}$ recovers the expression $x + y + z$.

## Solution 3.2

**(i)** Let $\mathbf{u} = \begin{pmatrix} \frac{x}{2} \\ \frac{y}{3} \\ \frac{z}{5} \end{pmatrix}$. We are given $|\mathbf{u}|^2 = \frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{25} = 1$, so $|\mathbf{u}| = 1$. We want to maximize $x + y + z$. We observe:

$$x + y + z = \frac{x}{2}(2) + \frac{y}{3}(3) + \frac{z}{5}(5)$$

This suggests setting $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$. Apply Cauchy-Schwarz:

$$\mathbf{u} \cdot \mathbf{v} \leq |\mathbf{u}||\mathbf{v}|$$

$$x + y + z \leq (1)\sqrt{2^2 + 3^2 + 5^2}$$

$$x + y + z \leq \sqrt{4 + 9 + 25} = \sqrt{38}$$

**(ii)** Equality occurs when $\mathbf{u} = k\mathbf{v}$, i.e., $\frac{x}{2} = 2k, \frac{y}{3} = 3k, \frac{z}{5} = 5k$. So $x = 4k, y = 9k, z = 25k$. Substitute into the plane eq $x + y + z = \sqrt{38}$:

$$4k + 9k + 25k = \sqrt{38} \implies 38k = \sqrt{38} \implies k = \frac{1}{\sqrt{38}}$$

Point $P$: $\left(\frac{4}{\sqrt{38}}, \frac{9}{\sqrt{38}}, \frac{25}{\sqrt{38}}\right)$.

**Takeaways 3.2**

1. **Normalization:** If given a constraint like $Ax^2 + By^2 = C$, define vector components as $\sqrt{A}x$ and $\sqrt{B}y$.
2. **The "Canceling" Vector:** The second vector is chosen specifically to cancel out the denominators introduced by the normalization step.

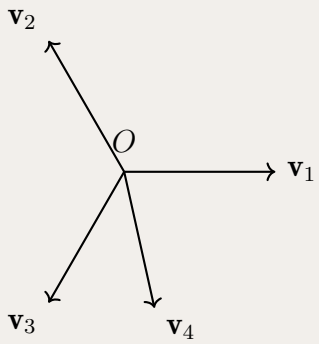
**Problem 3.3: Unit Vector Cosine Sum**

Four unit vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ originate from the origin $O$. It is given that their vector sum is zero:

$$\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \mathbf{v}_4 = \mathbf{0}$$

Let $\theta_{ij}$ be the angle between vectors $\mathbf{v}_i$ and $\mathbf{v}_j$. Show that the sum of the cosines of the angles between all distinct pairs is $-2$.

$$\sum_{1 \le i < j \le 4} \cos\theta_{ij} = -2$$




**Hint:** Consider the squared magnitude of the vector sum. Since $\sum \mathbf{v}_i = \mathbf{0}$, it follows that $|\sum \mathbf{v}_i|^2 = 0$. Expand the dot product $(\mathbf{v}_1 + \cdots + \mathbf{v}_4) \cdot (\mathbf{v}_1 + \cdots + \mathbf{v}_4)$. Separate the expansion into diagonal terms $(\mathbf{v}_i \cdot \mathbf{v}_i)$ and mixed terms $(\mathbf{v}_i \cdot \mathbf{v}_j$ for $i \neq j)$.

## Solution 3.3

Let $S = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \mathbf{v}_4 = \mathbf{0}$.

$$|S|^2 = S \cdot S = 0$$

Expanding the dot product:

$$\sum_{i=1}^{4} |\mathbf{v}_i|^2 + 2 \sum_{1 \le i < j \le 4} (\mathbf{v}_i \cdot \mathbf{v}_j) = 0$$

We are given that vectors are unit vectors, so $|\mathbf{v}_i|^2 = 1$ for all $i = 1..4$. There are 4 such terms.

$$4 + 2 \sum_{1 \le i < j \le 4} (\mathbf{v}_i \cdot \mathbf{v}_j) = 0$$

Using the definition of dot product: $\mathbf{v}_i \cdot \mathbf{v}_j = |\mathbf{v}_i||\mathbf{v}_j| \cos\theta_{ij} = 1 \cdot 1 \cdot \cos\theta_{ij}$.

$$4 + 2 \sum_{1 \le i < j \le 4} \cos\theta_{ij} = 0$$

$$2 \sum \cos\theta_{ij} = -4$$

$$\sum_{1 \le i < j \le 4} \cos\theta_{ij} = -2$$

## Takeaways 3.3

1. **Squaring the Sum:** The most powerful tool for analyzing vector sums equal to zero (equilibrium) is to take the dot product of the sum with itself.
2. **Counting Terms:** When expanding $(\sum_{i=1}^{n} a_i)^2$, there are $n$ squared terms and $n(n-1)$ cross terms. Since dot product is commutative, this groups into $n$ squared terms and $2 \times$ (distinct pairs).

## Problem 3.4: Complex Numbers Forming Triangle

Three complex numbers $z_1, z_2, z_3$ satisfy:

$$\begin{cases} |z_1| = |z_2| = |z_3| = r \quad (r > 0) \\ z_1 + z_2 + z_3 = 0 \end{cases}$$

Show that $z_1, z_2, z_3$ represent the vertices of an equilateral triangle inscribed in the circle $|z| = r$.

**Hint:** For an algebraic approach, use $z_1 + z_2 = -z_3$ and apply the parallelogram law relating $|z_1 + z_2|^2$ and $|z_1 - z_2|^2$. For a geometric approach, where is the centroid of this triangle? Where is its circumcenter?

## Solution 3.4

**Method 1 (Algebraic - Parallelogram Law):** Since $z_1+z_2+z_3=0$, we have $z_1+z_2=-z_3$. Squaring the modulus of both sides:

$$|z_1+z_2|^2=|-z_3|^2 \implies |z_1+z_2|^2=r^2$$

Using the parallelogram law $|z_1+z_2|^2+|z_1-z_2|^2=2(|z_1|^2+|z_2|^2)$:

$$r^2+|z_1-z_2|^2=2(r^2+r^2)$$

$$|z_1-z_2|^2=3r^2$$

By symmetry, $|z_2-z_3|^2=3r^2$ and $|z_3-z_1|^2=3r^2$. Since all squared side lengths are equal to $3r^2$, the triangle is equilateral.

**Method 2 (Geometric):** The centroid of the triangle is given by $\frac{z_1+z_2+z_3}{3}=0$, which is the origin. The circumcenter is also the origin, since all vertices lie on the circle $|z|=r$. In any triangle, if the centroid and the circumcenter coincide, the triangle must be equilateral.

## Takeaways 3.4

1. **Vector Addition:** $z_1+z_2+z_3=0$ means the centroid is the origin. If the circumcenter (origin) and centroid coincide, the triangle is equilateral.
2. **Modulus Algebra:** Using $|z_1+z_2|^2=|-z_3|^2$ is much faster than expanding everything.

## Problem 3.5: Minimum Distance Between Moving Particles

Two particles $A$ and $B$ move in space such that their position vectors at time $t\geq 0$ are:

$$\mathbf{r}_A=\begin{pmatrix}1\\0\\2\end{pmatrix}+t\begin{pmatrix}1\\1\\0\end{pmatrix} \quad \text{and} \quad \mathbf{r}_B=\begin{pmatrix}4\\2\\0\end{pmatrix}+t\begin{pmatrix}0\\1\\1\end{pmatrix}$$

(i) Express the squared distance $S(t)=|\mathbf{r}_A-\mathbf{r}_B|^2$ as a quadratic in $t$.
(ii) Find the minimum distance between the particles and the time at which this occurs.

**Hint:** Find the displacement vector $\mathbf{d}(t)=\mathbf{r}_A-\mathbf{r}_B$. Compute the dot product $\mathbf{d}\cdot\mathbf{d}$ to get the squared magnitude. You will get a quadratic expression $At^2+Bt+C$. Find the vertex of this parabola ($t=-B/2A$).

## Solution 3.5

**(i)** The vector connecting them is:

$$\mathbf{d} = \mathbf{r}_A - \mathbf{r}_B = \begin{pmatrix} 1-4 \\ 0-2 \\ 2-0 \end{pmatrix} + t\begin{pmatrix} 1-0 \\ 1-1 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -3+t \\ -2 \\ 2-t \end{pmatrix}$$

Squared distance:

$$S(t) = (-3+t)^2 + (-2)^2 + (2-t)^2$$
$$S(t) = (t^2 - 6t + 9) + 4 + (t^2 - 4t + 4)$$
$$S(t) = 2t^2 - 10t + 17$$

**(ii)** To minimize $S(t)$, we can use calculus or algebra.
**Method 1 (Calculus):** Find $S'(t) = 4t - 10$. Set $4t - 10 = 0 \implies t = 2.5$. Substitute $t = 2.5$ back into $S(t)$:

$$S(2.5) = 2(6.25) - 25 + 17 = 12.5 - 25 + 17 = 4.5$$

**Method 2 (Algebra - Completing the Square):** Rewrite $S(t)$ by completing the square:

$$\begin{aligned} S(t) &= 2(t^2 - 5t) + 17 \\ &= 2\left(t^2 - 5t + 6.25\right) - 2(6.25) + 17 \\ &= 2(t-2.5)^2 - 12.5 + 17 \\ &= 2(t-2.5)^2 + 4.5 \end{aligned}$$

This instantly yields the vertex at $t = 2.5$ and the minimum squared distance of 4.5.
In both cases, the minimum distance is $\sqrt{4.5} = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$.

## Takeaways 3.5

1. **Distance Squared:** Always minimize distance *squared* ($|\mathbf{d}|^2$) rather than distance ($|\mathbf{d}|$). It avoids square roots and simplifies the calculus derivatives.
2. **Kinematics Connection:** If velocities are constant vectors, the distance function is always a quadratic (convex parabola), ensuring a unique minimum.

## Problem 3.6: Complex System via Newton Sums

Find all complex numbers $z_1, z_2, z_3$ that satisfy the following system of equations simultaneously:

$$\begin{cases} z_1 + z_2 + z_3 = 3 \\ z_1^2 + z_2^2 + z_3^2 = 3 \\ z_1^3 + z_2^3 + z_3^3 = 3 \end{cases}$$

**Hint:** Do not try to substitute variables directly. Instead, assume $z_1, z_2, z_3$ are the roots of a cubic equation:

$$P(z) = z^3 - e_1 z^2 + e_2 z - e_3 = 0$$

Use the identity for the sum of squares: $\sum z^2 = (\sum z)^2 - 2\sum z_1 z_2$. Then use Newton's Sums to find the sum of cubes relation:

$$S_3 - e_1 S_2 + e_2 S_1 - 3e_3 = 0$$

where $S_k = z_1^k + z_2^k + z_3^k$.

## Solution 3.6

Let $e_1, e_2, e_3$ be the elementary symmetric polynomials. Then $z_1, z_2, z_3$ are the roots of

$$P(z) = z^3 - e_1 z^2 + e_2 z - e_3.$$

**Step 1: Find $e_1$.** From the first condition, $e_1 = z_1 + z_2 + z_3 = 3$.
**Step 2: Find $e_2$.** Use the identity $\sum z^2 = (\sum z)^2 - 2\sum z_1 z_2$:

$$3 = 3^2 - 2e_2 = 9 - 2e_2$$

$$2e_2 = 6 \implies e_2 = 3$$

**Step 3: Find $e_3$.** By Newton's Sums, $S_3 - e_1 S_2 + e_2 S_1 - 3e_3 = 0$, where $S_k = \sum z_i^k$:

$$3 - 3 \cdot 3 + 3 \cdot 3 - 3e_3 = 0$$

$$3 - 9 + 9 - 3e_3 = 0 \implies 3e_3 = 3 \implies e_3 = 1$$

**Step 4: Solve the cubic.** The polynomial is $P(z) = z^3 - 3z^2 + 3z - 1$. Recognize that this is $(z-1)^3 = z^3 - 3z^2 + 3z - 1$. Thus, the only root is $z = 1$ with multiplicity 3.
**Solution:** $z_1 = z_2 = z_3 = 1$.

## Takeaways 3.6

1. **Newton's Identities:** When dealing with sums of powers, always think about elementary symmetric polynomials and Newton's Sums relations.
2. **Binomial Recognition:** The polynomial $z^3 - 3z^2 + 3z - 1$ matches the binomial expansion of $(z-1)^3$.
3. **Symmetric System Strategy:** Systems involving $\sum z_i, \sum z_i^2, \sum z_i^3$ are often easiest when approached via symmetric polynomials.

## Problem 3.7: Proof that $e$ is Irrational

You are given the series definition

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots.$$

(i) Let $S_k = \sum_{n=0}^{k} \frac{1}{n!}$. Show that for any integer $k \geq 1$,

$$0 < e - S_k < \frac{1}{k \cdot k!}.$$

(ii) Suppose for contradiction that $e = \frac{p}{q}$ with positive integers $p, q$. Consider $X = q!\,(e - S_q)$ and show that $X$ is an integer.

(iii) Using (i) with $k = q$, prove that $0 < X < 1$.

(iv) Conclude that $e$ is irrational.

**Hint:(i)** Write the tail $e - S_k = \frac{1}{(k+1)!} + \frac{1}{(k+2)!} + \cdots$ and bound it by a geometric progression with ratio $\frac{1}{k+1}$.
**(ii)** Note $q!/n! \in \mathbb{Z}$ for all $0 \leq n \leq q$, so $q!S_q$ is an integer while $q!e = (q-1)!\,p$.
**(iii)** Multiply the inequality from (i) by $q!$.
**(iv)** There is no integer strictly between 0 and 1.

## Solution 3.7

**(i)** We have $e - S_k = \sum_{j=k+1}^{\infty} \frac{1}{j!} = \frac{1}{(k+1)!}\left(1 + \frac{1}{k+2} + \frac{1}{(k+2)(k+3)} + \cdots\right)$. This tail is strictly less than the geometric series with ratio $\frac{1}{k+1}$, so

$$0 < e - S_k < \frac{1}{(k+1)!} \cdot \frac{1}{1 - \frac{1}{k+1}} = \frac{1}{(k+1)!} \cdot \frac{k+1}{k} = \frac{1}{k \cdot k!}.$$

**(ii)** Define $X = q!\,(e - S_q) = q!e - q!\sum_{n=0}^{q} \frac{1}{n!}$. If $e = \frac{p}{q}$, then $q!e = p\,(q-1)! \in \mathbb{Z}$, and $q!/n! \in \mathbb{Z}$ for $n \leq q$, hence $q!S_q \in \mathbb{Z}$. Therefore $X \in \mathbb{Z}$.
**(iii)** Using (i) with $k = q$ and multiplying by $q!$ yields $0 < q!(e - S_q) < \frac{1}{q}$, i.e., $0 < X < 1$.
**(iv)** There is no integer strictly between 0 and 1, contradiction. Hence $e$ is irrational.

**Takeaways 3.7**

1. **Integrality Trick:** Multiplying by $q!$ clears denominators in partial sums, turning analysis into integer arithmetic.
2. **Series Bounding:** Bound factorial tails by simple geometric estimates to obtain sharp inequalities.
3. Similar techniques can be used to show $\frac{1}{e} = e^{-1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ is irrational as well. The product of $e$ and $e^{-1}$ is 1, so if one were rational, the other would be rational too.
4. Known fact:$e^r$ is irrational for any non-zero rational number $r$, published by Johann Heinrich Lambert in 1768 using continued fractions.

## 3.2 Advanced Last Resort Problems

The most challenging problems requiring multiple advanced techniques and mathematical maturity.

**Problem 3.8: Complex Ellipsoid Optimization**

Consider the ellipsoid $E$ defined by:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

where $a > b > c > 0$.
Find the point $P(x_0, y_0, z_0)$ on $E$ that maximizes the distance to the plane $\pi : x+y+z = 0$, and compute this maximum distance.

**Solution 3.8**

We maximize the distance $d = \dfrac{x+y+z}{\sqrt{3}}$ under $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$ with $x + y + z > 0$.
By Cauchy–Schwarz,

$$(x+y+z)^2 = \left(\tfrac{x}{a}a + \tfrac{y}{b}b + \tfrac{z}{c}c\right)^2 \le \left(\tfrac{x^2}{a^2} + \tfrac{y^2}{b^2} + \tfrac{z^2}{c^2}\right)(a^2+b^2+c^2) = a^2+b^2+c^2,$$

so $x + y + z \le \sqrt{a^2+b^2+c^2}$ with equality exactly when $\frac{x}{a}/a = \frac{y}{b}/b = \frac{z}{c}/c = k > 0$. Writing $x = ka^2$, $y = kb^2$, $z = kc^2$ in the constraint gives $k^2(a^2+b^2+c^2) = 1$, hence $k = 1/\sqrt{a^2+b^2+c^2}$ and

$$P = \left(\frac{a^2}{\sqrt{a^2+b^2+c^2}}, \frac{b^2}{\sqrt{a^2+b^2+c^2}}, \frac{c^2}{\sqrt{a^2+b^2+c^2}}\right), \quad d_{\max} = \frac{\sqrt{a^2+b^2+c^2}}{\sqrt{3}}.$$

**Takeaways 3.8**

1. **Cauchy-Schwarz Strategy:** To bound a linear expression (like $x+y+z$) given a quadratic constraint, carefully split the linear terms into a dot product that utilizes the constraint.
2. **Equality Condition:** The maximum or minimum in Cauchy-Schwarz is strictly achieved when the proportional vectors are aligned, instantly giving us the coordinate ratios.

**Remark 0.6: Distance Between Shapes**

In general, the distance between two shapes $A$ and $B$ is defined as

$$d(A,B) = \inf\{\|p-q\| : p \in A,\ q \in B\}.$$

For an ellipsoid and a plane, this reduces to finding the point on the ellipsoid whose normal is aligned with the plane's normal, yielding the maximum/minimum perpendicular separation between the surfaces. Here, inf means the greatest lower bound (or minimum if it exists).

**Problem 3.9: Cauchy's Root Bound via Triangle Inequality**

Let $P(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$ be a monic complex polynomial. Show that every root $\zeta$ of $P$ satisfies

$$|\zeta| \le 1 + M, \quad \text{where } M = \max_{0\le k<n} |a_k|.$$

Hint: Assume $|\zeta| > 1+M$ and compare $|\zeta|^n$ with $\sum_{k=0}^{n-1} |a_k|\,|\zeta|^k$ via the triangle inequality.

**Solution 3.9**

Assume for contradiction that $|\zeta| > 1 + M$. Since $P(\zeta) = 0$, we have $\zeta^n = -\sum_{k=0}^{n-1} a_k \zeta^k$. Divide by $\zeta^n$ to get:

$$1 = \left| -\sum_{k=0}^{n-1} a_k \zeta^{k-n} \right| \le \sum_{k=0}^{n-1} |a_k||\zeta|^{k-n}$$

Replacing $|a_k|$ with the maximum $M$ and letting $j = n - k$, we get:

$$1 \le M \sum_{j=1}^{n} |\zeta|^{-j}$$

Since all terms are positive, this finite sum is strictly less than the infinite geometric series:

$$1 < M \sum_{j=1}^{\infty} |\zeta|^{-j} = M\left(\frac{|\zeta|^{-1}}{1-|\zeta|^{-1}}\right) = \frac{M}{|\zeta|-1}$$

From our assumption that $|\zeta| > 1 + M$, we have $|\zeta| - 1 > M$. Therefore:

$$\frac{M}{|\zeta|-1} < \frac{M}{M} = 1$$

This implies $1 < 1$, which is a contradiction. Thus, all roots must satisfy $|\zeta| \le 1 + M$.

**Takeaways 3.9**

1. **Cauchy Bound:** Every root of a monic polynomial lies in the disk $|z| \leq 1 + \max|a_k|$.
2. **Triangle Inequality Tool:** Comparing $|z|^n$ with coefficient-weighted sums yields robust bounds.
3. **Rescaling Remarks:** Stronger bounds exist (e.g., Fujiwara's), derivable by rescaling or sharper estimates.

**Problem 3.10: Distance Between Skew Lines**

Consider two skew lines $L_1$ and $L_2$ in $\mathbb{R}^3$:

$$L_1: \quad \mathbf{r}_1(s) = (1,2,3) + s(2,-1,1) \tag{1}$$
$$L_2: \quad \mathbf{r}_2(t) = (0,1,-1) + t(1,1,-2) \tag{2}$$

(a) Verify that $L_1$ and $L_2$ are skew (neither parallel nor intersecting).
(b) Find the shortest distance between the two lines.
(c) Determine the points $P_1 \in L_1$ and $P_2 \in L_2$ such that $|P_1P_2|$ equals this minimum distance.
(d) Show that the line segment $P_1P_2$ is perpendicular to both $L_1$ and $L_2$.

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## Solution 3.10

**(a) Verifying skew lines:** Direction vectors: $\mathbf{d}_1 = (2, -1, 1)$ and $\mathbf{d}_2 = (1, 1, -2)$.
To test parallelism, compare component ratios:

$$\frac{2}{1} = 2, \qquad \frac{-1}{1} = -1, \qquad \frac{1}{-2} = -\frac{1}{2}.$$

These are not equal, so the lines are not parallel.
To test intersection, set

$$(1, 2, 3) + s(2, -1, 1) = (0, 1, -1) + t(1, 1, -2).$$

Equating coordinates gives

$$1 + 2s = t, \qquad 2 - s = 1 + t, \qquad 3 + s = -1 - 2t.$$

From the first two equations, $t = 1 + 2s = 1 - s$, so $s = 0$ and $t = 1$. Substituting into the third gives $3 = -3$, a contradiction. Therefore, the lines are skew.
**Finding the shortest distance and closest points:** Let $\mathbf{w} = \mathbf{r}_2(t) - \mathbf{r}_1(s)$. For the shortest distance, $\mathbf{w}$ must satisfy

$$\mathbf{w} \cdot \mathbf{d}_1 = 0, \qquad \mathbf{w} \cdot \mathbf{d}_2 = 0.$$

$$\mathbf{w} = (0, 1, -1) + t(1, 1, -2) - (1, 2, 3) - s(2, -1, 1) = (-1 + t - 2s, -1 + t + s, -4 - 2t - s).$$

So

$$\mathbf{w} \cdot \mathbf{d}_1 = 0 \implies t + 6s = -5,$$
$$\mathbf{w} \cdot \mathbf{d}_2 = 0 \implies 6t + s = -6,$$

giving

$$t = -\frac{31}{35}, \qquad s = -\frac{24}{35}.$$

**Part (c) Points:**

$$P_1 = \left(-\frac{13}{35}, \frac{94}{35}, \frac{81}{35}\right), \qquad P_2 = \left(-\frac{31}{35}, \frac{4}{35}, \frac{27}{35}\right).$$

**Part (b) Distance:**

$$\overrightarrow{P_1P_2} = P_2 - P_1 = \left(-\frac{18}{35}, -\frac{90}{35}, -\frac{54}{35}\right),$$

so

$$|P_1P_2| = \frac{18}{35}\sqrt{1 + 25 + 9} = \frac{18}{\sqrt{35}} = \frac{18\sqrt{35}}{35}.$$

**(d) Perpendicularity verification:** Because we found $P_1$ and $P_2$ specifically by enforcing $\mathbf{w} \cdot \mathbf{d}_1 = 0$ and $\mathbf{w} \cdot \mathbf{d}_2 = 0$, the segment $P_1P_2$ is perpendicular to both $L_1$ and $L_2$ by definition of our algebraic setup.

## Takeaways 3.10

1. **Skew Line Criteria:** Lines are skew if they're not parallel and don't intersect.
2. **Shortest-Distance Strategy:** Join general points on the two lines and impose perpendicularity to both direction vectors.
3. **Perpendicularity Condition:** Minimum distance occurs when connecting segment is perpendicular to both lines.

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## Problem 3.11: Powers of Roots and Recurrence Relations

Consider the polynomial $P(x) = x^4 - 6x^3 + 11x^2 - 6x + 1$ with roots $r_1, r_2, r_3, r_4$.
(i) Find the elementary symmetric polynomials $e_1, e_2, e_3, e_4$ in terms of the coefficients.
(ii) Let $S_k = r_1^k + r_2^k + r_3^k + r_4^k$ be the $k$-th power sum. Use Newton's identities to find $S_1, S_2, S_3, S_4$.
(iii) Establish the recurrence relation for $S_k$ when $k \geq 4$.
(iv) Without finding the actual roots, determine $S_5$ and $S_6$ using the recurrence relation.

## Solution 3.11

**(i) Elementary symmetric polynomials:** By Viète's formulas for $P(x) = x^4 - 6x^3 + 11x^2 - 6x + 1$,

$$e_1 = r_1 + r_2 + r_3 + r_4 = 6, \quad e_2 = \sum_{i<j} r_i r_j = 11, \quad e_3 = \sum_{i<j<k} r_i r_j r_k = 6,$$

$$e_4 = r_1 r_2 r_3 r_4 = 1.$$

**(ii) Newton's identities:** Let $S_k = r_1^k + r_2^k + r_3^k + r_4^k$. Newton's identities give

$$\begin{aligned} S_1 - e_1 = 0 \quad &\Rightarrow \quad S_1 = 6, \\ S_2 - e_1 S_1 + 2e_2 = 0 \quad &\Rightarrow \quad S_2 = 6 \cdot 6 - 2 \cdot 11 = 14, \\ S_3 - e_1 S_2 + e_2 S_1 - 3e_3 = 0 \quad &\Rightarrow \quad S_3 = 6 \cdot 14 - 11 \cdot 6 + 3 \cdot 6 = 36, \\ S_4 - e_1 S_3 + e_2 S_2 - e_3 S_1 + 4e_4 = 0 \quad &\Rightarrow \quad S_4 = 6 \cdot 36 - 11 \cdot 14 + 6 \cdot 6 - 4 = 94. \end{aligned}$$

**(iii) Recurrence relation:** For $k \geq 4$,

$$S_k - e_1 S_{k-1} + e_2 S_{k-2} - e_3 S_{k-3} + e_4 S_{k-4} = 0,$$

so

$$S_k - 6S_{k-1} + 11S_{k-2} - 6S_{k-3} + S_{k-4} = 0, \qquad S_k = 6S_{k-1} - 11S_{k-2} + 6S_{k-3} - S_{k-4}.$$

**(iv) Computing $S_5$ and $S_6$:** Using the recurrence,

$$S_5 = 6S_4 - 11S_3 + 6S_2 - S_1 = 6 \cdot 94 - 11 \cdot 36 + 6 \cdot 14 - 6 = 246,$$

and

$$S_6 = 6S_5 - 11S_4 + 6S_3 - S_2 = 6 \cdot 246 - 11 \cdot 94 + 6 \cdot 36 - 14 = 644.$$

**Alternative Method (Self-Reciprocal Trick):** Notice $P(x)$ is self-reciprocal. Divide the equation by $x^2$:

$$x^2 - 6x + 11 - \frac{6}{x} + \frac{1}{x^2} = 0$$

Substitute $u = x + \frac{1}{x}$, which implies $x^2 + \frac{1}{x^2} = u^2 - 2$:

$$(u^2 - 2) - 6u + 11 = 0 \implies u^2 - 6u + 9 = 0 \implies (u - 3)^2 = 0$$

Since every root of $P(x)$ satisfies $u = 3$, they must satisfy $r^2 - 3r + 1 = 0$. Thus, the power sums satisfy the much simpler second-order recurrence: $S_k = 3S_{k-1} - S_{k-2}$. We can recompute $S_5$ and $S_6$ trivially:

$$S_5 = 3S_4 - S_3 = 3(94) - 36 = 246$$

$$S_6 = 3S_5 - S_4 = 3(246) - 94 = 644$$

**Takeaways 3.11**

1. **Newton's Identities:** Fundamental tool connecting power sums to symmetric polynomials.
2. **Recurrence Relations:** Enable computation of higher power sums without finding roots.
3. **Self-Reciprocal Substitution:** Self-reciprocal polynomials of even degree $2n$ can always be reduced to polynomials of degree $n$ using the substitution $u = x + \frac{1}{x}$. This often collapses complex 4th-order recurrence relations into 2nd-order ones.
4. **Coefficient Relationships:** Direct connection between polynomial coefficients and root properties.

## Problem 3.12: The Irrationality of $\zeta(3)$

Let

$$\zeta(3) = 1 + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \cdots = \sum_{n=1}^{\infty} \frac{1}{n^3}$$

denote Apéry's constant. You can assume (without proof) that the sequence converges, and that $\zeta(3)$ is known to be approximately $1.2020569\ldots$ and is not known to be rational or irrational.
In this problem you will show $\zeta(3)$ is irrational using a sequence of integrals and bounds (you may use the stated facts without proof).

(a) **Irrationality Criterion.** Let $\alpha = \frac{p}{q}$ be rational with integers $p, q > 0$. Show that for any integers $A, B$ with $A\alpha - B \neq 0$,

$$|A\alpha - B| \geq \frac{1}{q}.$$

(b) **Optimization bound.** For $0 \leq x, y, z \leq 1$, consider

$$g(x, y, z) = \frac{x(1-x)\, y(1-y)\, z(1-z)}{1 - (1-xy)z}.$$

Show that $g$ attains a global maximum value $g_{\max} = (\sqrt{2} - 1)^4$ (you may assume the maximum occurs when $x = y$ and optimize in $z$ first).

(c) **Integral construction.** Let $d_n = \operatorname{lcm}(1, 2, \ldots, n)$ and

$$I_n = \int_0^1 \int_0^1 \int_0^1 \Big(g(x, y, z)\Big)^n \frac{dx\, dy\, dz}{1 - (1-xy)z}.$$

here, lcm means least common multiple. You are given:

- $I_n > 0$ for all $n \geq 1$;
- $2d_n^3 I_n = A_n\, \zeta(3) + B_n$ for some integers $A_n, B_n$;
- for large $n$, $d_n < 3^n$.

Do NOT prove these facts, just use them.

Using (b), deduce a bound of the form

$$0 < I_n < C\, (\sqrt{2} - 1)^{4n},$$

where $C$ is independent of $n$.

(d) **Conclusion.** Assuming $\zeta(3)$ is rational, apply (a) to $X_n = A_n\, \zeta(3) + B_n$ and use (c) together with $d_n < 3^n$ to derive a contradiction as $n \to \infty$. Conclude that $\zeta(3)$ is irrational.

**Hint:(a)** $A\frac{p}{q} - B = \frac{Ap-Bq}{q}$ and use that a nonzero integer has magnitude at least 1.
**(b)** Set $y = x$, maximize in $z$ to get $z = \frac{1}{1+x}$, then maximize in $x \in [0, 1]$.
**(c)** Bound the integrand by $g^n_{\max}$ and pull it outside the integral.
**(d)** Compare the fixed lower bound $1/q$ with the upper bound decaying like $\left(27(\sqrt{2}-1)^4\right)^n$.

## Solution 3.12

**(a)** If $\alpha = \frac{p}{q}$, then $A\alpha - B = \frac{Ap-Bq}{q}$. The numerator is a nonzero integer, so $|Ap - Bq| \geq 1$, hence $|A\alpha - B| \geq \frac{1}{q}$.
**(b)** Setting $y = x$ and optimizing first in $z$ yields $z = \frac{1}{1+x}$. Substituting and maximizing over $x \in [0, 1]$ gives $\max g = (\sqrt{2} - 1)^4$.
**(c)** Since $g(x, y, z) \leq (\sqrt{2} - 1)^4$ on $[0, 1]^3$, we have

$$\begin{aligned} 0 < I_n &= \iiint g(x, y, z)^n \frac{dx\, dy\, dz}{1 - (1 - xy)z} \\ &< \left((\sqrt{2} - 1)^4\right)^n \iiint \frac{dx\, dy\, dz}{1 - (1 - xy)z} \\ &= C\,(\sqrt{2} - 1)^{4n}. \end{aligned}$$

Note that the remaining integral $C = \iiint \frac{dx\, dy\, dz}{1-(1-xy)z}$ converges to a finite positive constant (it actually evaluates to $2\zeta(3)$). This confirms the upper bound does not blow up to infinity, and $C$ is indeed a valid, independent constant.
**(d)** Write $X_n = A_n\,\zeta(3) + B_n = 2d_n^3 I_n$. By (a), $|X_n| \geq \frac{1}{q}$ if $\zeta(3)$ were rational. But from (c) and $d_n < 3^n$ we get

$$|X_n| = 2d_n^3 I_n < 2\,(3^n)^3\,C\,(\sqrt{2} - 1)^{4n} = 2C\left(27(\sqrt{2} - 1)^4\right)^n,$$

whose right-hand side tends to 0 as $n \to \infty$. This contradicts $|X_n| \geq \frac{1}{q} > 0$. Therefore $\zeta(3)$ is irrational.

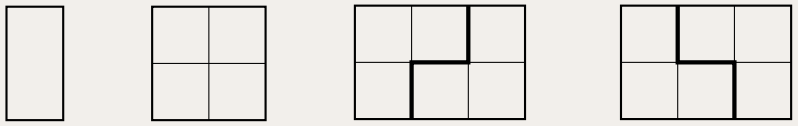
## Takeaways 3.12

1. **Logic of Irrationality:** Use integer linear forms and uniform lower bounds $(1/q)$ vs exponentially decaying upper bounds to force contradictions.
2. **Bounding Integrals:** Maximize the integrand to bound the whole integral without evaluation.
3. **Assumption Management:** Treat given constructions (LCM growth, integral identity) as black boxes and connect them via inequalities.

## Problem 3.13: Tetris Tile Tiling

A puzzle game requires you to completely tile a $2 \times n$ rectangular grid using three specific types of polyomino blocks:

1. **The I-Domino:** A $2 \times 1$ vertical tile.
2. **The O-Tetromino:** A $2 \times 2$ square tile.
3. **The L-Pairs:** A $2 \times 3$ rectangular block formed by interlocking two L-trominoes. There are exactly two distinct patterns to form this $2 \times 3$ block, shown below.



I-Domino O-Tetromino L-Pair (Pattern 1) L-Pair (Pattern 2)

Let $W_n$ be the total number of valid ways to tile a $2 \times n$ grid using only these blocks.

(i) By considering the possible rightmost blocks of the grid, explain why the sequence satisfies the recurrence relation $W_n = W_{n-1} + W_{n-2} + 2W_{n-3}$ for all $n \geq 3$.

(ii) The sequence can be expressed in the explicit form $W_n = A\alpha^n + B\beta^n + C\gamma^n$, where $\alpha, \beta, \gamma$ are the roots of the characteristic polynomial $P(z) = z^3 - z^2 - z - 2 = 0$. Show that the roots are $z = 2$ and $z = \text{cis}\left(\pm\dfrac{2\pi}{3}\right)$.

(iii) Given $W_0 = 1$ *(Note: By mathematical convention for the empty grid, $W_0 = 1$)*, $W_1 = 1$, $W_2 = 2$, establish a system of equations for $n = 0, 1, 2$. By summing these equations and utilising the properties of the complex cube roots of unity, prove that $A = \dfrac{4}{7}$.

(iv) Hence, deduce the exact closed-form expression for $W_n$ purely in terms of real numbers, and formally show that $\lim\limits_{n\to\infty} \dfrac{W_{n+1}}{W_n} = 2$.

**Hint:(i)** Think about how any valid tiling *must* end. It can only end in one of the defined fundamental blocks. How much width does each block take up? Are the ending scenarios mutually exclusive?

**(ii)** Use the Rational Root Theorem to guess a small integer root (try factors of 2). Once you factor out the linear term, use the quadratic formula for the remaining roots. Do you recognise the resulting complex numbers?

**(iii)** Let the complex roots be $\omega$ and $\bar{\omega}$. Add the three equations for $n = 0, 1, 2$ together vertically. What happens to the $B$ and $C$ coefficients when you factor out $(1 + \omega + \omega^2)$?

**(iv)** Since $W_n$ is a sequence of real integers, the coefficients $B$ and $C$ must be complex conjugates. Let $B = u + vi$ and $C = u - vi$, and solve for $u$ and $v$ using the $W_0$ and $W_1$ equations. For the limit, identify which term dominates as $u \to \infty$.

## Solution 3.13

**(i)** Any valid tiling must end with one of the three block types (mutually exclusive and exhaustive): an I-Domino (width 1) leaves $W_{n-1}$ tilings; an O-Tetromino (width 2) leaves $W_{n-2}$; an L-Pair (width 3) has 2 patterns, contributing $2W_{n-3}$. Hence $W_n = W_{n-1} + W_{n-2} + 2W_{n-3}$.

**(ii)** $P(2) = 8 - 4 - 2 - 2 = 0$, so $(z-2)$ is a factor, giving $P(z) = (z-2)(z^2+z+1)$. The roots of $z^2+z+1=0$ are the primitive cube roots of unity $\omega = \operatorname{cis}\frac{2\pi}{3}$ and $\bar{\omega} = \operatorname{cis}\left(-\frac{2\pi}{3}\right)$, so the three roots are $2$, $\operatorname{cis}\frac{2\pi}{3}$, $\operatorname{cis}\left(-\frac{2\pi}{3}\right)$.

**(iii)** Write $W_n = A \cdot 2^n + B\omega^n + C\bar{\omega}^n$ and substitute $n = 0, 1, 2$:

$$A + B + C = 1, \quad 2A + B\omega + C\bar{\omega} = 1, \quad 4A + B\omega^2 + C\bar{\omega}^2 = 2.$$

Summing: $7A + B(1+\omega+\omega^2) + C(1+\bar{\omega}+\bar{\omega}^2) = 4$. Since $1+\omega+\omega^2 = 0$, this gives $7A = 4$, so $A = \frac{4}{7}$.

**(iv)** Since $W_n \in \mathbb{R}$, write $B = u + vi$, $C = u - vi$. From $n=0$: $\frac{4}{7} + 2u = 1 \Rightarrow u = \frac{3}{14}$. From $n=1$: $B\omega + C\bar{\omega} = 2\operatorname{Re}(B\omega) = -u - v\sqrt{3} = -\frac{1}{7}$, so $v = -\frac{1}{14\sqrt{3}}$.

To convert to real trigonometric functions, we explicitly expand the complex conjugate terms using De Moivre's Theorem:

$$\begin{aligned} B\omega^n + C\bar{\omega}^n &= 2\operatorname{Re}\left((u+vi)\left(\cos\frac{2n\pi}{3} + i\sin\frac{2n\pi}{3}\right)\right) \\ &= 2\left(u\cos\frac{2n\pi}{3} - v\sin\frac{2n\pi}{3}\right) \end{aligned}$$

Showing the $i^2 = -1$ explains why the imaginary components produce a $-v\sin\frac{2n\pi}{3}$ term. Substituting $u = \frac{3}{14}$ and $v = -\frac{1}{14\sqrt{3}}$, the negative $v$ value creates a positive sine term in the final answer:

$$\boxed{W_n = \frac{4}{7}\cdot 2^n + \frac{3}{7}\cos\frac{2n\pi}{3} + \frac{\sqrt{3}}{21}\sin\frac{2n\pi}{3}.}$$

The trig terms are bounded while $2^n \to \infty$, so $W_n \sim \frac{4}{7}\cdot 2^n$ and $\lim_{n\to\infty}\frac{W_{n+1}}{W_n} = \lim_{n\to\infty}\frac{\frac{4}{7}\cdot 2^{n+1}}{\frac{4}{7}\cdot 2^n} = 2$.

## Takeaways 3.13

1. **Combinatorics as Linear Recurrences:** Any tiling problem where tiles have fixed widths generates a linear recurrence; the characteristic polynomial technique converts geometric counting into root-finding.
2. **The Magic of Roots of Unity:** Whenever a characteristic polynomial factors to expose cube roots of unity, the identity $1 + \omega + \omega^2 = 0$ is a powerful tool for eliminating unwanted variables from systems of equations.
3. **Dominant Term Determines Growth Rate:** In a sequence whose closed form is a sum of exponentials, the term with the largest base dominates; this immediately yields the ratio limit without computing the exact coefficients of the subdominant terms.

# 4 Part 2: Problems with Hints and Solutions (Concise)

Part 2 presents the remaining 44 problems with upside-down hints followed by solution sketches. These problems provide comprehensive practice across the full spectrum of Problem 16 topics and difficulty levels.

## 4.1 Easy Last Resort Problems

Foundation problems that introduce key concepts with straightforward applications.

**Problem 4.1: Distance Ratio Regions**

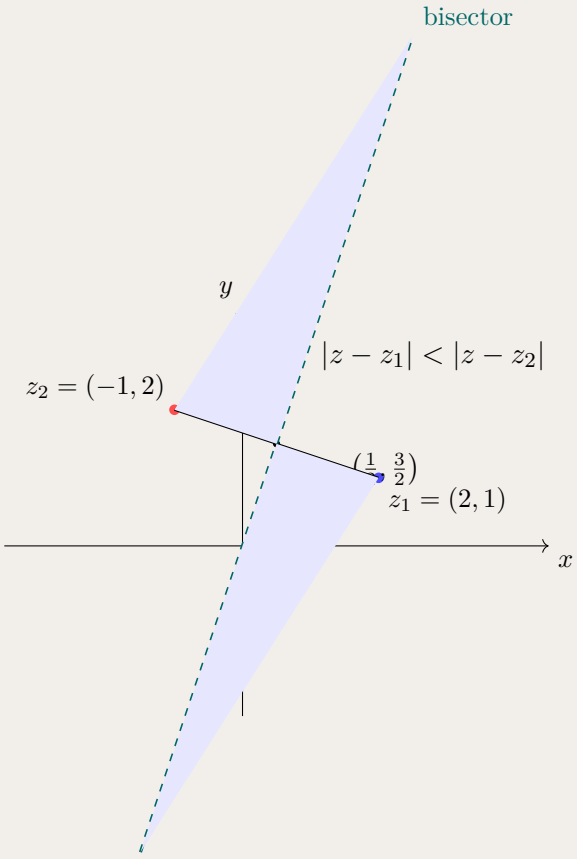
Let $z_1 = 2 + i$ and $z_2 = -1 + 2i$. Sketch the region in the complex plane where:

$$\frac{|z - z_1|}{|z - z_2|} < 1$$

**Hint:** Consider the geometric interpretation of $|z - z_1|/|z - z_2| < 1$. This represents the locus where one distance is smaller than another.

## Solution 4.1

The inequality $\frac{|z-z_1|}{|z-z_2|} < 1$ is equivalent to $|z-z_1| < |z-z_2|$. This means point $z$ is closer to $z_1 = 2+i$ than to $z_2 = -1+2i$. The locus of points equidistant from two points is the perpendicular bisector of the line segment joining them. The midpoint of $z_1 z_2$ is $\frac{(2+i)+(-1+2i)}{2} = \frac{1+3i}{2}$, which is $M\left(\frac{1}{2}, \frac{3}{2}\right)$. The direction vector is $z_2 - z_1 = (-1+2i) - (2+i) = -3+i$. The perpendicular bisector has normal vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$. The Cartesian equation of the line is $-3(x-0.5) + 1(y-1.5) = 0 \implies y = 3x$. Because the region inequality is strictly less than ($<$), the boundary line $y = 3x$ must be drawn as a **dashed/dotted line**. The region where $|z-z_1| < |z-z_2|$ is the half-plane containing $z_1$.




## Takeaways 4.1

1. **Distance Comparison:** The locus $|z-a| = |z-b|$ is the perpendicular bisector of segment $ab$.
2. **Half-Plane Regions:** Inequality $|z-a| < |z-b|$ defines the half-plane containing $a$.

## Problem 4.2: Orthocenter Vector Identity

Let $ABC$ be a triangle with circumcentre $O$, and set

$$\vec{OA} = \mathbf{a}, \qquad \vec{OB} = \mathbf{b}, \qquad \vec{OC} = \mathbf{c}.$$

Define the point $H$ by

$$\vec{OH} = \mathbf{a} + \mathbf{b} + \mathbf{c}.$$

Show that $\vec{BH} \perp \vec{AC}$.

**Hint:** Express $\vec{BH}$ and $\vec{AC}$ in terms of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and compute their dot product. Use that $|\mathbf{a}| = |\mathbf{c}|$ since $O$ is the circumcentre.

## Solution 4.2

Compute

$$\vec{BH} = \vec{OH} - \vec{OB} = (\mathbf{a} + \mathbf{b} + \mathbf{c}) - \mathbf{b} = \mathbf{a} + \mathbf{c},$$

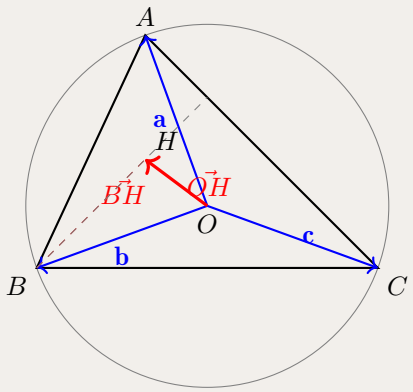
and

$$\vec{AC} = \vec{OC} - \vec{OA} = \mathbf{c} - \mathbf{a}.$$

Their dot product is

$$\begin{aligned}(\mathbf{a} + \mathbf{c}) \cdot (\mathbf{c} - \mathbf{a}) &= \mathbf{a} \cdot \mathbf{c} - \mathbf{a} \cdot \mathbf{a} + \mathbf{c} \cdot \mathbf{c} - \mathbf{c} \cdot \mathbf{a} \\ &= |\mathbf{c}|^2 - |\mathbf{a}|^2 \\ &= 0,\end{aligned}$$

since $|\mathbf{a}| = |\mathbf{c}|$ (both equal the circumradius). Hence $\vec{BH} \perp \vec{AC}$.




## Takeaways 4.2

1. **Vector orthocenter:** The identity $\vec{OH} = \vec{OA} + \vec{OB} + \vec{OC}$ characterises the triangle's orthocentre relative to the circumcentre.
2. **Dot-product trick:** Use $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{v} - \mathbf{u}) = |\mathbf{v}|^2 - |\mathbf{u}|^2$ when equal magnitudes appear.

## Problem 4.3: Imaginary Part Constraints

Sketch the region in the complex plane where:

$$\text{Im}(2z + 3\bar{z}) \geq 1$$

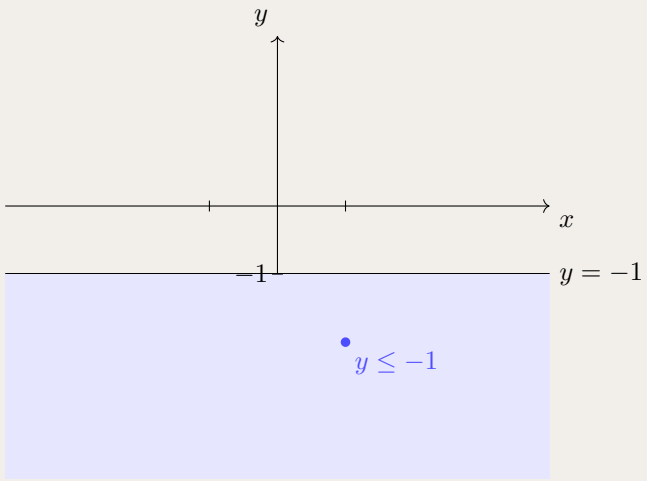
**Hint:** For a linear combination $az + b\bar{z}$, the imaginary part has a specific geometric interpretation in the complex plane.

## Solution 4.3

Let $z = x + iy$, so $\bar{z} = x - iy$.

$$\begin{aligned} 2z + 3\bar{z} &= 2(x + iy) + 3(x - iy) \\ &= 2x + 2iy + 3x - 3iy \\ &= 5x - iy \end{aligned}$$

Therefore, $\text{Im}(2z + 3\bar{z}) = -y$. The constraint becomes $-y \geq 1$, which is equivalent to $y \leq -1$. This is the region below the horizontal line $y = -1$. Because the original inequality is inclusive ($\geq$), the boundary line $y = -1$ must be drawn as a **solid line** (unlike the strict inequality in Problem 4.1).




## Takeaways 4.3

1. **Linear Combinations:** For $az + b\bar{z}$ with $z = x + iy$, substitute and collect real/imaginary parts.
2. **Half-Plane Regions:** Constraints on Im(...) or Re(...) typically define half-planes.
3. **Boundary Inclusion:** Always check the inequality symbol; inclusive inequalities ($\leq, \geq$) require a solid boundary line, while strict inequalities ($<, >$) require a dashed line.

## Problem 4.4: Leibniz Formula for $\pi$

(a) Show that for any integer $n \geq 0$ and $x \in \mathbb{R}$:

$$\frac{1}{1+x^2} = \sum_{k=0}^{n} (-1)^k x^{2k} + \frac{(-1)^{n+1} x^{2n+2}}{1+x^2}$$

(b) Deduce that:

$$\int_0^1 \frac{1}{1+x^2}\,dx = \sum_{k=0}^{n} \frac{(-1)^k}{2k+1} + (-1)^{n+1} \int_0^1 \frac{x^{2n+2}}{1+x^2}\,dx$$

(c) Show that for $n \geq 0$:

$$0 \leq \int_0^1 \frac{x^{2n+2}}{1+x^2}\,dx \leq \frac{1}{2n+3}$$

(d) Hence, prove the Leibniz formula for $\pi$:

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$$

**Hint:**

- **Part (a):** Consider the sum of a Geometric Progression with first term $a = 1$, common ratio $r = -x^2$, and $n+1$ terms. The sum formula is $S_N = \frac{a(1-r^N)}{1-r}$.
- **Part (b):** Integrate both sides of the identity established in part (a) with respect to $x$ over the interval $[0, 1]$. Recall standard integrals involving inverse trigonometric functions.
- **Part (c):** Analyze the integrand $\frac{x^{2n+2}}{1+x^2}$ over the interval $0 \leq x \leq 1$. Notice that $1 + x^2 \geq 1$, which implies $\frac{1}{1+x^2} \leq 1$.
- **Part (d):** Apply the limit as $n \to \infty$ to the equation derived in part (b). Use the Squeeze Theorem (Sandwich Theorem) on the inequality from part (c).

## Solution 4.4

**(a)** The expression $\sum_{k=0}^{n}(-1)^k x^{2k} = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n}$ is a geometric series. Note that the sum from $k = 0$ to $n$ contains exactly $n + 1$ terms, which is why the exponent in the numerator is $n + 1$. Using the sum formula $S = \frac{a(1-r^{n+1})}{1-r}$:

$$\sum_{k=0}^{n}(-1)^k x^{2k} = \frac{1 \cdot (1 - (-x^2)^{n+1})}{1 - (-x^2)} = \frac{1 - (-1)^{n+1}x^{2n+2}}{1 + x^2}$$

Rearranging terms:

$$\frac{1}{1 + x^2} = \sum_{k=0}^{n}(-1)^k x^{2k} + \frac{(-1)^{n+1}x^{2n+2}}{1 + x^2}$$

**(b)** Integrating both sides from $x = 0$ to $x = 1$:

$$\int_0^1 \frac{1}{1 + x^2}\,dx = \int_0^1 \left(\sum_{k=0}^{n}(-1)^k x^{2k}\right) dx + \int_0^1 \frac{(-1)^{n+1}x^{2n+2}}{1 + x^2}\,dx$$

Evaluating the left side and the summation term:

$$\left[\tan^{-1} x\right]_0^1 = \sum_{k=0}^{n}(-1)^k \left[\frac{x^{2k+1}}{2k + 1}\right]_0^1 + (-1)^{n+1} \int_0^1 \frac{x^{2n+2}}{1 + x^2}\,dx$$

$$\frac{\pi}{4} = \sum_{k=0}^{n} \frac{(-1)^k}{2k + 1} + (-1)^{n+1} \int_0^1 \frac{x^{2n+2}}{1 + x^2}\,dx$$

**(c)** For $x \in [0, 1]$, we have $1 \leq 1+x^2$. Therefore, $\frac{1}{1+x^2} \leq 1$. Multiplying by $x^{2n+2}$ (which is non-negative):

$$\frac{x^{2n+2}}{1 + x^2} \leq x^{2n+2}$$

Integrating from 0 to 1:

$$\int_0^1 \frac{x^{2n+2}}{1 + x^2}\,dx \leq \int_0^1 x^{2n+2}\,dx = \left[\frac{x^{2n+3}}{2n + 3}\right]_0^1 = \frac{1}{2n + 3}$$

Since the integrand is non-negative, the lower bound is 0. Thus:

$$0 \leq \int_0^1 \frac{x^{2n+2}}{1 + x^2}\,dx \leq \frac{1}{2n + 3}$$

**(d)** By explicitly writing the limits for the inequality from (c):

$$\lim_{n\to\infty} 0 \leq \lim_{n\to\infty} \int_0^1 \frac{x^{2n+2}}{1 + x^2}\,dx \leq \lim_{n\to\infty} \frac{1}{2n + 3} \implies 0 \leq \lim_{n\to\infty} \int_0^1 \frac{x^{2n+2}}{1 + x^2}\,dx \leq 0$$

Thus, by the Squeeze Theorem, the remainder integral $\int_0^1 \frac{x^{2n+2}}{1+x^2}\,dx \to 0$. Taking the limit $n \to \infty$ in the result from (b):

$$\frac{\pi}{4} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k + 1}$$

## Takeaways 4.4

1. **Series Expansion integration:** A powerful method to evaluate constants (like $\pi$) involves expanding a function into a series and integrating term by term.
2. **Error Estimation:** In Extension 2 Mathematics, proving the validity of an infinite series often requires estimating the "remainder" or "error" term and showing it approaches zero.
3. **Inequalities:** Bounding a complex integral by a simpler one (e.g., $\int \frac{x^n}{1+x^2} \leq \int x^n$) is a standard technique for convergence proofs.

## 4.2 Medium Last Resort Problems

Problems that develop technique combinations and multi-step reasoning.

### Problem 4.5: Arithmetic–Geometric Mean and an Elliptic Integral

Let $a$ and $b$ be positive real numbers with $a > b$. Define

$$I(a,b) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}}.$$

The AGM sequences are defined by $a_1 = a$, $b_1 = b$ and for $n \geq 1$:

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = \sqrt{a_n b_n}.$$

(i) By using the substitution $\sin\theta = \frac{2b\sin\phi}{(a+b)+(a-b)\sin^2\phi}$, show that

$$I(a,b) = I\Big(\frac{a+b}{2}, \sqrt{ab}\Big).$$

(ii) Prove that $a_n$ and $b_n$ converge to the same limit $L$, and deduce

$$I(a,b) = \frac{\pi}{2L}.$$

(iii) Show that for $a > b > 0$,

$$\frac{\pi}{2a} < I(a,b) < \frac{\pi}{2b},$$

and hence $b < L < a$.

(iv) By substituting $x = a\tan\theta$ show that

$$\int_0^{\infty} \frac{dx}{\sqrt{(x^2+a^2)(x^2+b^2)}} = I(a,b).$$

**Hint:**

- For (i) use Gauss's transformation (a standard substitution for elliptic integrals); if short on time you may accept the invariance $I(a_n,b_n) = I(a_{n+1},b_{n+1})$ and proceed.
- For (ii) use monotonicity: $a_{n+1} \leq a_n$, $b_{n+1} \geq b_n$, and $b_n \leq a_n$ for all $n$, so both converge to the same limit $L$.
- For (iii) bound the integrand by using $b^2 < a^2\cos^2\theta + b^2\sin^2\theta < a^2$ on $(0,\pi/2)$.
- For (iv) set $x = a\tan\theta$ and simplify the radical; then apply the AGM iteration numerically to find $L$.

## Solution 4.5

**(i)** The substitution

$$\sin\theta = \frac{2b\sin\phi}{(a+b)+(a-b)\sin^2\phi}$$

is Gauss's transformation and (after simplifying differentials) converts the integrand to the same form with parameters $(\frac{a+b}{2}, \sqrt{ab})$, proving the invariance.

**(ii)** The AGM sequences satisfy

$$b_1 \le b_2 \le \cdots \le a_2 \le a_1,$$

so they converge to a common limit $L$. Using (i) repeatedly gives

$$I(a,b) = I(a_2,b_2) = \cdots = I(L,L) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{L^2}} = \frac{\pi}{2L}.$$

**(iii)** For $0 \le \theta \le \pi/2$ we have $b^2 \le a^2\cos^2\theta + b^2\sin^2\theta \le a^2$, so

$$\frac{1}{a} \le \frac{1}{\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}} \le \frac{1}{b}.$$

Integrate to obtain the stated bounds; combining with (ii) gives $b < L < a$.

**(iv)** Let $x = a\tan\theta$ so $dx = a\sec^2\theta\,d\theta$. Then

$$\begin{aligned}\int_0^\infty \frac{dx}{\sqrt{(x^2+a^2)(x^2+b^2)}} &= \int_0^{\pi/2} \frac{a\sec^2\theta\,d\theta}{\sqrt{a^2\tan^2\theta + a^2}\sqrt{a^2\tan^2\theta + b^2}} \\ &= \int_0^{\pi/2} \frac{d\theta}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}\end{aligned}$$

To match the exact definition of $I(a,b)$, use the complementary angle substitution $\alpha = \frac{\pi}{2} - \theta \implies d\theta = -d\alpha$. The limits 0 to $\frac{\pi}{2}$ become $\frac{\pi}{2}$ to 0, and sine and cosine swap:

$$\begin{aligned}&= \int_{\pi/2}^0 \frac{-d\alpha}{\sqrt{a^2\sin^2(\frac{\pi}{2}-\alpha) + b^2\cos^2(\frac{\pi}{2}-\alpha)}} \\ &= \int_0^{\pi/2} \frac{d\alpha}{\sqrt{a^2\cos^2\alpha + b^2\sin^2\alpha}} = I(a,b).\end{aligned}$$

## Takeaways 4.5

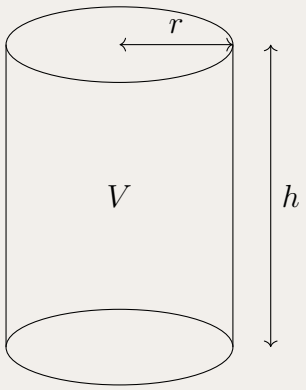
1. The arithmetic–geometric mean iteration quickly reduces certain elliptic integrals to an elementary value $\pi/(2L)$; convergence is typically quadratic.
2. Bounding integrals by simple constants is a reliable way to obtain order estimates and locate limits.
3. The substitution $x = a\tan\theta$ is useful for converting improper integrals with symmetric quadratic factors into trigonometric elliptic forms.

## Problem 4.6: AM-GM with Weighted Constraints

(a) Given positive real numbers $p$ and $q$, show that:

$$2p + q \geq 3\sqrt[3]{p^2 q}$$

(b) A closed cylindrical can has radius $r$, height $h$, and a fixed Total Surface Area $A$. Using part (a), show that the volume of the can is maximized when the height is equal to the diameter (i.e., $h = 2r$).




**Hint:** Use the weighted AM-GM inequality where the weights correspond to the constraint structure.

## Solution 4.6

**(a) Proof:** Consider the three positive numbers $p, p, q$. Applying AM-GM:

$$\frac{p+p+q}{3} \geq \sqrt[3]{p \cdot p \cdot q}$$
$$\frac{2p+q}{3} \geq \sqrt[3]{p^2q}$$
$$2p+q \geq 3\sqrt[3]{p^2q}$$

**(b) Application:** Let Surface Area $A$ be constant, and we maximize Volume $V$.

$$A = 2\pi r^2 + 2\pi rh$$
$$V = \pi r^2 h$$

Split the curved surface area term $2\pi rh$ into two equal parts: $\pi rh$ and $\pi rh$. Apply AM-GM to the three terms: $2\pi r^2$, $\pi rh$, and $\pi rh$.

$$\text{Sum} = 2\pi r^2 + \pi rh + \pi rh = A \quad \text{(Constant)}$$
$$\text{Product} = (2\pi r^2)(\pi rh)(\pi rh) = 2\pi^3 r^4 h^2 = 2\pi(\pi r^2 h)^2 = 2\pi V^2$$

Using AM-GM:

$$\frac{2\pi r^2 + \pi rh + \pi rh}{3} \geq \sqrt[3]{2\pi V^2}$$
$$\frac{A}{3} \geq \sqrt[3]{2\pi V^2}$$

Since $A$ is fixed, the maximum Volume $V$ occurs when equality holds. Equality holds when the three terms are equal:

$$2\pi r^2 = \pi rh$$
$$2r = h$$

Thus, the volume is maximized when the height equals the diameter. □

## Takeaways 4.6

1. **Weighted AM-GM:** When surface area terms have different coefficients, split larger terms to create equal weights in the AM-GM application.
2. **Strategic Grouping:** Choose terms that when multiplied together yield a power of the volume expression to be maximized.

## Problem 4.7: Viète's Formula and Infinite Cosine Product

(i) Prove that, for all integers $n \geq 1$ and $\theta \neq m\pi$ (where $m$ is any integer):

$$\prod_{k=1}^{n} \cos\left(\frac{\theta}{2^k}\right) = \frac{\sin\theta}{2^n \sin\left(\frac{\theta}{2^n}\right)}.$$

(ii) Hence show that for $\theta \neq 0$:

$$\prod_{k=1}^{\infty} \cos\left(\frac{\theta}{2^k}\right) = \frac{\sin\theta}{\theta}.$$

(iii) Let $u_1 = \sqrt{2}$ and $u_{n+1} = \sqrt{2 + u_n}$ for $n \geq 1$. Prove that for all $n \geq 1$:

$$u_n = 2\cos\left(\frac{\pi}{2^{n+1}}\right).$$

(iv) Using (ii) and (iii), prove Viète's Formula:

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdots$$

**Hint:** For (i) use induction; for the inductive step apply the double-angle identity $\sin 2A = 2\sin A\cos A$. For (ii) take the limit as $n \to \infty$ and use the small-angle limit $\sin x \sim x$. For (iii) use the identity $1 + \cos 2A = 2\cos^2 A$ to rewrite the recursion. For (iv) substitute $\theta = \pi/2$ into (ii) and relate the cosine factors to $u_n/2$ from (iii).

## Solution 4.7

**(i)** Base case $n = 1$ is immediate from $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$. Assume the formula holds for $n = k$. Then

$$\prod_{r=1}^{k+1} \cos\left(\frac{\theta}{2^r}\right) = \left(\prod_{r=1}^{k} \cos\left(\frac{\theta}{2^r}\right)\right) \cos\left(\frac{\theta}{2^{k+1}}\right)$$
$$= \frac{\sin\theta}{2^k \sin\left(\frac{\theta}{2^k}\right)} \cos\left(\frac{\theta}{2^{k+1}}\right).$$

Use $\sin(2A) = 2\sin A\cos A$ with $A = \frac{\theta}{2^{k+1}}$ to write $\sin\left(\frac{\theta}{2^k}\right) = 2\sin\left(\frac{\theta}{2^{k+1}}\right)\cos\left(\frac{\theta}{2^{k+1}}\right)$. Canceling the cosine gives the formula for $n = k + 1$.

**(ii)** From (i):

$$\prod_{k=1}^{n} \cos\left(\frac{\theta}{2^k}\right) = \frac{\sin\theta}{2^n \sin(\theta/2^n)}.$$

Let $x = \theta/2^n$. As $n \to \infty$, $x \to 0$ and $2^n = \theta/x$, so the right-hand side becomes

$$\frac{\sin\theta}{\theta} \cdot \frac{x}{\sin x} \to \frac{\sin\theta}{\theta} \cdot 1 = \frac{\sin\theta}{\theta}.$$

**(iii)** For $n = 1$ we have $u_1 = \sqrt{2} = 2\cos(\pi/4)$. Assume $u_k = 2\cos(\pi/2^{k+1})$. Then

$$u_{k+1} = \sqrt{2 + u_k} = \sqrt{2 + 2\cos\left(\frac{\pi}{2^{k+1}}\right)}$$
$$= \sqrt{4\cos^2\left(\frac{\pi}{2^{k+2}}\right)} = 2\cos\left(\frac{\pi}{2^{k+2}}\right),$$

using $1 + \cos 2A = 2\cos^2 A$ with $2A = \frac{\pi}{2^{k+1}}$. (Note: We take the positive root since the angle $\frac{\pi}{2^{k+2}}$ is acute for all $k \geq 1$).

**(iv)** Substitute $\theta = \pi/2$ in (ii):

$$\prod_{k=1}^{\infty} \cos\left(\frac{\pi}{2^{k+1}}\right) = \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi}.$$

Each factor $\cos\left(\frac{\pi}{2^{k+1}}\right) = \frac{u_k}{2}$ by (iii), so the infinite product equals

$$\prod_{k=1}^{\infty} \frac{u_k}{2} = \frac{2}{\pi},$$

which rearranges to the stated nested radical product (Viète's formula).

## Takeaways 4.7

1. Infinite products can often be obtained by taking limits of finite trigonometric product identities.
2. Recursive nested radicals are closely linked to half-angle cosine values.
3. Viète's Formula provides a beautiful product representation of $2/\pi$ and connects classical geometry with analysis. The formula is named after François Viète, who published it in 1593.

## Problem 4.8: Pendulum Motion and the AGM

A simple pendulum of length $l$ is released from rest at the horizontal position ($\theta = \frac{\pi}{2}$). Let $\theta$ be the angle the pendulum makes with the downward vertical. The angular velocity satisfies:

$$\left(\frac{d\theta}{dt}\right)^2 = \frac{2g}{l}\cos\theta,$$

where $g$ is the acceleration due to gravity. Let $T$ be the time taken for the pendulum to swing from the horizontal position ($\theta = \frac{\pi}{2}$) to the vertical position ($\theta = 0$).

(i) By using the substitution $\sin(\frac{\theta}{2}) = \frac{1}{\sqrt{2}}\sin\phi$, show that:

$$T = \sqrt{\frac{l}{g}}\int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - \frac{1}{2}\sin^2\phi}}$$

(ii) Let the elliptic-type integral

$$I(a,b) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{a^2\cos^2\phi + b^2\sin^2\phi}}.$$

Show the expression for $T$ can be written as

$$T = \sqrt{\frac{l}{g}}\, I\left(1, \tfrac{1}{\sqrt{2}}\right).$$

(iii) Define sequences $a_1 = 1$, $b_1 = \frac{1}{\sqrt{2}}$ and for $n \geq 1$:

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = \sqrt{a_n b_n}.$$

Show that $I(a_n, b_n)$ is invariant in $n$, and the sequences $a_n, b_n$ converge to the same limit. Let $M = \lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. By taking limits show that the full period $P = 4T$ satisfies

$$P = \frac{2\pi\sqrt{l/g}}{M}.$$

(iv) The small-angle (harmonic) approximation gives $P_0 = 2\pi\sqrt{l/g}$. Using $b_1 < M < a_1$, deduce

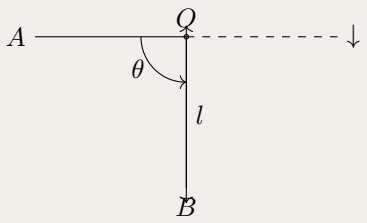
$$P_0 < P < \sqrt{2}\, P_0.$$

Briefly explain why $P > P_0$.

**Hint:**

- For (i): Separate variables from the energy relation to get $dt$ in terms of $d\theta$, pick the correct sign for the motion, then substitute $\sin(\theta/2) = \frac{1}{\sqrt{2}}\sin\phi$ and change limits.
- For (ii): Use $\cos^2\phi + \sin^2\phi = 1$ to rewrite the integrand into the form $a^2\cos^2\phi + b^2\sin^2\phi$.
- For (iii): Use the invariance to equate $I(1, 1/\sqrt{2}) = I(M, M)$ and evaluate $I(M, M) = \pi/(2M)$.
- For (iv): Invert the inequalities to bound $1/M$, then multiply by $2\pi\sqrt{l/g}$; explain that large amplitude slows the motion compared to the small-angle linearisation.

## Solution 4.8




**(i)** From energy, $d\theta/dt = -\sqrt{(2g/l)\cos\theta}$ for the downward swing. Thus

$$T = \sqrt{\tfrac{l}{2g}}\int_0^{\pi/2}\frac{d\theta}{\sqrt{\cos\theta}}.$$

With $\sin(\theta/2) = \frac{1}{\sqrt{2}}\sin\phi$, we have $\cos\theta = 1 - 2\sin^2(\theta/2) = 1 - \sin^2\phi = \cos^2\phi$. Differentiating the substitution gives $\frac{1}{2}\cos\left(\frac{\theta}{2}\right)d\theta = \frac{1}{\sqrt{2}}\cos\phi\,d\phi$. Since $\cos\left(\frac{\theta}{2}\right) = \sqrt{1-\sin^2\left(\frac{\theta}{2}\right)} = \sqrt{1-\frac{1}{2}\sin^2\phi}$, we rearrange to find:

$$d\theta = \frac{\sqrt{2}\cos\phi}{\sqrt{1-\frac{1}{2}\sin^2\phi}}\,d\phi,$$

which yields the stated integral.
**(ii)** Note $1 - \frac{1}{2}\sin^2\phi = \cos^2\phi + \frac{1}{2}\sin^2\phi$, so the integrand is of the form in $I(a,b)$ with $a = 1$, $b = 1/\sqrt{2}$.
**(iii)** By invariance $I(1, 1/\sqrt{2}) = I(M, M) = \int_0^{\pi/2} d\phi/(M) = \pi/(2M)$. Hence

$$T = \sqrt{\tfrac{l}{g}}\cdot\frac{\pi}{2M}, \qquad P = 4T = \frac{2\pi\sqrt{l/g}}{M}.$$

**(iv)** Since $b_1 < M < a_1$ we have $1/\sqrt{2} < M < 1$, so $1 < 1/M < \sqrt{2}$. Multiplying by $2\pi\sqrt{l/g}$ gives $P_0 < P < \sqrt{2}\,P_0$. Physically, for large amplitude the restoring tangential acceleration is not proportional to $\theta$, so the motion is slower than the small-angle linear approximation.

**Takeaways 4.8**

1. The pendulum period for large amplitude leads to elliptic integrals; the AGM gives an efficient exact representation.
2. Recursive arithmetic–geometric mean sequences converge rapidly and can be used to evaluate certain definite integrals.
3. Small-angle linearisation underestimates the true period; inequalities give sharp bounds on the error.

**Problem 4.9: Linear Growth Coefficients**

Let $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z$ be a polynomial with complex coefficients satisfying $|a_k| \leq k$ for all $1 \leq k \leq n$.

(i) Show that for real $x$ with $|x| < 1$:

$$\sum_{k=1}^{\infty} k x^k = \frac{x}{(1-x)^2}.$$

(ii) Using (i), prove that for $|z| < 1$:

$$|P(z)| < \frac{|z|}{(1-|z|)^2}.$$

(iii) Deduce that if $P(z) = 1$ and $|z| < 1$ then

$$|z| > \frac{3-\sqrt{5}}{2}.$$

(iv) If instead $Q(z) = b_n z^n + \cdots + b_1 z$ with $|b_k| \leq 1$, show any root of $Q(z) = 1$ with $|z| < 1$ satisfies $|z| > \frac{1}{2}$.

**Hint:** Differentiate the geometric series to obtain (i). For (ii) apply the triangle inequality and compare the finite sum with the infinite series from (i). For (iii) set $r = |z|$ and solve the inequality $1 < r/(1-r)^2$. For (iv) repeat with the bound $|b_k| \leq 1$ and the standard geometric series.

## Solution 4.9

**(i)** Differentiate $\sum_{k=0}^{\infty} x^k = 1/(1-x)$ (valid for $|x| < 1$) to get

$$\sum_{k=1}^{\infty} k x^{k-1} = \frac{1}{(1-x)^2},$$

then multiply by $x$ to obtain the stated formula.

**(ii)** For $|z| < 1$ the triangle inequality gives

$$|P(z)| \le \sum_{k=1}^{n} |a_k||z|^k \le \sum_{k=1}^{n} k|z|^k < \sum_{k=1}^{\infty} k|z|^k = \frac{|z|}{(1-|z|)^2}.$$

**(iii)** If $P(z) = 1$ then $1 < \dfrac{r}{(1-r)^2}$ with $r = |z|$. Rearranging yields

$$r^2 - 3r + 1 < 0,$$

whose roots are $(3 \pm \sqrt{5})/2$. Since $r < 1$ we obtain $r > \dfrac{3-\sqrt{5}}{2}$.

**(iv)** With $|b_k| \le 1$ we have

$$1 = |Q(z)| \le \sum_{k=1}^{n} |z|^k < \sum_{k=1}^{\infty} r^k = \frac{r}{1-r},$$

so $1 < r/(1-r)$, giving $r > 1/2$.

## Takeaways 4.9

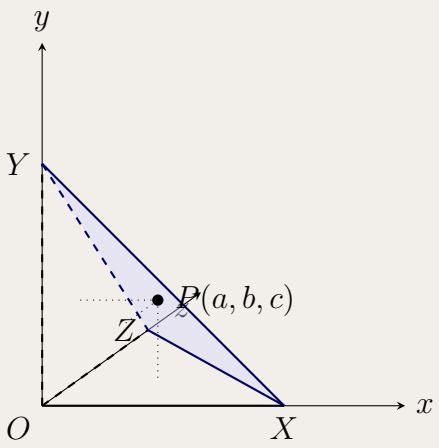
1. Differentiate power series to obtain weighted sums like $\sum k x^k$.
2. Bounding $|P(z)|$ by a real series is a reliable method to locate roots in the complex plane.
3. Coefficient growth affects how close roots can lie to the origin: larger coefficients allow roots nearer 0.

## Problem 4.10: Cauchy-Schwarz and Plane Intersections

(a) Let $x, y, z$ be positive real numbers. Prove that:

$$(x+y+z)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right) \geq 9$$

(b) A plane passes through the fixed point $P(a,b,c)$ where $a,b,c>0$. The plane cuts the positive coordinate axes at $X(x_0,0,0), Y(0,y_0,0), Z(0,0,z_0)$. Show that the minimum sum of the intercepts $S = x_0+y_0+z_0$ is $(\sqrt{a}+\sqrt{b}+\sqrt{c})^2$.




**Hint:** Apply Cauchy-Schwarz to vectors formed by coordinates and reciprocals for part (a). For part (b), apply Cauchy-Schwarz directly to the constraint equation to minimize the sum.

## Solution 4.10

**(a) Proof:** By Cauchy–Schwarz on vectors $(\sqrt{x}, \sqrt{y}, \sqrt{z})$ and $(\frac{1}{\sqrt{x}}, \frac{1}{\sqrt{y}}, \frac{1}{\sqrt{z}})$:

$$(x+y+z)\Big(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\Big) \geq \left(\sqrt{x}\cdot\frac{1}{\sqrt{x}}+\sqrt{y}\cdot\frac{1}{\sqrt{y}}+\sqrt{z}\cdot\frac{1}{\sqrt{z}}\right)^2 = (1+1+1)^2 = 9$$

with equality when $x=y=z$.

**(b) Application:** The equation of the plane is $\frac{x}{x_0}+\frac{y}{y_0}+\frac{z}{z_0}=1$. Since the plane passes through $P(a,b,c)$:

$$\frac{a}{x_0}+\frac{b}{y_0}+\frac{c}{z_0}=1$$

We want to minimize the sum $S = x_0+y_0+z_0$. Applying Cauchy-Schwarz to the vectors $(\sqrt{x_0}, \sqrt{y_0}, \sqrt{z_0})$ and $(\sqrt{\frac{a}{x_0}}, \sqrt{\frac{b}{y_0}}, \sqrt{\frac{c}{z_0}})$:

$$(x_0+y_0+z_0)\left(\frac{a}{x_0}+\frac{b}{y_0}+\frac{c}{z_0}\right) \geq (\sqrt{a}+\sqrt{b}+\sqrt{c})^2$$

$$S\cdot(1) \geq (\sqrt{a}+\sqrt{b}+\sqrt{c})^2$$

Thus, the minimum sum of the intercepts is $(\sqrt{a}+\sqrt{b}+\sqrt{c})^2$. □

**Takeaways 4.10**

1. **Multiplying Inequalities:** When all terms are positive, inequalities can be multiplied directly to achieve stronger bounds.
2. **Constraint Optimization:** Use the constraint equation to express the objective function, then apply AM-GM to the constraint terms.

**Remark 0.7: Alternate proof for part (a)**

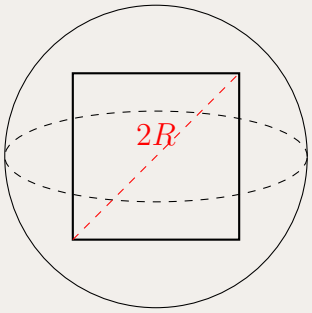
By AM-GM on $(x+y+z)$ and $(\frac{1}{x}+\frac{1}{y}+\frac{1}{z})$ separately:

$$(x+y+z)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right) \geq (3\sqrt[3]{xyz})\left(\frac{3}{\sqrt[3]{xyz}}\right) = 9$$

This gives the same bound using products instead of vector projections.

**Problem 4.11: Cube in Sphere Optimization**

(a) Establish the inequality $u^2+v^2+w^2 \geq 3(uvw)^{\frac{2}{3}}$ for positive numbers $u, v, w$.
(b) A rectangular prism is inscribed inside a sphere of fixed radius $R$. Show that the prism has the maximum volume when it is a cube.




Cross-section through diagonal

**Hint:** Establish the relationship between edge length and sphere radius, then apply AM-GM to the constraint.

## Solution 4.11

**(a) Proof:** Let the three terms be $u^2, v^2, w^2$. Applying AM-GM:

$$\frac{u^2+v^2+w^2}{3} \geq \sqrt[3]{u^2v^2w^2}$$
$$u^2+v^2+w^2 \geq 3(uvw)^{2/3}$$

**(b) Application:** Let the dimensions of the prism be $x, y, z$. The prism is inscribed in a sphere of radius $R$, meaning the space diagonal of the prism equals the diameter of the sphere ($2R$). By applying the Pythagorean theorem twice (once for the base diagonal, and once using the height to find the space diagonal), the square of the space diagonal is $x^2+y^2+z^2$.

$$x^2+y^2+z^2 = (2R)^2 = 4R^2 \quad \text{(Constant)}$$

We wish to maximize the Volume $V = xyz$. Substitute $u = x, v = y, w = z$ into the inequality from part (a):

$$x^2+y^2+z^2 \geq 3(xyz)^{2/3}$$
$$4R^2 \geq 3V^{2/3}$$

Rearranging for $V$:

$$\frac{4R^2}{3} \geq V^{2/3}$$
$$\left(\frac{4R^2}{3}\right)^{3/2} \geq V$$

The Volume $V$ is bounded by a constant. The maximum occurs when equality holds in the AM-GM step. Equality requires:

$$x^2 = y^2 = z^2 \implies x = y = z$$

Therefore, the rectangular prism must be a cube to maximize the volume. □

## Takeaways 4.11

1. **Constraint Transformation:** Convert geometric constraints (sphere inscribed in cube) into algebraic relationships between variables.
2. **Substitution Strategy:** Identify which form of AM-GM to use based on the powers appearing in your objective function.

## Problem 4.12: De Moivre's Theorem and Geometric Series

Let $\alpha = \cos\theta + i\sin\theta$ and consider the series

$$C = \alpha^{-n} + \alpha^{-n+1} + \cdots + \alpha^{-1} + \alpha^{0} + \alpha^{1} + \cdots + \alpha^{n}$$

for a positive integer $n$.

(i) Show that $\alpha^k + \alpha^{-k} = 2\cos k\theta$.

(ii) Prove that

$$C = \frac{\alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})}{(1-\alpha)(1-\bar{\alpha})}$$

where $\bar{\alpha}$ is the complex conjugate of $\alpha$.

(iii) Deduce that

$$1 + 2\sum_{k=1}^{n} \cos k\theta = \frac{\cos n\theta - \cos(n+1)\theta}{1-\cos\theta}.$$

(iv) Show that

$$\sum_{k=1}^{n} \cos\frac{k\pi}{n} = -1 \quad \text{(independent of } n\text{)}.$$

**Hint:** Use De Moivre's theorem to express $\sin(n\theta)$ in terms of $z = e^{i\theta}$, then sum the resulting geometric series.

## Solution 4.12

(i) Since $\alpha = \cos\theta + i\sin\theta$, by De Moivre's Theorem:

$$\alpha^k = \cos k\theta + i\sin k\theta, \quad \alpha^{-k} = \cos k\theta - i\sin k\theta$$

Adding: $\alpha^k + \alpha^{-k} = 2\cos k\theta$.

(ii) The series $C$ is geometric with first term $\alpha^{-n}$, ratio $\alpha$, and $(2n+1)$ terms:

$$C = \frac{\alpha^{-n}(\alpha^{2n+1}-1)}{\alpha - 1} = \frac{\alpha^{n+1} - \alpha^{-n}}{\alpha - 1}$$

Multiply by $\frac{\bar{\alpha}-1}{\bar{\alpha}-1}$:

$$C = \frac{(\alpha^{n+1} - \alpha^{-n})(\bar{\alpha} - 1)}{(\alpha - 1)(\bar{\alpha} - 1)}$$

Since $|\alpha| = 1$, we know $\bar{\alpha} = \alpha^{-1}$. Expanding the numerator yields:

$$(\alpha^{n+1} - \alpha^{-n})(\alpha^{-1} - 1) = \alpha^n - \alpha^{n+1} - \alpha^{-n-1} + \alpha^{-n} = \alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})$$

For the denominator, expanding yields:

$$(\alpha - 1)(\bar{\alpha} - 1) = \alpha\bar{\alpha} - \alpha - \bar{\alpha} + 1 = |\alpha|^2 - (\alpha + \bar{\alpha}) + 1 = 1 - 2\cos\theta + 1 = 2(1 - \cos\theta)$$

Substituting these back into the expression for $C$:

$$C = \frac{\alpha^n + \alpha^{-n} - (\alpha^{n+1} + \alpha^{-(n+1)})}{(1-\alpha)(1-\bar{\alpha})}$$

(iii) From the definition: $C = 1 + \sum_{k=1}^{n}(\alpha^k + \alpha^{-k}) = 1 + 2\sum_{k=1}^{n}\cos k\theta$ Using part (ii) with $\alpha^k + \alpha^{-k} = 2\cos k\theta$:

$$1 + 2\sum_{k=1}^{n}\cos k\theta = \frac{\cos n\theta - \cos(n+1)\theta}{1 - \cos\theta}$$

(iv) Substitute $\theta = \frac{\pi}{n}$:

$$1 + 2\sum_{k=1}^{n}\cos\frac{k\pi}{n} = \frac{\cos\pi - \cos\left(\pi + \frac{\pi}{n}\right)}{1 - \cos\frac{\pi}{n}}$$

Since $\cos\pi = -1$ and $\cos(\pi + x) = -\cos x$:

$$1 + 2\sum_{k=1}^{n}\cos\frac{k\pi}{n} = \frac{-1 + \cos\frac{\pi}{n}}{1 - \cos\frac{\pi}{n}} = -1$$

Therefore: $\sum_{k=1}^{n}\cos\frac{k\pi}{n} = -1$.

## Takeaways 4.12

1. **Geometric Series with Complex Numbers:** Apply standard formulas but manipulate using conjugates when needed.
2. **Trigonometric Identities:** De Moivre's theorem connects complex exponentials to trigonometric sums.

## Problem 4.13: Complex Number Perpendicularity

Let $n$ be a positive integer, and let $z_k = e^{i\frac{2k\pi}{n}}$ for $k \in \{1, 2, \ldots, n\}$.
Let $z_a, z_b$, and $z_c$ be three distinct complex numbers from this set. Prove that if they satisfy the equation:

$$\frac{z_a}{z_b} = -\frac{z_c}{z_a}$$

then the vector represented by the sum $z_b + z_c$ is perpendicular to the vector $z_a$.

**Hint:**

1. Use the given equation to express $\frac{z_c}{z_a}$ in terms of $\frac{z_b}{z_a}$.
2. Recall that two complex numbers $w_1$ and $w_2$ are perpendicular if their ratio $\frac{w_1}{w_2}$ is purely imaginary.
3. Form the ratio $Z = \frac{z_b+z_c}{z_a}$ and simplify using the given condition.
4. For any complex number $u$ on the unit circle, $u - \frac{1}{u} = u - \overline{u} = 2i\text{Im}(u)$ is purely imaginary.

**Solution 4.13**

To prove perpendicularity, we show that $Z = \frac{z_b+z_c}{z_a}$ is purely imaginary.

**Step 1:** Express the ratio:

$$Z = \frac{z_b + z_c}{z_a} = \frac{z_b}{z_a} + \frac{z_c}{z_a}$$

**Step 2:** From the given condition $\frac{z_a}{z_b} = -\frac{z_c}{z_a}$, rearrange:

$$\frac{z_c}{z_a} = -\frac{z_a}{z_b}$$

**Step 3:** Substitute into $Z$:

$$Z = \frac{z_b}{z_a} - \frac{z_a}{z_b}$$

**Step 4:** Let $u = \frac{z_b}{z_a}$. Since both $z_a$ and $z_b$ are roots of unity, $|u| = 1$, so $\overline{u} = \frac{1}{u}$.

$$Z = u - \frac{1}{u} = u - \overline{u} = 2i \cdot \text{Im}(u)$$

Since $Z$ is purely imaginary, the vectors $z_b + z_c$ and $z_a$ are perpendicular.

**Takeaways 4.13**

1. **Perpendicularity Test:** For complex numbers, perpendicularity is equivalent to having a purely imaginary ratio.
2. **Unit Circle Property:** When $|z| = 1$, we have $\overline{z} = \frac{1}{z}$, which simplifies many algebraic manipulations.
3. **Conjugate Differences:** The expression $z - \overline{z}$ is always purely imaginary, while $z + \overline{z}$ is purely real.

## Problem 4.14: Converging Rectangles — Newton's Method for $\sqrt{2}$

Consider a sequence of rectangles with side lengths $a_n$ and $b_n$. The first rectangle has $a_1 = 2$ and $b_1 = 1$. For integers $n \geq 1$ define

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = \frac{2}{a_{n+1}}.$$

(i) Show that every rectangle in the sequence has area 2.

(ii) Prove that $a_n \geq \sqrt{2}$ for all $n \geq 1$.

(iii) Show by induction that

$$a_n - \sqrt{2} \leq \frac{1}{2^{n-1}}(2 - \sqrt{2}), \qquad n \geq 1.$$

(iv) Deduce that the rectangles approach a square as $n \to \infty$.

(v) (Optional, no points) Use a calculator to find the first five terms of the sequence $a_n$ and comment on the speed of convergence to the actual value of $\sqrt{2}$.

**Hint:** For (i) compute $A_{n+1} = a_{n+1}b_{n+1}$ directly. For (ii) apply AM–GM to $a_n$ and $b_n$ and use the area from (i). For (iii) express $a_{k+1} - \sqrt{2}$ in terms of $(a_k - \sqrt{2})^2$ and bound the multiplicative factor by $\frac{1}{2}$. For (iv) use the squeeze theorem together with the bound in (iii).

## Solution 4.14

**(i)** The area is $A_n = a_n b_n$. Using the recursion,

$$A_{n+1} = a_{n+1} b_{n+1} = a_{n+1} \cdot \frac{2}{a_{n+1}} = 2,$$

so $A_n \equiv 2$ for all $n$.
**(ii)** By AM–GM,

$$a_{n+1} = \frac{a_n + b_n}{2} \geq \sqrt{a_n b_n} = \sqrt{2},$$

and the base case $a_1 = 2 \geq \sqrt{2}$ holds, so $a_n \geq \sqrt{2}$ for all $n$.
**(iii)** From $b_k = 2/a_k$ we get

$$a_{k+1} = \frac{a_k + \frac{2}{a_k}}{2} = \frac{a_k^2 + 2}{2a_k}.$$

Subtracting $\sqrt{2}$ and expanding the numerator to form a perfect square yields:

$$a_{k+1} - \sqrt{2} = \frac{a_k^2 + 2}{2a_k} - \frac{2\sqrt{2}a_k}{2a_k} = \frac{a_k^2 - 2\sqrt{2}a_k + 2}{2a_k} = \frac{(a_k - \sqrt{2})^2}{2a_k}.$$

Since $a_k \geq \sqrt{2}$, the factor $1/(2a_k) \leq 1/(2\sqrt{2}) < \frac{1}{2}$, hence

$$a_{k+1} - \sqrt{2} \leq \tfrac{1}{2}(a_k - \sqrt{2}).$$

Applying the inductive hypothesis gives the claimed bound.
**(iv)** From (ii) and (iii) we have

$$0 \leq a_n - \sqrt{2} \leq \frac{2 - \sqrt{2}}{2^{n-1}} \to 0 \quad (n \to \infty).$$

Since $a_n b_n = 2$ for all $n$, it follows $b_n \to \sqrt{2}$ as well. Therefore the side lengths both tend to $\sqrt{2}$ and the rectangles approach a square.

## Takeaways 4.14

1. Newton's Method: This problem is actually a geometric visualization of Newton's Method for estimating square roots. The recurrence $a_{n+1} = \frac{1}{2}(a_n + \frac{2}{a_n})$ is the standard algorithm for calculating $\sqrt{2}$.

2. Convergence Speed: While the problem asks to prove linear convergence (error halves each time), Newton's method actually converges quadratically (the number of correct digits doubles with every step), which is much faster than the bound proven in part (iii).

3. AM-GM Utility: The arithmetic mean of a number and its reciprocal (scaled) is a classic way to generate a value closer to the root, bounded below by the root itself.

Table 1: Convergence of sequence $a_n$ to $\sqrt{2}$ (Newton's Method)

| n | Exact Fraction ($a_n$) | Decimal Approx. | Error ($\lvert a_n - \sqrt{2}\rvert$) | Correct Digits |
|---|---|---|---|---|
| 1 | 2 | 2.0000000000 | $5.86 \times 10^{-1}$ | 1 |
| 2 | $\frac{3}{2}$ | 1.5000000000 | $8.58 \times 10^{-2}$ | 1 |
| 3 | $\frac{17}{12}$ | 1.4166666667 | $2.45 \times 10^{-3}$ | 3 |
| 4 | $\frac{577}{408}$ | 1.4142156863 | $2.12 \times 10^{-6}$ | 6 |
| 5 | $\frac{665857}{470832}$ | 1.4142135624 | $1.59 \times 10^{-12}$ | 12 |
| 6 | $\frac{886731088897}{627013566048}$ | 1.4142135623 | $8.86 \times 10^{-25}$ | 24 |
| 7 | (too large to display) | 1.4142135623 | $2.75 \times 10^{-49}$ | 49 |
| 8 | (too large to display) | 1.4142135623 | $2.64 \times 10^{-98}$ | 98 |
| 9 | (too large to display) | 1.4142135623 | $2.44 \times 10^{-196}$ | 196 |
| 10 | (too large to display) | 1.4142135623 | $2.08 \times 10^{-392}$ | 392 |

**Performance Evaluation:** The sequence exhibits *Quadratic Convergence*, meaning the number of correct digits roughly doubles with every iteration. By $n = 10$, the value is accurate to nearly 400 decimal places.

## Problem 4.15: Integrals and a Combinatorial Identity

Let $I_n$ be defined for integers $n \geq 0$ by

$$I_n = \int_{-1}^{0} x^n \sqrt{x+1}\, dx.$$

(i) Use integration by parts to show that, for $n \geq 1$,

$$I_n = \frac{-2n}{2n+3} I_{n-1}.$$

(ii) Hence prove

$$I_n = (-1)^n \frac{2^{n+1} n!}{3 \times 5 \times \cdots \times (2n+3)}.$$

(iii) By substituting $u = \sqrt{x+1}$, show that

$$I_n = 2 \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} \frac{1}{2k+3}.$$

(iv) Deduce the combinatorial identity

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{1}{2k+3} = \frac{2^n n!}{3 \times 5 \times \cdots \times (2n+3)}.$$

**Hint:** For (i) take $u = x^n$ and $dv = (x+1)^{1/2}dx$ and use $x = (x+1) - 1$ to relate $I_n$ and $I_{n-1}$.
For (ii) evaluate $I_0$ and iterate the reduction.
For (iii) substitute $u = \sqrt{x+1}$ so $x = u^2 - 1$ and expand by the binomial theorem before integrating.
For (iv) equate the two expressions for $I_n$ and simplify.

## Solution 4.15

**(i)** Use integration by parts with $u = x^n$ and $dv = (x+1)^{1/2}\,dx$, which gives $du = nx^{n-1}\,dx$ and $v = \frac{2}{3}(x+1)^{3/2}$.

$$\begin{aligned} I_n &= \left[\frac{2}{3}x^n(x+1)^{3/2}\right]_{-1}^{0} - \int_{-1}^{0}\frac{2n}{3}x^{n-1}(x+1)^{3/2}\,dx \\ &= 0 - \frac{2n}{3}\int_{-1}^{0} x^{n-1}(x+1)\sqrt{x+1}\,dx \\ &= -\frac{2n}{3}\int_{-1}^{0}(x^n\sqrt{x+1} + x^{n-1}\sqrt{x+1})\,dx \\ &= -\frac{2n}{3}(I_n + I_{n-1}) \end{aligned}$$

Multiplying by 3 and gathering the $I_n$ terms to the left side:

$$\begin{aligned} 3I_n &= -2nI_n - 2nI_{n-1} \\ (2n+3)I_n &= -2nI_{n-1} \\ I_n &= \frac{-2n}{2n+3}I_{n-1}. \end{aligned}$$

**(ii)** $I_0 = \int_{-1}^{0}(x+1)^{1/2}dx = \frac{2}{3}$. Repeated application of the reduction gives the stated closed form.

$$I_n = \left(\frac{-2n}{2n+3}\right)\left(\frac{-2(n-1)}{2n+1}\right)\cdots\left(\frac{-2(1)}{5}\right)I_0$$

$$I_n = (-1)^n\frac{2^n n!}{(2n+3)(2n+1)\ldots(5)}\cdot\frac{2}{3} = (-1)^n\frac{2^{n+1}n!}{3\times 5\times\cdots\times(2n+3)}$$

**(iii)** With $u = \sqrt{x+1}$ we have $dx = 2u\,du$, $x = u^2 - 1$ and using the binomial theorem,

$$u^2(u^2-1)^n = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}u^{2k+2}.$$

hence

$$I_n = 2\int_0^1 u^2(u^2-1)^n\,du = 2\sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}\frac{1}{2k+3}.$$

**(iv)** Equating the expressions in (ii) and (iii) and cancelling $2(-1)^n$ yields the required identity.

**Takeaways 4.15**

1. **The Power of Two Methods:** Reduction formulas and substitutions give two complementary approaches to definite integrals; equating them often produces useful combinatorial identities.

2. **Algebraic Manipulation:** The trick in step (i) of splitting $(x+1)$ to recover $I_n + I_{n-1}$ is a recurring motif in Extension 2 integration problems.

3. **Double Factorial:** It is often helpful to express products of integers using double factorial notation to simplify expressions and recognize patterns.

   For odd numbers, $(2n+1)!! = 1 \times 3 \times 5 \times \cdots \times (2n+1) = \frac{(2n+1)!}{2^n n!}$.

   For even numbers, $(2n)!! = 2 \times 4 \times 6 \times \cdots \times (2n) = 2^n n!$.

   And we can write the equation in a beautifully compact form:

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{1}{2k+3} = \frac{(2n)!!}{(2n+3)!!}.$$

## Problem 4.16: Mandelbrot Escape Criterion

**Topic: Complex Dynamics, Inequalities, Mathematical Induction**

For a given complex constant $c \in \mathbb{C}$, define the Mandelbrot sequence by

$$z_0 = 0, \qquad z_{n+1} = z_n^2 + c \quad (n \geq 0).$$

The Mandelbrot set $\mathcal{M}$ is the set of all $c \in \mathbb{C}$ for which the sequence of moduli $(|z_n|)_{n\geq 0}$ remains bounded.

(i) Suppose $c \in \mathbb{R}$ and $c > \frac{1}{4}$. By induction, prove that for all $n \geq 0$, $z_n \in \mathbb{R}$ and

$$z_{n+1} - z_n = z_n^2 - z_n + c > 0.$$

Hence, deduce that if $c > \frac{1}{4}$ then $c \notin \mathcal{M}$.

(ii) Recall the reverse triangle inequality $|u - v| \geq \big||u| - |v|\big|$. Let $z \in \mathbb{C}$ with $|z| \geq 1$. Prove that

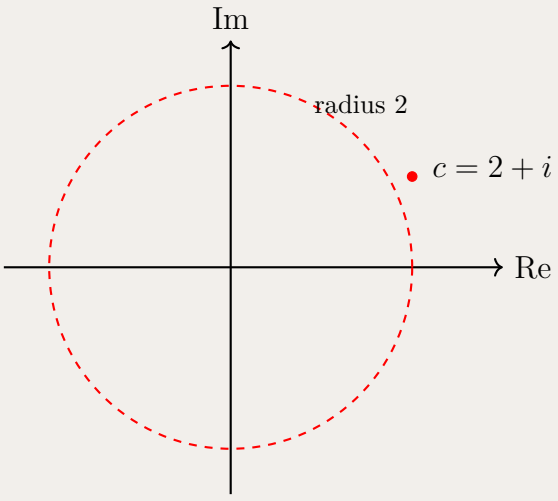
$$|z^2 - z| \geq |z|(|z| - 1).$$

(iii) Let $c \in \mathbb{C}$ with $|c| > 2$, and define $z_0 = 0, z_{n+1} = z_n^2 + c$. Note that $|z_1| = |c| > 2$. Using the estimate in (ii) as the base idea, prove by induction that for all $n \geq 1$ with $|z_n| \geq 2$ we have

$$|z_{n+1}| \geq |z_n|^2 - |c| \geq |z_n|^2 - |z_n| = |z_n|(|z_n| - 1).$$

Hence, conclude that if $|c| > 2$, then $c \notin \mathcal{M}$.

(iv) Determine, using the results above, whether $c = 2 + i$ belongs to $\mathcal{M}$. Justify your answer rigorously.



**Hint:**

- **(i)** Try completing the square for the quadratic $x^2-x+c$. What does $c > \frac{1}{4}$ guarantee about its minimum value?
- **(ii)** Factorize $z^2 - z$ into $z(z-1)$. Apply the multiplicative property of absolute values, then use the reverse triangle inequality on $|z-1|$.
- **(iii)** Use $|z_{n+1}| = |z_n^2 + c| \geq |z_n|^2 - |c|$ (reverse triangle inequality). When $|z_n| \geq 2$ and $|c| > 2$, show $|z_n| \geq |c|$ to get $|z_n|^2 - |c| \geq |z_n|(|z_n| - 1)$. Conclude $|z_{n+1}| \geq |z_n|$ so $|z_n| \geq 2$ for all $n \geq 1$ and the sequence grows.
- **(iv)** Compute $|2+i|$. Which result from (i)–(iii) applies?

## Solution 4.16

**(i)** Since $z_0 = 0 \in \mathbb{R}$ and $z_{n+1} = z_n^2 + c \in \mathbb{R}$ whenever $z_n \in \mathbb{R}$, induction gives $z_n \in \mathbb{R}$ for all $n$. Also

$$z_{n+1} - z_n = z_n^2 - z_n + c = \left(z_n - \frac{1}{2}\right)^2 + c - \frac{1}{4}.$$

If $c > \frac{1}{4}$, then $z_{n+1} - z_n > 0$ for every real $z_n$, so $(z_n)$ is strictly increasing. If it were bounded, it would converge to some real limit $L$ with

$$L = L^2 + c \qquad \Rightarrow \qquad L^2 - L + c = 0.$$

But for $c > \frac{1}{4}$, the discriminant is $1 - 4c < 0$, impossible for a real $L$. Hence $(z_n)$ is unbounded, so $c \notin \mathcal{M}$.
**(ii)** Since

$$z^2 - z = z(z-1),$$

we have

$$|z^2 - z| = |z|\,|z-1| \geq |z|\,\big||z| - 1\big|.$$

If $|z| \geq 1$, then $\big||z| - 1\big| = |z| - 1$, hence

$$|z^2 - z| \geq |z|(|z| - 1).$$

**(iii)** Assume $|c| > 2$. Then $|z_1| = |c| > 2$, and

$$|z_{n+1}| = |z_n^2 + c| \geq |z_n|^2 - |c|.$$

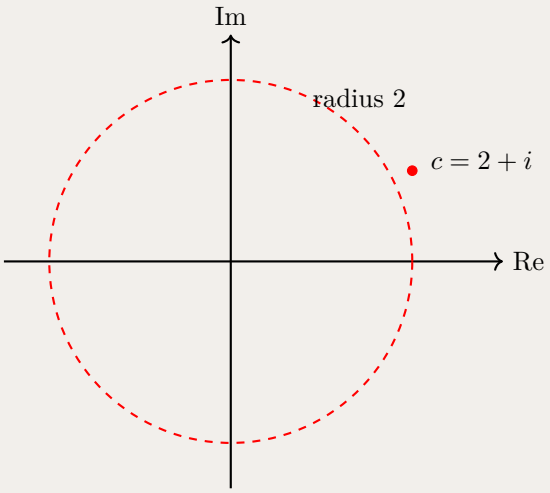
Assume $|z_k| \geq |c|$. Since we are given $|c| > 2$, it follows that $|z_k| > 2$. Substituting this into our inequality gives:

$$|z_{k+1}| \geq |z_k|^2 - |c| \geq |z_k|^2 - |z_k| = |z_k|(|z_k| - 1) > |z_k|(2-1) = |z_k| \geq |c|.$$

This proves $|z_{k+1}| \geq |c|$, completing the induction (since $|z_1| = |c| \geq |c|$ for the base case). Furthermore, because $|z_{k+1}| > |z_k|$, it proves the sequence of moduli strictly increases with every step, and thus grows without bound. Therefore, $c \notin \mathcal{M}$.
**(iv)** For $c = 2 + i$,

$$|c| = |2 + i| = \sqrt{5} > 2.$$

By part (iii), $2 + i \notin \mathcal{M}$.



**Takeaways 4.16**

1. **Escape Criterion:** If $|c| > 2$, the Mandelbrot sequence escapes to infinity; such $c$ lie outside the Mandelbrot set. The diagram shows $c = 2 + i$ outside the circle of radius 2.
2. **Reverse Triangle Inequality:** $|u - v| \geq \big||u| - |v|\big|$ is essential for modulus estimates in complex dynamics.
3. **Completing the Square:** For real $c > \frac{1}{4}$, the quadratic $x^2 - x + c$ has no real roots, so no fixed point exists and the sequence diverges.
4. **Induction on Modulus:** When $|z_n| \geq 2$, the recurrence $z_{n+1} = z_n^2 + c$ amplifies the modulus; induction shows unbounded growth.

## 4.3 Advanced Last Resort Problems

Complex problems that push the boundaries of high school mathematics.

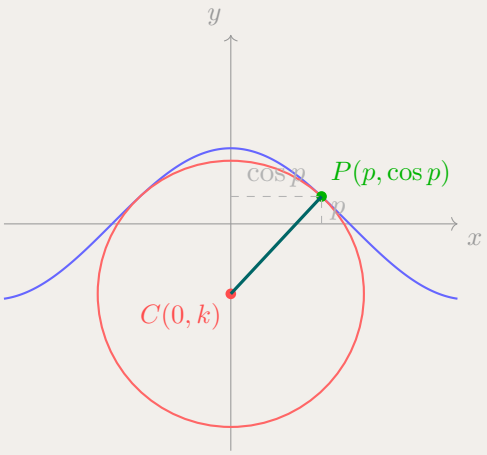
**Problem 4.17: Normal Lines and Curve Tangency**

A circle with center $C(0, k)$ on the $y$-axis is tangent to the curve $y = \cos x$ at point $P(p, \cos p)$. Find the value of $k$ in terms of $p$.

**Hint:** Find the normal line to the curve at a general point. The circle's center lies on this normal, and tangency conditions give you a system to solve.

**Solution 4.17**

Gradient of $y = \cos x$ at $P$ is $m_T = -\sin p$. Gradient of Normal is $m_N = \frac{1}{\sin p} = \csc p$ (assuming $p \neq n\pi$). Because the circle and the curve share a tangent at $P$, the circle's radius $CP$ must be perpendicular to that tangent line. Therefore, $CP$ lies precisely on the normal line. The line $CP$ connects $(0, k)$ and $(p, \cos p)$, so its slope is $m_{CP} = \frac{\cos p - k}{p - 0}$. Equating slopes: $\frac{\cos p - k}{p} = \frac{1}{\sin p}$. $\cos p - k = \frac{p}{\sin p} \implies k = \cos p - p \csc p$.



**Takeaways 4.17**

1. **Normal Line Property:** In optimization problems involving distance to a curve, the shortest/critical distance is always along the normal line.
2. **Tangency Conditions:** For circle tangent to curve, the line from center to tangent point is normal to the curve.

**Problem 4.18: Binomial Integrals and Wallis' Product for $\pi$**

Consider the quadratic equation

$$z^2 - 2z\sin\theta + 1 = 0, \qquad 0 < \theta < \frac{\pi}{2}.$$

(i) Find the roots $\alpha, \beta$ in the exponential form $e^{i\psi}$

(ii) By expanding $(\alpha+\beta)^{2n}$ and grouping conjugate terms, prove the identity

$$2^{2n-1}\sin^{2n}\theta = \frac{1}{2}\binom{2n}{n} + \sum_{k=0}^{n-1}(-1)^{n-k}\binom{2n}{k}\cos(2(n-k)\theta).$$

(iii) Let

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n\theta\, d\theta.$$

You are given the recurrence form: $I_n = \frac{n-1}{n}I_{n-2}$ (do NOT prove this).

Hence evaluate the following definite integral in term of $n$:

$$I_{2n} = \int_0^{\frac{\pi}{2}} \sin^{2n}\theta\, d\theta.$$

(iv) Use the result of (iii), or otherwise, to prove the limit

$$\lim_{n\to\infty}\left[\frac{2\cdot 4\cdot 6\cdots(2n)}{1\cdot 3\cdot 5\cdots(2n-1)}\cdot\frac{1}{\sqrt{n}}\right] = \sqrt{\pi}.$$

You are given the Wallis inequality: $\frac{2n}{2n+1} < \frac{I_{2n+1}}{I_{2n}} < 1$ (do NOT prove this).

**Hint:** For (i) use the quadratic formula and rewrite the root in polar form. For (ii) note $\alpha\beta = 1$ and group terms $k$ with $2n-k$. For (iii) integrate the identity from (ii) on $[0, \pi/2]$ and use that cosine terms integrate to zero. For (iv) express the central binomial coefficient using factorials and rearrange into products of even and odd integers; apply Stirling/Wallis asymptotics.

## Solution 4.18

**(i)** The quadratic formula gives

$$z = \sin\theta \pm i\cos\theta = \cos\left(\tfrac{\pi}{2} - \theta\right) \pm i\sin\left(\tfrac{\pi}{2} - \theta\right) = e^{\pm i(\frac{\pi}{2} - \theta)}.$$

**(ii)** Since $\alpha + \beta = 2\sin\theta$ and $\alpha\beta = 1$, expand $(\alpha+\beta)^{2n}$ by the Binomial Theorem and group conjugate pairs to obtain the stated identity after simplifying terms.
Using $(\alpha+\beta)^{2n} = (2\sin\theta)^{2n}$. Expanding the LHS:

$$\sum_{k=0}^{2n} \binom{2n}{k} \alpha^{2n-k}\beta^k = \binom{2n}{n} + \sum_{k=0}^{n-1} \binom{2n}{k} (\alpha^{2n-2k} + \beta^{2n-2k})$$

Using $\alpha^m + \beta^m = 2\cos(m(\frac{\pi}{2} - \theta))$ and simplified parity logic leads to the result.
**(iii)** Integrate the identity in (ii) from 0 to $\pi/2$.
All cosine terms $\cos(2(n-k)\theta)$ integrate to zero over this interval, leaving

$$2^{2n-1}\int_0^{\pi/2} \sin^{2n}\theta\, d\theta = \frac{\pi}{4}\binom{2n}{n}.$$

Hence

$$\int_0^{\pi/2} \sin^{2n}\theta\, d\theta = \frac{\pi}{2^{2n+1}}\binom{2n}{n} = \frac{\pi(2n)!}{2^{2n+1}(n!)^2}.$$

and so,

$$2^{2n-1}I_{2n} = \frac{1}{2}\binom{2n}{n}\cdot\frac{\pi}{2} \implies I_{2n} = \frac{\pi(2n)!}{2^{2n+1}(n!)^2}$$

**(iv)** Rearranging the factorials in the expression from (iii)

$$I_{2n} = \int_0^{\frac{\pi}{2}} \sin^{2n}\theta\, d\theta = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}\cdot\frac{\pi}{2}$$

$$I_{2n+1} = \int_0^{\frac{\pi}{2}} \sin^{2n+1}\theta\, d\theta = \frac{2\cdot 4\cdot 6\cdots(2n)}{1\cdot 3\cdot 5\cdots(2n+1)}$$

Let $W_n$ be the expression inside the limit we are trying to find (the "Wallis Product" term):

$$W_n = \frac{2\cdot 4\cdot 6\cdots(2n)}{1\cdot 3\cdot 5\cdots(2n-1)}$$

Using the provided hint (or we can prove it within 10 - 15 minutes easily), we can show that $\frac{2n}{2n+1} < \frac{I_{2n+1}}{I_{2n}} < 1$.
Use this inequality, we can write:

$$\frac{I_{2n+1}}{I_{2n}} = \frac{W_n}{2n+1}\cdot\frac{2W_n}{\pi} = \frac{2W_n^2}{\pi(2n+1)}$$

Substitute back to the inequality:

$$n\pi < W_n^2 < \frac{\pi(2n+1)}{2}$$

and use the Squeeze Theorem to conclude the proof.

$$\lim_{n\to\infty}\frac{W_n}{\sqrt{n}} = \sqrt{\pi}$$

**Takeaways 4.18**

1. Connects complex roots, binomial expansions and definite integrals — a useful synthesis for Problem 16 style questions.
2. Wallis derived this result (heuristically) before Newton and Leibniz fully developed calculus.
3. Grouping conjugate terms is a recurring trick when expanding symmetric binomial expressions involving roots on the unit circle.
4. **Wallis' Product for $\pi$:** Wallis' Product: (published by John Wallis in 1656) is a famous mathematical formula that expresses the value of $\pi$ as an infinite product of rational numbers. It is typically written as:

$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \left( \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \right) = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots$$

   Or, equivalently:

$$\pi = 2 \cdot \frac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdots}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots}$$

5. Prove the recurrence relation $I_n = \frac{n-1}{n} I_{n-2}$ using integration by parts when you are bothered.
6. Prove the Wallis inequality $\frac{2n}{2n+1} < \frac{I_{2n+1}}{I_{2n}} < 1$ using induction and integration by parts when you are bothered.

## Problem 4.19: Wallis Integrals and Asymptotic Approximations

Let

$$U_n = \int_0^1 \frac{x^n}{\sqrt{1-x^2}}\,dx, \qquad n \geq 0.$$

(i) Prove the reduction formula

$$U_n = \frac{n-1}{n}\,U_{n-2} \quad (n \geq 2),$$

(ii) Evaluate $U_0$ and $U_1$ and deduce the closed forms for $U_{2n}$ and $U_{2n+1}$
(iii) Prove, then use the inequalities

$$U_{2n+2} < U_{2n+1} < U_{2n}$$

to obtain the bounds

$$\frac{2n+1}{2n+2} \cdot \frac{\pi}{2} < \frac{16^n}{(2n+1)\binom{2n}{n}^2} < \frac{\pi}{2}$$

(iv) Hence, using the inequality in part (iii), show that:

$$\lim_{n\to\infty} \left[ \binom{2n}{n} \div \frac{4^n}{\sqrt{n\pi}} \right] = 1$$

Here, $\binom{2n}{n} = \dfrac{(2n)!}{(n!)^2}$ is the central binomial coefficient.

**Hint:** Use integration by parts for the reduction formula (split $x^n = x^{n-1} \cdot x$). Evaluate $U_0 = \int_0^1 (1-x^2)^{-1/2} dx = \frac{\pi}{2}$ and $U_1 = 1$. Reduce even and odd indices down to these base cases to get factorial/binomial expressions. For the inequality chain compare powers of $x$ on $(0,1)$ and integrate; convert the resulting bounds into expressions involving central binomial coefficients and apply the Squeeze Theorem.

## Solution 4.19

**(i) Reduction:** Write

$$U_n = \int_0^1 x^{n-1} \cdot \frac{x}{\sqrt{1-x^2}}\, dx.$$

Let $u = x^{n-1}$ and $dv = \dfrac{x}{\sqrt{1-x^2}}dx$, so $du = (n-1)x^{n-2}dx$ and $v = -\sqrt{1-x^2}$. Integration by parts gives

$$U_n = \left[-x^{n-1}\sqrt{1-x^2}\right]_0^1 + (n-1)\int_0^1 x^{n-2}\sqrt{1-x^2}\, dx.$$

The boundary term evaluates to 0 at $x = 1$ because of the $\sqrt{1-x^2}$ factor, and it evaluates to 0 at $x = 0$ specifically because $n \geq 2$, meaning the $x^{n-1}$ term is at least $x^1$. Rewrite $\sqrt{1-x^2} = \dfrac{1-x^2}{\sqrt{1-x^2}}$ in the remaining integral to obtain

$$U_n = (n-1)(U_{n-2} - U_n),$$

hence $U_n = \dfrac{n-1}{n}U_{n-2}$.

**(ii) Base values and closed forms:** $U_0 = \int_0^1 (1-x^2)^{-1/2}dx = \frac{\pi}{2}$ (substitute $x = \sin\theta$) and $U_1 = \int_0^1 \dfrac{x}{\sqrt{1-x^2}}dx = 1$. Repeated application of the reduction yields

$$U_{2n} = \frac{(2n-1)(2n-3)\cdots 1}{(2n)(2n-2)\cdots 2}\, U_0 = \frac{\binom{2n}{n}\pi}{2^{2n+1}},$$

and

$$U_{2n+1} = \frac{(2n)(2n-2)\cdots 2}{(2n+1)(2n-1)\cdots 3}\, U_1 = \frac{2^{2n}}{(2n+1)\binom{2n}{n}}.$$

**(iii) Inequalities and bounds:** For $x \in (0,1)$ we have $x^{2n+2} < x^{2n+1} < x^{2n}$; integrating against the positive weight $(1-x^2)^{-1/2}$ preserves the inequalities, giving $U_{2n+2} < U_{2n+1} < U_{2n}$. Substituting the closed forms and rearranging yields

$$\frac{2^{2n}}{(2n+1)\binom{2n}{n}} < \frac{\binom{2n}{n}\pi}{2^{2n+1}},$$

and the companion inequality coming from $U_{2n+2} < U_{2n+1}$ gives the lower bound. Combining both bounds produces

$$\frac{2n+1}{2n+2}\cdot\frac{\pi}{2} < \frac{16^n}{(2n+1)\binom{2n}{n}^2} < \frac{\pi}{2}.$$

**(iv) Asymptotics:** Let $A_n = \dfrac{16^n}{(2n+1)\binom{2n}{n}^2}$. The bounds show $A_n \to \dfrac{\pi}{2}$ as $n \to \infty$, so

$$\binom{2n}{n}^2 \sim \frac{16^n}{(2n)\cdot(\pi/2)} \sim \frac{8\cdot 4^{2n}}{\pi n},$$

and taking square roots yields the familiar central binomial asymptotic

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}.$$

**Takeaways 4.19**

1. **Reduction formulas:** Integrals of the form $\int x^n(1-x^2)^{-1/2}\,dx$ often reduce by two degrees; IBP is the systematic tool.
2. **Wallis connection:** Comparing integrals of even and odd powers of cosine gives precise bounds involving $\pi$ and leads to Wallis-type asymptotics.
3. **Central binomial asymptotics:** The estimate $\binom{2n}{n} \sim 4^n/\sqrt{\pi n}$ follows from simple integral comparisons and is a core asymptotic used throughout combinatorics and analysis.
4. This result establishes Stirling's approximation for the central binomial coefficient.
5. **The Trigonometric Isomorphism:** While algebraic Integration by Parts works, substituting $x = \sin\theta$ instantly converts this problem into the standard Wallis integral $\int_0^{\pi/2} \sin^n\theta\,d\theta$. Try to link this problem with Chebyshev polynomials for deeper insights.
6. **The Weight Function:** The term $\frac{1}{\sqrt{1-x^2}}$ appearing in the integrand is the characteristic weight that makes Chebyshev polynomials orthogonal.

$$\int_{-1}^{1} \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}}\,dx = 0 \quad \text{for } n \neq m$$

**Problem 4.20: Triangle Inequality and Coefficient Analysis**

Let $P(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z$. It is given that the coefficients satisfy $|a_k| \leq 2$ for all $1 \leq k \leq n$. Prove that if $z$ is a solution to $P(z) = 1$, then $|z| > \frac{1}{3}$.

**Hint:** Use the triangle inequality to bound the polynomial by the sum of absolute values of its terms. Convert this to a geometric series bound.

## Solution 4.20

First, note that if $z = 0$, then $P(0) = 0$, which contradicts $P(z) = 1$. Therefore $z \neq 0$, meaning $|z| > 0$. Assume $|z| \leq \frac{1}{3}$.

$$1 = |P(z)| = \left|\sum_{k=1}^{n} a_k z^k\right|$$

$$1 \leq \sum_{k=1}^{n} |a_k||z|^k$$

Using $|a_k| \leq 2$ and $|z| \leq \frac{1}{3}$:

$$1 \leq 2\sum_{k=1}^{n} \left(\frac{1}{3}\right)^k$$

Consider the infinite geometric series sum to establish a strict bound (since terms are positive):

$$\sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^k = \frac{1/3}{1 - 1/3} = \frac{1/3}{2/3} = \frac{1}{2}$$

Thus, the finite sum is strictly less than $\frac{1}{2}$.

$$1 \leq 2 \times (\text{something } < 0.5)$$

$$1 < 1$$

Contradiction. Thus $|z| > \frac{1}{3}$.

## Takeaways 4.20

1. **Infinite vs Finite Sums:** Often, calculating the infinite geometric series sum is easier and sufficient. If the infinite sum creates a contradiction ($< 1$), then the finite sum definitely will too.
2. **Strict Inequalities:** Geometric series of positive terms are always strictly less than their limit at infinity.

## Problem 4.21: Polynomial Root Bounds

Consider the polynomial equation:

$$z^n + c_{n-1}z^{n-1} + \cdots + c_1 z + c_0 = 0$$

Let $M = \max\{|c_0|, |c_1|, \ldots, |c_{n-1}|\}$. Prove that all roots of this equation satisfy $|z| < 1 + M$.

**Hint:** This is a direct application of Cauchy's bound theorem. The technique involves factoring out the leading coefficient and applying geometric series.

## Solution 4.21

Assume $|z| \geq 1 + M$. Rearranging: $z^n = -(c_{n-1}z^{n-1} + \cdots + c_0)$.

$$|z|^n \leq |c_{n-1}||z|^{n-1} + \cdots + |c_1||z| + |c_0|$$

Replace $|c_k|$ with $M$:

$$|z|^n \leq M(|z|^{n-1} + \cdots + |z| + 1)$$

Sum the geometric progression (ratio $|z| > 1$):

$$|z|^n \leq M\frac{|z|^n - 1}{|z| - 1}$$

Since $|z|^n - 1 < |z|^n$ (which is strictly true because $|z| \geq 1 > 0$):

$$|z|^n < M\frac{|z|^n}{|z| - 1}$$

Divide by $|z|^n$ (which is non-zero):

$$1 < \frac{M}{|z| - 1}$$

$$|z| - 1 < M \implies |z| < M + 1$$

This contradicts the assumption $|z| \geq M + 1$. Therefore, $|z| < 1 + M$.

## Takeaways 4.21

1. **Isolation:** Always isolate the highest power ($z^n$) because it grows the fastest. You want to show it "overpowers" the sum of the rest.
2. **Strict Inequality Trick:** Replacing ($|z|^n - 1$) with $|z|^n$ is a valid step to create a strict inequality ($<$) which is crucial for the contradiction.

## Problem 4.22: Polynomial Solutions with Trigonometry

Solve the cubic equation $4x^3 - 3x = \frac{1}{2}$ using trigonometric substitution.

**Hint:** Use trigonometric substitution $x = 2\cos\theta$ to transform the cubic equation. Exploit the identity $4\cos^3\theta - 3\cos\theta = \cos(3\theta)$.

## Solution 4.22

$$\text{Let } x = \cos\theta.$$

Then the equation becomes

$$4\cos^3\theta - 3\cos\theta = \frac{1}{2}.$$

Using the triple-angle identity

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta,$$

we obtain

$$\cos(3\theta) = \frac{1}{2}.$$

Hence

$$3\theta = \pm\frac{\pi}{3} + 2\pi k, \qquad k \in \mathbb{Z},$$

so

$$\theta = \pm\frac{\pi}{9} + \frac{2\pi k}{3}.$$

Taking the three distinct values of $\theta$ modulo $2\pi$ gives

$$\theta = \frac{\pi}{9}, \quad \frac{5\pi}{9}, \quad \frac{7\pi}{9}.$$

Therefore,

$$x = \cos\left(\frac{\pi}{9}\right), \quad \cos\left(\frac{5\pi}{9}\right), \quad \cos\left(\frac{7\pi}{9}\right).$$

Since a polynomial of degree 3 can have at most 3 roots in the complex plane, and we have found 3 distinct real roots, these are all the solutions.

## Takeaways 4.22

1. **Trigonometric Substitution:** For cubic equations with specific forms, trigonometric identities can provide exact solutions.
2. **Chebyshev Link ($T_3$):** The Chebyshev polynomial of the first kind $T_3(x)$ satisfies $T_3(x) = 4x^3 - 3x$. Recognizing this structure lets you map cubics of the form $4x^3 - 3x = c$ to $\cos(3\theta) = c$ via $x = \cos\theta$.
3. **Multiple Angle Formulas:** The identity $4\cos^3\theta - 3\cos\theta = \cos(3\theta)$ is particularly useful for solving cubics that match the structure of the $3^{\text{rd}}$ degree Chebyshev polynomial.

## Problem 4.23: Complex Series and Geometric Bounds

Suppose that a complex number $z$ satisfies:

$$\sum_{k=1}^{\infty} a_k z^k = 1$$

where the coefficients satisfy $|a_k| \leq \frac{1}{3^k}$ for all $k \geq 1$. Prove that $|z| \geq \frac{3}{2}$.

**Hint:** Assume the contrary: $|z| < \frac{3}{2}$. Apply the triangle inequality to the sum: $1 \leq \sum |a_k||z|^k$. Substitute the bound for $|a_k|$ to get a geometric series with ratio $r = \frac{3}{|z|}$. Calculate the sum to infinity and check if the inequality holds.

## Solution 4.23

Assume $|z| < \frac{3}{2}$.

$$1 = \left|\sum_{k=1}^{\infty} a_k z^k\right| \leq \sum_{k=1}^{\infty} |a_k||z|^k$$

Using $|a_k| \leq 3^{-k}$:

$$1 \leq \sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^k |z|^k = \sum_{k=1}^{\infty} \left(\frac{|z|}{3}\right)^k$$

This is a geometric series with ratio $r = \frac{|z|}{3}$. Since $|z| < 1.5$, $r < 0.5$, so the series converges. Sum formula $S = \frac{r}{1-r}$:

$$1 \leq \frac{|z|/3}{1 - |z|/3} = \frac{|z|}{3 - |z|}$$

Rearrange the inequality $1 \leq \frac{|z|}{3-|z|}$:

$$3 - |z| \leq |z|$$

$$3 \leq 2|z| \implies |z| \geq \frac{3}{2}$$

This contradicts the assumption $|z| < \frac{3}{2}$. Thus, $|z| \geq \frac{3}{2}$.

## Takeaways 4.23

1. **Convergence Check:** When dealing with infinite series, always quickly check that the common ratio ($|z|/3$) is less than 1 to ensure the sum formula is valid.
2. **Algebraic Rearrangement:** The contradiction often appears only after rearranging the final inequality (e.g., $3 \leq 2|z|$), not immediately upon summing.

## Problem 4.24: Bounds of Factorials

(i) By considering the area under the curve $y = \ln x$, show that for any integer $k \geq 1$:

$$\int_k^{k+1} \ln x \, dx > \ln k$$

(ii) Hence, using part (i), prove that for all integers $n \geq 2$:

$$n! < e\left(\frac{n+1}{e}\right)^{n+1}$$

**Hint:** For part (i), visualize the graph of $y = \ln x$. Is it increasing? Compare the area under the curve from $x = k$ to $x = k + 1$ with the area of a rectangle of width 1 and height $\ln k$ (a lower rectangle). For part (ii), sum the inequality from $k = 1$ to $n$. Recall that $\sum_{k=1}^{n} \ln k = \ln(n!)$.

## Solution 4.24

**(i)** Since $y = \ln x$ is strictly increasing, for $x \in [k, k+1]$, $\ln x > \ln k$.

$$\int_k^{k+1} \ln x \, dx > \int_k^{k+1} \ln k \, dx = [x \ln k]_k^{k+1} = \ln k$$

**(ii)** Sum both sides from $k = 1$ to $n$:

$$\sum_{k=1}^{n} \int_k^{k+1} \ln x \, dx > \sum_{k=1}^{n} \ln k$$

$$\int_1^{n+1} \ln x \, dx > \ln(n!)$$

Evaluate the integral: $[x \ln x - x]_1^{n+1} = (n+1)\ln(n+1) - (n+1) - (1 \ln 1 - 1)$.

$$(n+1)\ln(n+1) - n > \ln(n!)$$

Exponentiate both sides:

$$e^{(n+1)\ln(n+1)-n} > n! \implies \frac{(n+1)^{n+1}}{e^n} > n!$$

$$n! < e \cdot \frac{(n+1)^{n+1}}{e^{n+1}} = e\left(\frac{n+1}{e}\right)^{n+1}$$

## Takeaways 4.24

1. **Riemann Sums:** Any time you see $n!$ related to $(n/e)^n$, think about integrating $\ln x$.
2. **Rectangle Inequality:** $\int_k^{k+1} f(x)dx$ is often compared to $f(k)$ (lower rectangle) or $f(k+1)$ (upper rectangle).
3. **Stirling's Approximation:** These bounds on $\ln(n!)$ are key steps toward Stirling's formula $n! \sim \sqrt{2\pi n}\,(n/e)^n$.

### Remark 0.8: How good is Stirling's approximation?

For reference, the table below compares Stirling's approximation

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

with the exact value of $n!$ for $n = 1, 2, \ldots, 10$. The error shown is the absolute error

$$\left|\sqrt{2\pi n}(n/e)^n - n!\right|.$$

| $n$ | Stirling approximation | exact $n!$ | absolute error |
|---|---|---|---|
| 1 | 0.922137 | 1 | 0.077863 |
| 2 | 1.919004 | 2 | 0.080996 |
| 3 | 5.836210 | 6 | 0.163790 |
| 4 | 23.506175 | 24 | 0.493825 |
| 5 | 118.019168 | 120 | 1.980832 |
| 6 | 710.078185 | 720 | 9.921815 |
| 7 | 4980.395832 | 5040 | 59.604168 |
| 8 | 39902.395453 | 40320 | 417.604547 |
| 9 | 359536.872842 | 362880 | 3343.127158 |
| 10 | 3598695.618741 | 3628800 | 30104.381259 |

Although the absolute error grows, the *relative* accuracy improves quickly, which is why Stirling's formula is so useful for large $n$.

### Problem 4.25: The Logarithmic Bound

(i) Prove that for $x > 0$,

$$\frac{1}{x+1} < \ln(x+1) - \ln x < \frac{1}{x}$$

Let

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

denote the $n$-th harmonic number.
(ii) Hence, prove that for any integer $n > 1$:

$$\ln(n+1) < H_n < 1 + \ln n$$

(iii) Use parts (i) and (ii), or otherwise, to prove that the sequence

$$\gamma_n := H_n - \ln n$$

converges as $n \to \infty$. The limit is known as the Euler's constant $\gamma$.

**Hint:** For part (i), express $\ln(x+1) - \ln x$ as an integral $\int_x^{x+1} \frac{1}{t}\,dt$. Use the max/min value of $1/t$ on this interval. For part (ii), use the method of telescoping sums. Write the inequality for $k = 1, 2, \ldots$ and sum them up.

## Solution 4.25

**(i)** Consider $f(t) = 1/t$. For $t \in [x, x+1]$, we have $\frac{1}{x+1} < \frac{1}{t} < \frac{1}{x}$. Integrating from $x$ to $x+1$:

$$\int_x^{x+1} \frac{dt}{x+1} < \int_x^{x+1} \frac{dt}{t} < \int_x^{x+1} \frac{dt}{x}$$

$$\frac{1}{x+1} < [\ln t]_x^{x+1} < \frac{1}{x} \implies \frac{1}{x+1} < \ln(x+1) - \ln x < \frac{1}{x}$$

**(ii) Left inequality:** Sum $\ln(k+1) - \ln k < \frac{1}{k}$ from $k = 1$ to $n$:

$$\ln(n+1) - \ln 1 < \sum_{k=1}^{n} \frac{1}{k}$$

$$\ln(n+1) < 1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

**Right inequality:** Sum $\frac{1}{k+1} < \ln(k+1) - \ln k$ from $k = 1$ to $n-1$:

$$\sum_{k=2}^{n} \frac{1}{k} < \ln n - \ln 1$$

$$1 + \frac{1}{2} + \cdots + \frac{1}{n} < 1 + \ln n$$

**(iii)** Let $H_n = \sum_{k=1}^{n} \frac{1}{k}$ and set $a_n = H_n - \ln n$. By (i),

$$\frac{1}{n+1} < \ln(n+1) - \ln n,$$

so

$$a_{n+1} - a_n = \frac{1}{n+1} - (\ln(n+1) - \ln n) < 0,$$

hence $(a_n)$ is decreasing. By (ii), $\ln(n+1) < H_n$, thus

$$a_n = H_n - \ln n > \ln(n+1) - \ln n > 0,$$

so $(a_n)$ is bounded below. Therefore $a_n$ converges, and the limit

$$\gamma := \lim_{n\to\infty} \left( \sum_{k=1}^{n} \frac{1}{k} - \ln n \right)$$

exists (Euler's constant).

## Takeaways 4.25

1. **Integration Technique:** Expressing a discrete difference $(\ln(k+1) - \ln k)$ as an integral is a powerful way to bound reciprocals.
2. **Telescoping Sums:** Recognizing that $\sum(\ln(k+1) - \ln k) = \ln(n+1) - \ln 1$ is essential for Q16.

## Problem 4.26: Exponential Sequence Bounds

(i) Show that $e^x > 1 + x$ for all $x > 0$.
(ii) Hence, prove by substitution or otherwise that for any integer $n \geq 1$:

$$\left(1 + \frac{1}{n}\right)^n < e < \left(1 + \frac{1}{n}\right)^{n+1}$$

**Hint:** For part (i), let $f(x) = e^x - (1+x)$ and calculate $f'(x)$. For part (ii), this is a double inequality. Left side: Use part (i) with a substitution like $x = \frac{1}{n}$. Right side: Use part (i) with a substitution involving a negative exponent, e.g., $x = -\frac{1}{n+1}$, or rearrange the target to find the required $x$.

## Solution 4.26

**(i)** Let $f(x) = e^x - 1 - x$. $f'(x) = e^x - 1$. For $x > 0$, $e^x > 1 \implies f'(x) > 0$. Since $f(0) = 0$, $f(x) > 0$ for $x > 0$.
**(ii) Left Bound:** Let $x = \frac{1}{n}$. From (i):

$$e^{1/n} > 1 + \frac{1}{n}$$

Raise to power $n$: $e > (1 + \frac{1}{n})^n$.
**Right Bound:** Consider $x = -\frac{1}{n+1}$ in the inequality $e^x \geq 1 + x$ (valid for all real $x$):

$$e^{-\frac{1}{n+1}} > 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

Invert both sides:

$$e^{\frac{1}{n+1}} < \frac{n+1}{n} = 1 + \frac{1}{n}$$

Raise to power $n+1$:

$$e < \left(1 + \frac{1}{n}\right)^{n+1}$$

## Takeaways 4.26

1. **Standard Inequality:** $e^x \geq 1 + x$ is the convexity inequality for the exponential function. It is used constantly to convert sums $(1 + x)$ into products/exponentials $(e^x)$.
2. **Substitution:** The "Hard" part is choosing the correct $x$ (e.g., $1/n$ vs $-1/(n+1)$).

## Problem 4.27: Bounding Products

Let

$$P_n = \prod_{k=1}^{n} \left(1 + \frac{k}{n^2}\right)$$

.

(i) Prove that $x - \frac{x^2}{2} < \ln(1+x)$ for $x > 0$.

(ii) By using the inequality $x - \frac{x^2}{2} < \ln(1+x) < x$ for $x > 0$, show that:

$$e^{\frac{n(n+1)}{2n^2} - \frac{n(n+1)(2n+1)}{12n^4}} < P_n < e^{\frac{n(n+1)}{2n^2}}$$

(iii) By using the result from part (ii), determine the limit of the product as $n \to \infty$:

$$\lim_{n\to\infty} P_n$$

(iv) Use a Riemann sum argument, to show that:

$$\lim_{n\to\infty} \ln\left(P_n\right) = \int_0^1 x\,dx$$

**Hint:** Take the natural logarithm of the product in the middle. This converts the product into a sum: $\sum_{k=1}^{n} \ln(1 + \frac{k}{n^2})$. Use the inequality from part (i) by setting $x = \frac{k}{n^2}$. Recall the sum of squares formula $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.

## Solution 4.27

**(i)** Let $f(x) = \ln(1+x) - x + x^2/2$. $f'(x) = \frac{1}{1+x} - 1 + x = \frac{1-(1-x^2)}{1+x} = \frac{x^2}{1+x}$. Since $x > 0$, $f'(x) > 0$, so $f(x) > f(0) = 0$.
**(ii)** Let $P = \prod_{k=1}^{n}(1+\frac{k}{n^2})$. Then $\ln P = \sum_{k=1}^{n} \ln(1+\frac{k}{n^2})$. Using $x - x^2/2 < \ln(1+x) < x$ with $x = k/n^2$:

$$\sum \left(\frac{k}{n^2} - \frac{k^2}{2n^4}\right) < \ln P < \sum \frac{k}{n^2}$$

**Upper Bound:** $\frac{1}{n^2}\sum k = \frac{n(n+1)}{2n^2}$. So $P < e^{\frac{n(n+1)}{2n^2}}$.
**Lower Bound:** $\frac{1}{n^2}\sum k - \frac{1}{2n^4}\sum k^2$.

$$\frac{n(n+1)}{2n^2} - \frac{1}{2n^4} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)}{2n^2} - \frac{n(n+1)(2n+1)}{12n^4}$$

Exponentiating gives the result.
**(iii)** From part (ii), the bounds are:

$$e^{\frac{n(n+1)}{2n^2} - \frac{n(n+1)(2n+1)}{12n^4}} < \prod_{k=1}^{n}\left(1+\frac{k}{n^2}\right) < e^{\frac{n(n+1)}{2n^2}}$$

The exponents simplify as $n \to \infty$:

$$\frac{n(n+1)}{2n^2} = \frac{1}{2}\left(1+\frac{1}{n}\right) \to \frac{1}{2}$$

$$\frac{n(n+1)(2n+1)}{12n^4} = \frac{(1+\frac{1}{n})(2+\frac{1}{n})}{12n} \to 0$$

By the squeeze theorem, $\lim_{n\to\infty} \prod_{k=1}^{n}\left(1+\frac{k}{n^2}\right) = e^{1/2} = \sqrt{e}$.
**(iv)** Taking logarithms:

$$\ln\left(\prod_{k=1}^{n}\left(1+\frac{k}{n^2}\right)\right) = \sum_{k=1}^{n} \ln\left(1+\frac{k}{n^2}\right)$$

For small $x$ (see (i)), $\ln(1+x) \approx x$. With $x = k/n^2$:

$$\sum_{k=1}^{n} \ln\left(1+\frac{k}{n^2}\right) \approx \sum_{k=1}^{n} \frac{k}{n^2} = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}$$

This is a Riemann sum for $\int_0^1 x\,dx$ with partition width $\Delta x = 1/n$ and sample points $x_k = k/n$. As $n \to \infty$, the limit of the Riemann sum evaluates precisely to the definite integral: $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{k}{n} \cdot \frac{1}{n} = \int_0^1 x\,dx$.

$$\sum_{k=1}^{n} \frac{k}{n} \cdot \frac{1}{n} = \frac{1}{n^2}\sum_{k=1}^{n} k \to \int_0^1 x\,dx = \frac{1}{2}$$

Therefore, $\lim_{n\to\infty} \ln\left(\prod_{k=1}^{n}\left(1+\frac{k}{n^2}\right)\right) = \int_0^1 x\,dx = \frac{1}{2}$.

## Takeaways 4.27

1. **Logs turn Products to Sums:** This is the standard strategy for dealing with $\prod(1+a_k)$.
2. **Taylor Polynomial Bounds:** $\ln(1+x) \approx x - x^2/2$. Using the first few terms of a Taylor series as an inequality is a very common "Harder" topic.
3. $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. Still really useful to remember!
4. $\ln(1+x) < x$ for $x > 0$
5. $\ln(1+x) = x - x^2/2 + x^3/3 - \ldots$ (alternating series)

## Problem 4.28: Exponential Equation Analysis

Consider the equation $2^x = x^2$.

(i) Show that there are exactly three real solutions.

(ii) Find all integer solutions.

**Hint:** Take the natural logarithm of both sides to handle the variable exponent. Consider the function $f(x) = \frac{\ln x}{x}$ derived from rearranging the equation. Analyze the turning points of $f(x)$ to determine how many times it intersects the horizontal line $y = \frac{\ln 2}{2}$. Don't forget to check negative values of $x$ separately (where logs might not apply directly, but graphs do).

## Solution 4.28

**(i)** For $x > 0$: Take logs $\implies x \ln 2 = 2 \ln x \implies \frac{\ln x}{x} = \frac{\ln 2}{2}$. Let $g(x) = \frac{\ln x}{x}$. $g'(x) = \frac{1-\ln x}{x^2}$. Max at $x = e$. $g(e) = 1/e \approx 0.368$. The value $\frac{\ln 2}{2} \approx 0.346$. Since $0 < 0.346 < 0.368$, the line $y = \frac{\ln 2}{2}$ cuts the curve $y = g(x)$ twice for $x > 0$.
For $x < 0$: $2^x$ is positive, $x^2$ is positive. For $x \leq 0$, the function $y = 2^x$ is strictly increasing from 0 to 1, while $y = x^2$ is strictly decreasing from $\infty$ to 0. Since they are continuous, they must intersect exactly once by the Intermediate Value Theorem. Total = 3 solutions.
**(ii)** From the Log analysis ($x > 0$): One solution is obviously $x = 2$ ($2^2 = 2^2$). Check $x = 4$: $2^4 = 16$, $4^2 = 16$. Correct.
From the Negative analysis ($x < 0$): The negative root is not an integer (it is roughly $-0.766$). So integer solutions are $x = 2, x = 4$.

## Takeaways 4.28

1. **Auxiliary Functions:** Converting $a^x = x^a$ into $\frac{\ln x}{x} = \frac{\ln a}{a}$ is the standard method for solving "variable base and exponent" problems.

2. **Domain Checks:** Always separate positive and negative domains when taking logarithms.

## Problem 4.29: Complex Rotation and Imaginary Parts

Let $P, Q, R$ be vertices of an equilateral triangle in the complex plane with $P$ at the origin. Point $Q$ lies on the line $y = 1$ and point $R$ lies on the line $y = 3$. Find the complex number $z_R$ representing point $R$.

**Hint:** Let $z_P = 0$. Let $z_Q = a + i$ for some real $a$. Let $z_R = b + 3i$ for some real $b$. Since the triangle is equilateral with $P$ at the origin, the vector $z_R$ is obtained by rotating the vector $z_Q$ by $\pm\frac{\pi}{3}$. Use the rotation formula $z_R = z_Q e^{\pm i\pi/3}$. Equate the imaginary parts to solve for $a$.

## Solution 4.29

1. $z_R = z_Q(\cos\frac{\pi}{3} \pm i\sin\frac{\pi}{3}) = (a+i)(\frac{1}{2} \pm i\frac{\sqrt{3}}{2})$.
2. Expand the RHS:

$$\text{Im}(z_R) = a(\pm\frac{\sqrt{3}}{2}) + 1(\frac{1}{2})$$

3. We are given $\text{Im}(z_R) = 3$.

$$3 = \pm\frac{a\sqrt{3}}{2} + \frac{1}{2}$$

$$2.5 = \pm\frac{a\sqrt{3}}{2} \implies 5 = \pm a\sqrt{3} \implies a = \pm\frac{5}{\sqrt{3}}$$

4. Now find $z_R$. Since $z_R$ is just the rotated vector: $\text{Re}(z_R) = \frac{a}{2} \mp \frac{\sqrt{3}}{2}$.

Case 1 ($a = \frac{5}{\sqrt{3}}$): $\text{Re} = \frac{5}{2\sqrt{3}} - \frac{\sqrt{3}}{2} = \frac{5-3}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$.

Case 2 ($a = -\frac{5}{\sqrt{3}}$): $\text{Re} = -\frac{5}{2\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{-5+3}{2\sqrt{3}} = -\frac{1}{\sqrt{3}}$.

Solution: $z_R = \pm\frac{1}{\sqrt{3}} + 3i = \pm\frac{\sqrt{3}}{3} + 3i$.

## Takeaways 4.29

1. **Imaginary Part Matching:** When lines are horizontal ($y = k$), the most efficient algebraic step is usually Im(LHS) = Im(RHS).
2. **Origin Shift:** Placing one vertex at the origin simplifies the rotation formula from $(z_3 - z_1) = e^{i\theta}(z_2 - z_1)$ to just $z_3 = z_2 e^{i\theta}$.

## Problem 4.30: The 3-4-5 Triangle Construction

Let $ABC$ be an equilateral triangle in the Argand plane ordered anticlockwise. A point $P$ lies inside the triangle such that:

$$|z_P - z_A| = 3, \quad |z_P - z_B| = 4, \quad |z_P - z_C| = 5$$

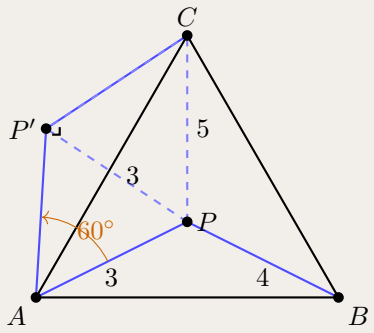
(i) By rotating the triangle $ABP$ by 60° about the vertex $A$ to a new position $ACP'$, or otherwise, show that the triangle $APP'$ is equilateral.

(ii) Hence, determine the area of the equilateral triangle $ABC$.

**Hint:** Consider the rotation of $P$ about $A$ by 60° to $P'$. Since $AP = AP'$ and the angle is 60°, $\triangle APP'$ is equilateral. This gives $|P'P| = 3$. Since $B$ rotates to $C$, the segment $PB$ rotates to $P'C$. Thus $|P'C| = |PB| = 4$. Now consider the triangle $PCP'$. You know sides 3, 4, 5.

## Solution 4.30

1. Rotate $\triangle ABP$ by $60°$ about $A$. $B \to C$. $P \to P'$.
2. Since rotation preserves length, $|P'C| = |PB| = 4$.




3. Since $AP$ rotates to $AP'$ by $60°$, $\triangle APP'$ is isosceles with $60°$, hence equilateral. Therefore, $|PP'| = |AP| = 3$.
4. Consider $\triangle PP'C$. The side lengths are $|PP'| = 3$, $|P'C| = 4$, and $|PC| = 5$. Since $3^2 + 4^2 = 5^2$, $\triangle PP'C$ is a right-angled triangle with $\angle PP'C = 90°$ (by the converse of the Pythagorean theorem).
5. We want the side length of $ABC$, which is $|AC|$. In $\triangle AP'C$, we have sides $AP' = 3$, $P'C = 4$. The angle $\angle AP'C = \angle AP'P + \angle PP'C = 60° + 90° = 150°$.
By Cosine Rule on $\triangle AP'C$: $|AC|^2 = 3^2 + 4^2 - 2(3)(4)\cos(150°)$. $|AC|^2 = 9 + 16 - 24(-\frac{\sqrt{3}}{2}) = 25 + 12\sqrt{3}$.
Area of equilateral triangle $= \frac{\sqrt{3}}{4}(\text{side})^2 = \frac{\sqrt{3}}{4}(25 + 12\sqrt{3})$.

## Takeaways 4.30

1. **Constructive Rotation:** If you are given three distances from a point to vertices of an equilateral triangle, **ALWAYS** rotate the whole setup by $60°$ around one vertex. It creates a new triangle with the known lengths as sides.

2. Rotating a point around a vertex of an equilateral triangle by $60°$ creates an equilateral triangle with the original point and its image.

## Problem 4.31: Polynomial-Trigonometric Identity

Let $w = e^{i2\pi/N} = \cos\left(\frac{2\pi}{N}\right) + i\sin\left(\frac{2\pi}{N}\right)$ where $N$ is an odd positive integer.
For $|\gamma| = 1$, let $P_N$ be the polynomial satisfying:

$$\gamma^N + \frac{1}{\gamma^N} = P_N\left(\gamma + \frac{1}{\gamma}\right)$$

(i) Show that $x = \gamma + \bar{\gamma}$ satisfies $P_N(x) = 2\cos\left(\frac{2\pi}{N}\right)$.

(ii) Find the roots of this polynomial equation in the form $x_k = 2\cos\theta_k$.

(iii) Given that $P_N(x)$ is monic with constant term 0 for odd $N$, prove:

$$\prod_{k=0}^{N-1} \cos\theta_k = \frac{1}{2^{N-1}}\cos\left(\frac{2\pi}{N}\right)$$

**Hint:**

- For **(i)**, substitute $z = \gamma$ into the given polynomial identity. Recall that if $|\gamma| = 1$, then $\bar{\gamma} = 1/\gamma$.
- For **(ii)**, solve $\gamma^N = e^{i2\pi/N}$ to find the distinct values of $\gamma$, then use the substitution $x = \gamma + 1/\gamma = 2\text{Re}(\gamma)$.
- For **(iii)**, use **Vieta's Formulas**. The product of the roots of a polynomial $a_n x^n + \cdots + a_0 = 0$ is equal to $(-1)^n \frac{a_0}{a_n}$. Be careful with the factor of 2 in $x = 2\cos\theta$.

## Solution 4.31

**(i)** Let $x = \gamma + \bar{\gamma}$. Since $|\gamma| = 1$, $\bar{\gamma} = 1/\gamma$. So $x = \gamma + 1/\gamma$. Using the given identity for $k = N$:

$$\gamma^N + \frac{1}{\gamma^N} = P_N\left(\gamma + \frac{1}{\gamma}\right) = P_N(x)$$

We are given $\gamma^N = w = e^{i2\pi/N}$. Thus $1/\gamma^N = \bar{w} = e^{-i2\pi/N}$.

$$P_N(x) = w + \bar{w} = 2\cos\left(\frac{2\pi}{N}\right)$$

**(ii)** Solving $\gamma^N = e^{i2\pi/N}$:

$$\gamma_k = e^{i\frac{2\pi(1+kN)}{N^2}} \quad \text{for } k = 0, 1, \ldots, N-1$$

The roots of the polynomial in $x$ are $x_k = \gamma_k + \bar{\gamma}_k = 2\cos\left(\frac{2\pi(1+kN)}{N^2}\right)$. Thus, $\theta_k = \frac{2\pi(1+kN)}{N^2}$.

**(iii)** The equation is $P_N(x) - 2\cos\left(\frac{2\pi}{N}\right) = 0$. The constant term of this equation is $0 - 2\cos(2\pi/N) = -2\cos(2\pi/N)$. By Vieta's formulas for degree $N$ (with $N$ odd, so $(-1)^N = -1$):

$$\text{Product of Roots} = (-1)^N \frac{\text{Constant Term}}{\text{Leading Coeff}} = (-1)\frac{-2\cos(2\pi/N)}{1} = 2\cos\left(\frac{2\pi}{N}\right)$$

Substitute $x_k = 2\cos(\theta_k)$:

$$\prod_{k=0}^{N-1}(2\cos\theta_k) = 2^N \prod_{k=0}^{N-1}\cos\theta_k = 2\cos\left(\frac{2\pi}{N}\right)$$

Divide by $2^N$:

$$\prod_{k=0}^{N-1}\cos\theta_k = \frac{1}{2^{N-1}}\cos\left(\frac{2\pi}{N}\right)$$

## Takeaways 4.31

1. **Polynomial Link:** The identity $z^n + 1/z^n = P_n(z + 1/z)$ links complex roots of unity to polynomials with real coefficients. These are related to Chebyshev polynomials ($T_n$).
2. **Roots to Coefficients:** Harder problems often ask you to evaluate a trigonometric product. This is almost always a signal to form a polynomial equation and use **Vieta's Product of Roots**.
3. **Generalization:** Identifying $N$ as odd was crucial. If $N$ were even, the constant term of $P_N(x)$ would be non-zero, changing the final result.
4. Try substitute $\gamma = e^{i\theta}$

### Problem 4.32: Cotangent Polynomial

Let $N = 2m + 1$ be an odd integer.

(i) Show that $\sin(N\theta)$ can be expressed as:

$$\sin(N\theta) = \sin^N \theta \cdot P(\cot^2 \theta)$$

where $P$ is a polynomial of degree $m$.

(ii) Find the roots of $P(x) = 0$ in terms of $\theta$.

(iii) Use Vieta's formulas to find:

$$\sum_{k=1}^{m} \cot^2\left(\frac{k\pi}{N}\right)$$

**Hint:**

- For (ii), recall $\sin(N\theta)$ is the imaginary part of $(\cos + i\sin)^N$.
- Group terms by powers of $i$. Factor out $\sin^N \theta$.
- Terms will look like $\binom{N}{1}\cot^{N-1}\theta - \binom{N}{3}\cot^{N-3}\theta \dots$
- For (iv), use $\textbf{Sum of Roots} = -\frac{\text{coeff of } x^{m-1}}{\text{coeff of } x^m}$.

## Solution 4.32

**(i) Derive the Polynomial $P(x)$**
Using De Moivre's Theorem and the binomial expansion for $N = 2m + 1$:

$$\begin{aligned}\sin(N\theta) &= \operatorname{Im}[(\cos\theta + i\sin\theta)^N] \\ &= \binom{N}{1}\cos^{N-1}\theta\sin\theta - \binom{N}{3}\cos^{N-3}\theta\sin^3\theta - \cdots + (-1)^m\binom{N}{N}\sin^N\theta\end{aligned}$$

Factor out $\sin^N\theta$:

$$\sin(N\theta) = \sin^N\theta\left[\binom{N}{1}\cot^{N-1}\theta - \binom{N}{3}\cot^{N-3}\theta - \cdots + (-1)^m\binom{N}{N}\right]$$

Let $x = \cot^2\theta$. Then $\cot^{N-1}\theta = (\cot^2\theta)^m = x^m$, and $\cot^{N-3}\theta = (\cot^2\theta)^{m-1} = x^{m-1}$, etc. The polynomial $P(x)$ is of degree $m$:

$$P(x) = \binom{N}{1}x^m - \binom{N}{3}x^{m-1} + \binom{N}{5}x^{m-2} - \cdots + (-1)^m\binom{N}{N}$$

**(ii) Find the Roots of $P(x) = 0$**
The roots of $P(\cot^2\theta) = 0$ occur when $\sin(N\theta) = 0$, provided $\sin\theta \neq 0$.

$$\sin(N\theta) = 0 \implies N\theta = k\pi \implies \theta = \frac{k\pi}{N}$$

Since $\cot^2\theta$ is periodic with period $\pi$ and is an even function, we need distinct values of $\cot^2\theta$. The values of $k = 1, 2, \ldots, m$ give distinct roots $x_k$:

$$x_k = \cot^2\left(\frac{k\pi}{N}\right) \quad \text{for } k = 1, 2, \ldots, m$$

Since $P(x)$ is a polynomial of degree $m$, these $m$ roots are all the roots of $P(x) = 0$.
**(iii) Use Vieta's Formulas to Find the Sum of Roots**
**Step 1: Identify the Coefficients.** The polynomial is $P(x) = a_m x^m + a_{m-1}x^{m-1} + \cdots + a_0$, where: (i) Coefficient of $x^m$: $a_m = \binom{N}{1}$ (ii) Coefficient of $x^{m-1}$: $a_{m-1} = -\binom{N}{3}$
**Step 2: Apply Vieta's Formula.** The sum of the roots $(\sum x_k)$ is given by Vieta's formula: $\sum x_k = -\frac{a_{m-1}}{a_m}$.

$$\sum_{k=1}^{m}\cot^2\left(\frac{k\pi}{N}\right) = -\frac{-\binom{N}{3}}{\binom{N}{1}} = \frac{\binom{N}{3}}{\binom{N}{1}}$$

**Step 3: Substitute Binomial Coefficients and Simplify.** Substitute the formulas for the binomial coefficients: $\binom{N}{1} = N$ and $\binom{N}{3} = \frac{N(N-1)(N-2)}{6}$.

$$\sum_{k=1}^{m}\cot^2\left(\frac{k\pi}{N}\right) = \frac{\frac{N(N-1)(N-2)}{6}}{N} = \frac{(N-1)(N-2)}{6}$$

## Takeaways 4.32

1. **Imaginary Part Extraction:** Equating the imaginary part of the expansion to $\sin(N\theta)$ is the standard way to generate polynomials in $\cot\theta$.
2. **Vieta's Sum:** The sum of trig squares usually simplifies to a quadratic in $N$.
3. The polynomial $P(x)$ links to Chebyshev polynomials of the second kind.

## Problem 4.33: Wallis-Type Integral and Pi Bounds

Let $I_n$ be the integral defined for integers $n \geq 0$ by:

$$I_n = \int_0^1 (1 - x^2)^n \, dx$$

(i) Use integration by parts to prove the reduction formula:

$$I_n = \frac{2n}{2n+1} I_{n-1}, \quad \text{for } n \geq 1$$

(ii) Hence, show that:

$$I_n = \frac{2^{2n}(n!)^2}{(2n+1)!}$$

(iii) Using the substitution $x = \sin\theta$, show that

$$I_n = \int_0^{\frac{\pi}{2}} \cos^{2n+1}\theta \, d\theta$$

(iv) Let

$$J_n = \int_0^{\frac{\pi}{2}} \cos^{2n}\theta \, d\theta$$

. You are given that

$$J_n = \frac{\pi(2n)!}{2^{2n+1}(n!)^2}$$

.

By considering the behaviour of $\cos\theta$ on the interval $(0, \frac{\pi}{2})$, explain why $I_n < J_n$.

(v) Deduce the inequality:

$$\frac{2^{4n}(n!)^4}{[(2n)!]^2(2n+1)} < \frac{\pi}{2}$$

**Hint:**

- **Part (i):** Do not integrate $1 \cdot (1-x^2)^n$. Instead, split the term:

  $$(1-x^2)^n = (1-x^2)(1-x^2)^{n-1} = (1-x^2)^{n-1} - x^2(1-x^2)^{n-1}$$

  Then apply Integration by Parts to $\int x \cdot [x(1-x^2)^{n-1}]\, dx$.

- **Part (ii):** Apply the formula recursively. Multiply the numerator and denominator by even terms to form factorials.

- **Part (iv):** If $0 < y < 1$, then $y^{k+1} < y^k$. Apply this to $y = \cos\theta$.

## Solution 4.33

**(i)** Write $I_n = \int_0^1 (1-x^2)^{n-1}dx - \int_0^1 x^2(1-x^2)^{n-1}dx$. Use IBP on the second integral: $u = x, dv = x(1-x^2)^{n-1}dx$. Then $du = dx, v = -\frac{1}{2n}(1-x^2)^n$.

$$\int_0^1 x^2(1-x^2)^{n-1}dx = \left[-\frac{x}{2n}(1-x^2)^n\right]_0^1 + \frac{1}{2n}\int_0^1 (1-x^2)^n dx = 0 + \frac{1}{2n}I_n$$

Substituting back: $I_n = I_{n-1} - \frac{1}{2n}I_n$. $I_n(1+\frac{1}{2n}) = I_{n-1} \implies I_n = \frac{2n}{2n+1}I_{n-1}$.
**(ii)** $I_n = \frac{2n}{2n+1}\cdot\frac{2n-2}{2n-1}\cdots\frac{2}{3}I_0$. Since $I_0 = 1$:

$$I_n = \frac{2n(2n-2)\cdots 2}{(2n+1)(2n-1)\cdots 1}$$

Multiply top and bottom by $P = 2n(2n-2)\cdots 2 = 2^n n!$:

$$I_n = \frac{P\cdot P}{(2n+1)!} = \frac{(2^n n!)^2}{(2n+1)!} = \frac{2^{2n}(n!)^2}{(2n+1)!}$$

**(iii)** $x = \sin\theta, dx = \cos\theta d\theta$. Limits $0\to 0, 1\to \pi/2$. $I_n = \int_0^{\pi/2}(\cos^2\theta)^n\cos\theta d\theta = \int_0^{\pi/2}\cos^{2n+1}\theta d\theta$.
**(iv)** For $\theta\in(0,\pi/2)$, $0<\cos\theta<1$. Thus $\cos^{2n+1}\theta < \cos^{2n}\theta$. Integrating over the interval preserves the inequality: $I_n < J_n$.
**(v)** Substitute the expressions:

$$\frac{2^{2n}(n!)^2}{(2n+1)!} < \frac{\pi(2n)!}{2^{2n+1}(n!)^2}$$

Multiply and rearrange:

$$\frac{2^{4n}(n!)^4}{[(2n)!]^2(2n+1)} < \frac{\pi}{2}$$

## Takeaways 4.33

1. **Algebraic Splitting:** The trick $(1-x^2)^n = (1-x^2)^{n-1} - x^2(1-x^2)^{n-1}$ is a high-yield technique for deriving reduction formulae.
2. **Wallis Product Logic:** Comparing $\int \cos^{odd}$ and $\int \cos^{even}$ is the standard method for establishing bounds for $\pi$.
3. **Factorial Identities:** The manipulation $(2n)!! = 2^n n!$ is crucial for Q16 combinatorics/integration.

## Problem 4.34: Logarithmic Differentiation of Polynomials

Let $\omega = e^{2\pi i/n} = \cos\left(\frac{2\pi}{n}\right) + i\sin\left(\frac{2\pi}{n}\right)$
be an $n$-th root of unity.

(i) Show that

$$z^n - 1 = \prod_{k=0}^{n-1}(z - \omega^k)$$

.

(ii) By taking logarithms and differentiating, prove:

$$\frac{nz^{n-1}}{z^n - 1} = \sum_{k=0}^{n-1}\frac{1}{z - \omega^k}$$

(iii) By taking the limit as $z \to 1$, show that:

$$\sum_{k=1}^{n-1}\frac{1}{1 - \omega^k} = \frac{n-1}{2}$$

**Hint:**

- **Part (i):** $z^n - 1 = \prod_{k=0}^{n-1}(z - \omega^k)$.
- **Part (ii):** Take the natural logarithm of both sides: $\ln(z^n - 1) = \sum \ln(z - \omega^k)$, then differentiate. Or use $\frac{P'(z)}{P(z)}$.
- **Part (iii):** This requires substituting $z = 1$, but $z = 1$ is a singularity. Separate the $k = 0$ term $(\frac{1}{z-1})$ from the RHS, then take the limit as $z \to 1$.

## Solution 4.34

**(i)** Since the $n$th roots of unity are $1, \omega, \omega^2, \ldots, \omega^{n-1}$, they are precisely the roots of $z^n - 1$. Hence

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}).$$

**(ii)** Take logarithms of the factorisation in part (i):

$$\ln(z^n - 1) = \sum_{k=0}^{n-1} \ln(z - \omega^k).$$

Differentiating both sides gives

$$\frac{nz^{n-1}}{z^n - 1} = \sum_{k=0}^{n-1} \frac{1}{z - \omega^k}.$$

**(iii)** Separate the $k = 0$ term from the sum:

$$\frac{nz^{n-1}}{z^n - 1} = \frac{1}{z-1} + \sum_{k=1}^{n-1} \frac{1}{z - \omega^k}.$$

Therefore,

$$\sum_{k=1}^{n-1} \frac{1}{z - \omega^k} = \frac{nz^{n-1}}{z^n - 1} - \frac{1}{z-1}.$$

Now let $z \to 1$. Then

$$\sum_{k=1}^{n-1} \frac{1}{1 - \omega^k} = \lim_{z \to 1} \left( \frac{nz^{n-1}}{z^n - 1} - \frac{1}{z-1} \right).$$

To evaluate this limit, write $z = 1 + h$. Then the expression becomes:

$$\frac{n(1+h)^{n-1}}{(1+h)^n - 1} - \frac{1}{h}.$$

Using the expansions

$$(1+h)^{n-1} = 1 + (n-1)h + O(h^2), \qquad (1+h)^n - 1 = nh + \frac{n(n-1)}{2}h^2 + O(h^3),$$

we obtain

$$\frac{n(1+h)^{n-1}}{(1+h)^n - 1} = \frac{1}{h} + \frac{n-1}{2} + O(h).$$

Hence

$$\lim_{z \to 1} \left( \frac{nz^{n-1}}{z^n - 1} - \frac{1}{z-1} \right) = \frac{n-1}{2}.$$

Therefore,

$$\sum_{k=1}^{n-1} \frac{1}{1 - \omega^k} = \frac{n-1}{2}.$$

## Takeaways 4.34

1. **Logarithmic Differentiation:** $\frac{P'(z)}{P(z)} = \sum \frac{1}{z - \alpha_i}$. This is a standard identity.
2. **Handling Singularities:** When asking for a sum involving roots of unity where one root makes the denominator zero, you usually need to take a limit.
3. **Power of Taylor Series:** Using the binomial expansion (or Taylor series) to evaluate the $0/0$ limit as $h \to 0$ is a brilliant choice. Attempting L'Hôpital's rule here would create a massive mess of quotient and product rules. Binomial expansion is vastly superior for limits involving $(1+h)^n$.

## Problem 4.35: Squaring the Circle — The Transcendence of $\pi$

**The History:**
For over 2,000 years, mathematicians chased the holy grail of ancient geometry: "Squaring the Circle." The challenge was to construct a square with the exact same area as a given circle using only a compass and a straightedge. In 1882, Ferdinand von Lindemann finally put the ancient problem to rest. He didn't find a way to do it; he proved it was mathematically impossible. To do this, he proved that $\pi$ is **transcendental**—meaning it can never be the root of any polynomial equation with rational coefficients. A number that is called **algebraic** is a root of some polynomial with rational coefficients (e.g., $\sqrt{2}$)

**The Problem:**
Assume, for the purpose of contradiction, that $\pi$ is algebraic. It follows that $i\pi$ is also algebraic. Let $\alpha_1 = i\pi$, and let $\alpha_2, \alpha_3, \ldots, \alpha_m$ be its conjugate roots from a polynomial with integer coefficients.

**(i)** Using Euler's Identity, briefly explain why:

$$(e^{\alpha_1} + 1)(e^{\alpha_2} + 1)\ldots(e^{\alpha_m} + 1) = 0$$

**(ii)** When the brackets in part (i) are expanded, the equation can be simplified into the form:

$$k + e^{\beta_1} + e^{\beta_2} + \cdots + e^{\beta_r} = 0$$

where $k$ is a positive integer, and $\beta_1, \ldots, \beta_r$ are the non-zero sums of various combinations of the $\alpha$ roots.
Let $p$ be a large prime number. We define the Niven-Lindemann polynomial $f(x)$ and its operator $F(x)$ as:

$$f(x) = \frac{x^{p-1}(x-\beta_1)^p(x-\beta_2)^p\ldots(x-\beta_r)^p}{(p-1)!}$$

$$F(x) = f(x) + f'(x) + f''(x) + \ldots$$

By differentiating, show that $\frac{d}{dx}\left[e^{-x}F(x)\right] = -e^{-x}f(x)$.

**(iii)** By integrating the result of part (ii) from 0 to $\beta_j$, deduce that:

$$-kF(0) - \sum_{j=1}^{r} F(\beta_j) = \sum_{j=1}^{r} e^{\beta_j} \int_0^{\beta_j} e^{-x} f(x)\, dx$$

**(iv)** You are given the following two advanced algebraic theorems. **(Do NOT prove these):**

- **Fact 1:** For a sufficiently large prime $p$, the left-hand side of the equation in part (iii) evaluates to a **non-zero integer**.
- **Fact 2:** Because of the $(p-1)!$ in the denominator of $f(x)$, as $p \to \infty$, the right-hand side of the equation approaches **zero**.

Using these facts, complete the proof by contradiction to deduce that $\pi$ is transcendental.

**Hint:**

- **(i)** What is $e^{i\pi}+1$? If one of the brackets in a massive product evaluates to exactly zero, what happens to the entire product?
- **(ii)** Use the product rule on $e^{-x}F(x)$. Remember that $F'(x)$ is just $f'(x)+f''(x)+f'''(x)\ldots$, which means $F(x)-F'(x)=f(x)$. Watch your negative signs carefully!
- **(iii)** Integrate both sides of the identity you found in part (ii) with respect to $x$. When you evaluate the bounds, you will get an equation linking $F(\beta_j)$ and $F(0)$. Rearrange this to make $F(\beta_j)$ the subject. Then, sum this across all roots from $j=1$ to $r$, and substitute the fact that $e^{\beta_1}+\cdots+e^{\beta_r}=-k$ from the prompt.
- **(iv)** You have an equation where LHS = RHS. Fact 1 tells you the LHS is an integer like $1,-2,7$, etc. Fact 2 tells you the RHS is getting crushed infinitely close to 0. What is the only integer that can equal 0? Why does this break the universe?

## Solution 4.35

**(i)** By Euler's identity,

$$e^{i\pi} + 1 = 0.$$

Since $\alpha_1 = i\pi$, the first factor is $e^{\alpha_1} + 1 = 0$. Hence the entire product is 0.

**(ii)** Let $y = e^{-x}F(x)$. Then

$$\frac{dy}{dx} = -e^{-x}F(x) + e^{-x}F'(x) = -e^{-x}\big(F(x) - F'(x)\big).$$

Since $F(x) = f(x) + f'(x) + f''(x) + \cdots$ and $F'(x) = f'(x) + f''(x) + \cdots$, we have

$$F(x) - F'(x) = f(x).$$

Therefore,

$$\frac{d}{dx}\big[e^{-x}F(x)\big] = -e^{-x}f(x).$$

**(iii)** Integrating from 0 to $\beta_j$,

$$\big[e^{-x}F(x)\big]_0^{\beta_j} = -\int_0^{\beta_j} e^{-x}f(x)\,dx,$$

$$e^{-\beta_j}F(\beta_j) - F(0) = -\int_0^{\beta_j} e^{-x}f(x)\,dx,$$

$$F(\beta_j) - e^{\beta_j}F(0) = -e^{\beta_j}\int_0^{\beta_j} e^{-x}f(x)\,dx.$$

Summing for $j = 1, 2, \ldots, r$ gives

$$\sum_{j=1}^{r} F(\beta_j) - F(0)\sum_{j=1}^{r} e^{\beta_j} = -\sum_{j=1}^{r} e^{\beta_j}\int_0^{\beta_j} e^{-x}f(x)\,dx.$$

Since $\sum_{j=1}^{r} e^{\beta_j} = -k$, this becomes

$$\sum_{j=1}^{r} F(\beta_j) + kF(0) = -\sum_{j=1}^{r} e^{\beta_j}\int_0^{\beta_j} e^{-x}f(x)\,dx.$$

Multiplying by $-1$,

$$-kF(0) - \sum_{j=1}^{r} F(\beta_j) = \sum_{j=1}^{r} e^{\beta_j}\int_0^{\beta_j} e^{-x}f(x)\,dx.$$

**(iv)** By Fact 1, the left-hand side is a non-zero integer. By Fact 2, the right-hand side tends to 0 as $p \to \infty$. Hence, for a sufficiently large prime $p$, the absolute value of the right-hand side is less than 1. So the left-hand side must be a non-zero integer whose absolute value is less than 1. That is impossible, because the only integer between $-1$ and 1 is 0. This contradiction shows that our original assumption was false. Therefore, $\pi$ is transcendental.

**Takeaways 4.35**

1. **The Power of the Black Box:** In high-level mathematics, you often use theorems proved by others as "black boxes" (like Fact 1 and Fact 2) to build a larger argument. Ext 2 examiners love doing this to test your logical reasoning without making you do three pages of abstract algebra.
2. **Telescoping Derivatives:** The Niven operator $F(x) - F'(x) = f(x)$ is the ultimate trick to bypass integration by parts.
3. **The Geometry Connection:** Because $\pi$ is transcendental, it is impossible to construct a line segment of length $\pi$ using just a compass and straightedge (which can only solve algebraic equations). This single calculus proof killed a 2,000-year-old Greek geometry problem!
4. **Historical Significance:** Lindemann's proof (1882) settled one of the oldest problems in mathematics. Before this proof, mathematicians spent millennia trying to square the circle.
5. **Proof Technique:** This is a masterclass in proof by contradiction combined with asymptotic analysis. The contradiction arises from the tension between algebraic properties (integer values) and analytic properties (limits going to zero).
6. **The Power of Powers:** $\pi^n$ is transcendental for any positive integer $n$. If $\pi^n$ were algebraic, we could substitute $x^n$ into its polynomial and accidentally prove $\pi$ is algebraic, which is impossible!
7. $e^n$ **is transcendental** for any positive integer $n$. This can be shown using a similar technique to the one used for $\pi$.

**Problem 4.36: Trigonometric Polynomials and Vieta's**

Let $n$ be an odd positive integer and $\theta$ be a real number such that $\frac{\pi}{\theta}$ is irrational. Define $a_k = \tan\left(\theta + \frac{k\pi}{n}\right)$ for $k = 1, 2, \ldots, n$.
**(i)** Show that the values $a_k$ are the roots of the equation $\tan(nx) = \tan(n\theta)$.
**(ii)** By considering the expansion of $\tan(nx)$ in terms of $\tan x$, show that the values $a_k$ are the roots of a polynomial equation of degree $n$ in the form:

$$t^n - \binom{n}{n-2}t^{n-2} + \cdots + (-1)^{(n-1)/2} = \tan(n\theta)\left[\binom{n}{1}t^{n-1} - \binom{n}{3}t^{n-3} + \ldots\right]$$

**(iii)** Use the relationship between the roots and coefficients (Vieta's formulas) to determine the sum $\sum_{k=1}^{n} a_k$ and the product $\prod_{k=1}^{n} a_k$ in terms of $\tan(n\theta)$.
**(iv)** Hence, prove that $\frac{\sum a_k}{\prod a_k}$ is a constant, and find its value in terms of $n$.

**Hint:**

- **Part (i):** Start with the definition $a_k = \tan(\theta + \frac{k\pi}{n})$. If you multiply the argument by $n$, what happens to the $k\pi$ term?
- **Part (ii):** Recall the standard expansion for $\tan(nx)$ using the binomial expansion of $(\cos x + i \sin x)^n$. Equate the real and imaginary parts.
- **Part (iii):** Remember that for a polynomial $P(t) = c_n t^n + c_{n-1}t^{n-1} + \cdots + c_0$, the sum of roots is $-c_{n-1}/c_n$ and the product of roots is $(-1)^n c_0/c_n$.
- **Part (iv):** You will need to carefully track the coefficients of the highest and lowest powers of $t$ in the equation derived in part (ii).

## Solution 4.36: Brief Solution

1. **Roots:** Since $\tan\left(n\left(\theta + \frac{k\pi}{n}\right)\right) = \tan(n\theta + k\pi) = \tan(n\theta)$, the values $a_k$ satisfy the equation $\tan(nx) = \tan(n\theta)$.
2. **Polynomial:** Using De Moivre's Theorem, $\tan(nx) = \frac{\binom{n}{1}t - \binom{n}{3}t^3 + \cdots}{\binom{n}{0} - \binom{n}{2}t^2 + \cdots}$. Setting this to $\tan(n\theta) = T$, we rearrange to form a polynomial in $t$.
3. **Vieta's:**
   - From part (ii), the polynomial is $c_n t^n + c_{n-1}t^{n-1} + \cdots + c_0 = 0$.
   - Comparing coefficients:
   - $c_n = 1$
   - $c_{n-1} = -\binom{n}{1}\tan(n\theta) = -n\tan(n\theta)$
   - $c_0 = (-1)^{(n-1)/2}\tan(n\theta)$ (since $n$ is odd, the constant term comes from the LHS bracket).
   - Therefore, $\sum a_k = -\frac{c_{n-1}}{c_n} = n\tan(n\theta)$.
   - The product $\prod a_k = \frac{(-1)^n c_0}{c_n} = (-1)^{(n-1)/2}\tan(n\theta)$ (since $n$ is odd, $(-1)^n = -1$, and multiplying it against the LHS constant sign gives this result).
4. **Result:** The ratio simplifies to a value dependent only on $n$. (For $n = 3$, the ratio is $-3$; the general result is $-n$).

## Takeaways 4.36

- **Polynomial Bridges:** Whenever you see trigonometric expressions set equal to roots (like $a_k = \tan(\dots)$), immediately think of forming a polynomial equation.
- **Vieta's is King:** In HSC Extension 2, if you are asked to find the sum or product of a set of complex trigonometric values, it is almost certainly a Vieta's Formula problem in disguise.
- **Expansion Mastery:** Knowing the binomial expansion of $(\cos x + i\sin x)^n$ is the fastest way to derive the identity for $\tan(nx)$. Don't try to derive it from scratch using $\tan(A + B)$ formulas; it takes too long.

## Problem 4.37: Wallis Integrals and the Gaussian Integral

Let $W_n = \int_0^{\pi/2} \sin^n x\,dx$ denote the Wallis Integrals. It is given that:

$$\lim_{n\to\infty} \sqrt{n}W_n = \sqrt{\pi/2}. \text{ (Do NOT prove this)}$$

(i) Prove that for all $t \geq 0$:

$$1 - t \leq e^{-t} \leq \frac{1}{1+t}$$

(ii) By substituting $t = \frac{x^2}{n}$, show that for all $n \in \mathbb{Z}^+$:

$$\int_0^{\sqrt{n}} \left(1 - \frac{x^2}{n}\right)^n dx \leq \int_0^{\sqrt{n}} e^{-x^2} dx \leq \int_0^{\infty} e^{-x^2} dx \leq \int_0^{\infty} \left(1 + \frac{x^2}{n}\right)^{-n} dx$$

(iii) Use the substitutions $x = \sqrt{n}\sin\theta$ and $x = \sqrt{n}\tan\theta$ to show that

$$\int_0^{\sqrt{n}} \left(1 - \frac{x^2}{n}\right)^n dx = \sqrt{n}W_{2n+1}, \text{ and } \int_0^{\infty} \left(1 + \frac{x^2}{n}\right)^{-n} dx = \sqrt{n}W_{2n-2}.$$

(iv) A continuous random variable $X$ follows the Normal Distribution $X \sim \mathcal{N}(0, \sigma^2)$ if its probability density function (PDF) is given by

$$f(x) = k \cdot e^{-\frac{x^2}{2\sigma^2}} \text{ for } x \in (-\infty, \infty)$$

.

Show that for $f(x)$ to be a valid PDF, the constant $k$ must be:

$$k = \frac{1}{\sigma\sqrt{2\pi}}$$

**Hint:**

- Use the inequality $e^t \geq 1 + t$ for $t \geq 0$. For the left side, consider the function $f(t) = e^{-t} - (1 - t)$.
- Recall that for $n > 0$, if $f(x) \leq g(x)$, then $f(x)^n \leq g(x)^n$. For the tangent substitution, remember that $1 + \tan^2\theta = \sec^2\theta$ and the limits for $x \in [0, \infty)$ map to $\theta \in [0, \pi/2)$.
- **Part (iii):** Use $x = \sqrt{n}\sin\theta$ and $x = \sqrt{n}\tan\theta$.
- **Part (iv):** Use the fact that $e^{-ax^2}$ is an even function. Apply the substitution $u = \frac{x}{\sigma\sqrt{2}}$ to transform the integral into the standard Gaussian form.

## Solution 4.37

- **(i)** Recall the Taylor series expansion: $e^t = 1+t+\frac{t^2}{2!}+\cdots \geq 1+t \implies e^{-t} \leq \frac{1}{1+t}$. For $1-t \leq e^{-t}$, let $f(t) = e^{-t} - 1 + t$. $f'(t) = 1 - e^{-t} \geq 0$ for $t \geq 0$. Since $f(0) = 0$, $f(t) \geq 0$. You can also consider the derevative of $e^{-t} + t - 1$ directly.
- **(ii)** From (i), $(1-x^2/n) \leq e^{-x^2/n} \leq (1+x^2/n)^{-1}$. Raising to power $n$ gives $(1-x^2/n)^n \leq e^{-x^2} \leq (1+x^2/n)^{-n}$. Integrating over the respective bounds yields the result.
- **(iii)** $I_{left} = \int_0^{\pi/2}(1-\sin^2\theta)^n\sqrt{n}\cos\theta\,d\theta = \sqrt{n}\int_0^{\pi/2}\cos^{2n+1}\theta\,d\theta = \sqrt{n}W_{2n+1}$ (since $\int_0^{\pi/2}\cos^m\theta\,d\theta = \int_0^{\pi/2}\sin^m\theta\,d\theta = W_m$ by symmetry). $I_{right} = \int_0^{\pi/2}(\sec^2\theta)^{-n}\sqrt{n}\sec^2\theta\,d\theta = \sqrt{n}W_{2n-2}$.
- **(iv)** $\int_{-\infty}^{\infty} ke^{-\frac{x^2}{2\sigma^2}}dx = 2k\int_0^{\infty} e^{-\frac{x^2}{2\sigma^2}}dx = 1$.
  Let $u = \frac{x}{\sigma\sqrt{2}} \implies dx = \sigma\sqrt{2}du$.
  $2k\sigma\sqrt{2}\int_0^{\infty} e^{-u^2}du = 2k\sigma\sqrt{2}\left(\frac{\sqrt{\pi}}{2}\right) = 1$.
  Solving for $k$ yields $k = \frac{1}{\sigma\sqrt{2\pi}}$.

## Takeaways 4.37

1. The Bernoulli Link: The expression $(1 + z/n)^n$ is the classic limit definition of $e^z$. This problem shows how that convergence behaves under an integral sign. It is kind of advance but useful to know.
2. **Normalization:** Normalization: This problem demonstrates why the "messy" constant $\frac{1}{\sqrt{2\pi}}$ exists in statistics; it is a mathematical necessity to ensure the total probability equals 1.
3. **Symmetry:** The use of even function properties is a vital shortcut in Extension 2 integration problems.
4. **Interdisciplinary Math:** This problem bridges the gap between pure integration techniques and the foundations of statistics.

## Problem 4.38: Complex Numbers and Polynomial Geometry

Let $n$ be a positive integer such that $n \geq 2$. Consider the complex equation $z^{2n} - 1 = 0$, whose roots are represented by the vertices $P_0, P_1, P_2, \ldots, P_{2n-1}$ of a regular $2n$-gon inscribed in the unit circle in the Argand diagram. Let $P_0$ correspond to the complex number 1.

**(i)** Let $\omega = e^{i\frac{\pi}{n}}$. Show that the roots of the equation are given by $z = \omega^k$ for $k = 0, 1, \ldots, 2n-1$.

**(ii)** Show that for any $z \neq 1$:

$$\sum_{j=0}^{2n-1} z^j = \prod_{k=1}^{2n-1} (z - \omega^k)$$

**(iii)** By considering the distance between the point $P_0$ and any other vertex $P_k$, show that the length of the chord $P_0P_k$ is given by:

$$|P_0P_k| = 2\sin\left(\frac{k\pi}{2n}\right)$$

**(iv)** Prove that the product of the lengths of all possible chords drawn from a single vertex to all other vertices in a regular $2n$-gon that inscribed in a unit circle is exactly $2n$. That is, show:

$$\prod_{k=1}^{2n-1} |P_0P_k| = 2n$$

**Hint:**

- **Part (i):** Recall the polar form of 1 is $e^{i2k\pi}$. Apply the $2n$-th root to find $z$.
- **Part (ii):** Use the fact that $z^{2n} - 1 = (z-1)(z-\omega)(z-\omega^2)\ldots(z-\omega^{2n-1})$ and equate this to the geometric series formula for $\frac{z^{2n}-1}{z-1}$.
- **Part (iii):** Use the formula $|z_1 - z_2| = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$ or $|1 - e^{i\theta}|^2 = (1-\cos\theta)^2 + \sin^2\theta$.
- **Part (iv):** Evaluate the identity in Part (ii) at $z = 1$ and use the property that the modulus of a product is the product of the moduli.

## Solution 4.38

**(i)** $z^{2n} = 1 = e^{i2k\pi} \implies z = e^{i\frac{2k\pi}{2n}} = e^{i\frac{k\pi}{n}}$. Let $\omega = e^{i\frac{\pi}{n}}$, then roots are $\omega^k$ for $k = 0, 1, \ldots, 2n-1$.
**(ii)** The polynomial $P(z) = z^{2n} - 1$ has roots $1, \omega, \omega^2, \ldots, \omega^{2n-1}$. Thus:

$$z^{2n} - 1 = (z-1)(z-\omega)(z-\omega^2)\ldots(z-\omega^{2n-1})$$

Dividing by $(z-1)$ for $z \neq 1$:

$$\frac{z^{2n}-1}{z-1} = \prod_{k=1}^{2n-1}(z-\omega^k)$$

Since $\frac{z^{2n}-1}{z-1} = 1 + z + z^2 + \cdots + z^{2n-1}$, the identity is proven.

**(iii)** Chord $|P_0P_k| = |1 - e^{i\frac{k\pi}{n}}|$. Using $|1 - \cos\theta - i\sin\theta| = \sqrt{(1-\cos\theta)^2 + \sin^2\theta} = \sqrt{2 - 2\cos\theta}$.
Applying $1 - \cos\theta = 2\sin^2(\frac{\theta}{2})$:

$$|P_0P_k| = \sqrt{4\sin^2\left(\frac{k\pi}{2n}\right)} = 2\sin\left(\frac{k\pi}{2n}\right) \text{ (as sin is positive in this range).}$$

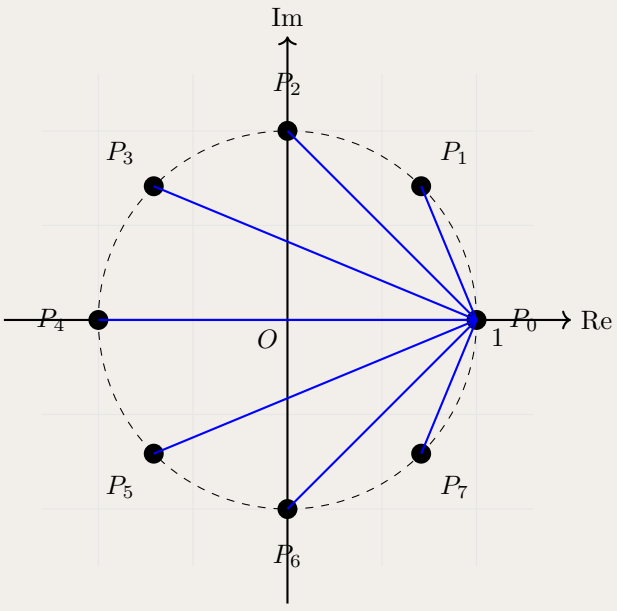
**(iv)** Let $z = 1$ in the identity from (ii):

$$1 + 1 + \cdots + 1 \ (2n \text{ terms}) = \prod_{k=1}^{2n-1}(1-\omega^k), \text{ i.e. } 2n = \prod_{k=1}^{2n-1}(1-\omega^k)$$

Taking the modulus of both sides:

$$|2n| = \prod_{k=1}^{2n-1}|1-\omega^k|$$

Since $|1-\omega^k|$ is the length of chord $|P_0P_k|$, we have $\prod_{k=1}^{2n-1}|P_0P_k| = 2n$.



**Takeaways 4.38**

1. Geometrically, $|1 - e^{i\frac{k\pi}{n}}|$ represents the distance between the vertex $P_0$ (at $1 + 0i$) and every other vertex $P_k$ of a regular $2n$-gon inscribed in the unit circle.
2. **Geometric Product:** The product of all chord lengths from one vertex in a regular $m$-gon is exactly $m$.
3. **Trigonometric Identity:** This problem proves $\prod_{k=1}^{2n-1} \sin\left(\frac{k\pi}{2n}\right) = \frac{2n}{2^{2n-1}}$.
4. **Complex-Geometric Connection:** Polynomial factorization over roots of unity directly yields geometric properties of regular polygons.
5. **Modulus of Products:** The identity $|z_1 z_2| = |z_1||z_2|$ is essential for connecting algebraic and geometric interpretations.
6. Try proving (iv) using induction as an alternative method (?) or without using complex numbers (?)

## Problem 4.39: Niven's Contradiction: The Irrationality of $\pi^2$

In this problem, we will prove that $\pi^2$ is irrational using a proof by contradiction inspired by Niven's approach. You may already know that $\pi$ itself is irrational yet the proof is somewhat cringe to high school students.

Proving the irrationality of $\pi^2$ makes the statement stronger, since if $\pi^2$ were irrational, then $\pi$ would also be irrational (as the square root of a irrational number) (do NOT prove this). Thus, by proving that $\pi^2$ is irrational, we also confirm the irrationality of $\pi$.

This problem will walk you through the steps to establish this result.

To prove that $\pi^2$ is irrational, we assume by way of contradiction that $\pi^2$ is rational. That is, we can write $\pi^2 = \frac{a}{b}$ where $a$ and $b$ are positive integers.

Let $n$ be a positive integer and define the polynomial $f(x)$ as:

$$f(x) = \frac{x^n(1-x)^n}{n!}$$

(i) Show $f(x)$ and its derivatives $f^{(k)}(x)$ take integer values at $x = 0$ and $x = 1$.

(ii) Define the function $G(x)$ by:

$$G(x) = b^n \sum_{k=0}^{n} (-1)^k \pi^{2n-2k} f^{(2k)}(x)$$

Using the assumption $\pi^2 = \frac{a}{b}$, show that $G(0)$ and $G(1)$ are integers.

(iii) Consider the function $H(x) = G'(x)\sin(\pi x) - \pi G(x)\cos(\pi x)$. Differentiate $H(x)$ with respect to $x$ and verify that:

$$H'(x) = \pi^2 a^n f(x)\sin(\pi x)$$

(iv) Consider the integral:

$$I_n = \int_0^1 \pi a^n f(x)\sin(\pi x)\,dx$$

Using the result from part (iii), show that $I_n = G(1) + G(0)$. Hence, explain why $I_n$ must be an integer.

Finally, by establishing an upper bound for $I_n$, derive a contradiction for sufficiently large $n$. Conclude that $\pi^2$ is irrational.

**Hint:**

- **Part (i):** Expand $(1-x)^n$ using the binomial theorem. Recall that the $k$-th derivative of $\frac{x^m}{n!}$ evaluated at $x=0$ is 0 if $m \neq k$, and is an integer if $m=k$. Use the symmetry $f(x)=f(1-x)$ for the $x=1$ case.
- **Part (ii):** Substitute $\pi^{2n-2k}=(\frac{a}{b})^{n-k}$. Combine this with $b^n$ outside the sum to clear the fractions. Observe that $\pi^{2m}=(a/b)^m$. Rewrite the sum to show the coefficients are integers.
- **Part (iii):** Use the product rule on $H(x)$. Group terms by $\sin(\pi x)$ and $\cos(\pi x)$. The cos terms should cancel. You may assume without proof the algebraic identity: $G''(x)+\pi^2 G(x)=\pi^{2n+2}b^n f(x)$. Note also that $\pi^{2n+2}b^n=\pi^2 a^n$.
- **Part (iv):** Integrate $H'(x)$ from 0 to 1. Note that $H(1)=\pi G(1)$ and $H(0)=-\pi G(0)$. Don't forget that $\lim_{n\to\infty}\frac{C^n}{n!}=0$.

## Solution 4.39

**(i) Integer Derivatives**

$f(x) = \frac{1}{n!}\sum_{j=0}^{n}\binom{n}{j}(-1)^j x^{n+j}$. For any $k$, $f^{(k)}(0)$ is determined by the term with power $x^k$. If $k < n$, the coefficient is 0. If $k \geq n$, the derivative of $\frac{x^k}{n!}$ is an integer multiple of $\frac{k!}{n!}$, which is an integer. Since $f(x) = f(1-x)$, by chain rule $f^{(k)}(1) = (-1)^k f^{(k)}(0)$, which is also an integer.

**(ii) G(x) Integers**

Substitute $\pi^2 = a/b$:

$$G(x) = \sum_{k=0}^{n}(-1)^k b^n \left(\frac{a}{b}\right)^{n-k} f^{(2k)}(x) = \sum_{k=0}^{n}(-1)^k a^{n-k} b^k f^{(2k)}(x)$$

Since $a, b, f^{(2k)}(0)$ and $f^{(2k)}(1)$ are all integers, $G(0)$ and $G(1)$ are integers.

**(iii) The Derivative**

$H'(x) = G''(x)\sin(\pi x) + \pi G'(x)\cos(\pi x) - \pi G'(x)\cos(\pi x) + \pi^2 G(x)\sin(\pi x)$

$$H'(x) = \sin(\pi x)[G''(x) + \pi^2 G(x)]$$

To see why this massive sum collapses, we explicitly evaluate $G''(x) + \pi^2 G(x)$:

$$\begin{aligned} G''(x) + \pi^2 G(x) &= b^n \sum_{k=0}^{n}(-1)^k \pi^{2n-2k} f^{(2k+2)}(x) + \pi^2 b^n \sum_{k=0}^{n}(-1)^k \pi^{2n-2k} f^{(2k)}(x) \\ &= b^n \sum_{k=0}^{n}(-1)^k \pi^{2n-2k} f^{(2k+2)}(x) + b^n \sum_{k=0}^{n}(-1)^k \pi^{2n-2k+2} f^{(2k)}(x) \end{aligned}$$

By expanding these sums, we see that they perfectly telescope. Every term in the first sum cancels with the subsequent term in the second sum. The only term that survives is the $k = 0$ term from the second sum (since $f^{(2n+2)}(x) = 0$, the upper boundary term vanishes). This leaves exactly:

$$G''(x) + \pi^2 G(x) = b^n (-1)^0 \pi^{2n+2} f^{(0)}(x) = \pi^{2n+2} b^n f(x)$$

Substituting this collapsed form back into $H'(x)$:

$$H'(x) = \sin(\pi x)[\pi^{2n+2} b^n f(x)]$$

Since $\pi^2 = a/b \implies b\pi^2 = a$, we have $\pi^{2n+2} b^n = \pi^2(\pi^2)^n b^n = \pi^2 (a/b)^n b^n = \pi^2 a^n$.

$$H'(x) = \pi^2 a^n f(x)\sin(\pi x)$$

**(iv) The Integral and Contradiction**

From (iii), $\int_0^1 \pi^2 a^n f(x)\sin(\pi x)dx = [H(x)]_0^1$. Evaluate RHS: $H(1) = G'(1)\sin\pi - \pi G(1)\cos\pi = \pi G(1)$. $H(0) = G'(0)\sin 0 - \pi G(0)\cos 0 = -\pi G(0)$. So, $\int_0^1 \pi^2 a^n f(x)\sin(\pi x)dx = \pi(G(1) + G(0))$. Divide both sides by $\pi$:

$$\int_0^1 \pi a^n f(x)\sin(\pi x)dx = G(1) + G(0)$$

The LHS is $I_n$. The RHS is the sum of integers (from part ii). Thus $I_n$ is an integer.

**Bounds:** On $(0,1)$, $0 < f(x) \leq \frac{1}{n!}(1/4)^n < \frac{1}{n!}$ and $0 < \sin(\pi x) \leq 1$.

$$0 < I_n < \frac{\pi a^n}{n!}$$

Since $\lim_{n\to\infty} \frac{\pi a^n}{n!} = 0$, for sufficiently large $n$, we have $0 < I_n < 1$. No integer exists strictly between 0 and 1. Contradiction. $\therefore \pi^2$ is irrational.

**Takeaways 4.39**

1. **Stronger Result**: Proving $\pi^2$ is irrational automatically proves $\pi$ is irrational, but the reverse is not true.
2. **Proof by Contradiction:** This classic technique assumes the opposite of what we want to prove, then derives an impossible conclusion. The contradiction forces our original assumption to be false.
3. **Auxiliary Functions:** The sophisticated choice of $f(x)$, $G(x)$, and $H(x)$ is characteristic of advanced proofs in analysis. Each function serves a specific purpose in the overall argument.
4. **Integer Forcing:** By carefully constructing $G(x)$ to have integer values at endpoints, we force the integral $I_n$ to be an integer, setting up the contradiction.
5. **Factorial Growth:** The key to the contradiction lies in the factorial in the denominator: $n!$ grows much faster than any exponential $a^n$, making the integral arbitrarily small.
6. **Historical Significance:** This problem represents one of the most elegant applications of calculus to number theory, demonstrating the deep connections between analysis and arithmetic.
7. **Transcendence of $\pi$:** $\pi$ is not only irrational but also transcendental, meaning it is not a root of any non-zero polynomial equation with rational coefficients. This was proven later by Ferdinand von Lindemann in 1882.

**Remark 0.9: Original HSC Mathematics Extension 2 (2003), Question 8**

If you enjoyed this style of proof, you are encouraged to challenge the **original** HSC Mathematics Extension 2 (2003) Question 8 (part (b))!

(b) Suppose that $\pi$ could be written in the form $\frac{p}{q}$, where $p$ and $q$ are positive integers. Define the family of integrals $I_n$ for $n = 0, 1, 2, \ldots$ by

$$I_n = \frac{q^{2n}}{n!} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\pi^2}{4} - x^2\right)^n \cos x \, dx.$$

You are given that $I_0 = 2$ and $I_1 = 4q^2$. (Do NOT prove this.)

(i) Use integration by parts twice to show that, for $n \geq 2$,

$$I_n = \frac{2q^{2n}}{(n-1)!} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\pi^2}{4} - x^2\right)^{n-1} \cos x \, dx - \frac{4q^{2n}}{(n-2)!} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \left(\frac{\pi^2}{4} - x^2\right)^{n-2} \cos x \, dx.$$

(ii) By writing $x^2$ as $\frac{\pi^2}{4} - \left(\frac{\pi^2}{4} - x^2\right)$ where appropriate, deduce that

$$I_n = (4n-2)q^2 I_{n-1} - p^2 q^2 I_{n-2}, \quad \text{for } n \geq 2.$$

(iii) Explain briefly why $I_n$ is an integer for $n = 0, 1, 2, \ldots$

(iv) Prove that

$$0 < I_n < \frac{p}{q} \left(\frac{p}{2}\right)^{2n} \frac{1}{n!} \quad \text{for } n = 0, 1, 2, \ldots$$

(v) Given that $\frac{p}{q} \left(\frac{p}{2}\right)^{2n} \frac{1}{n!} < 1$, if $n$ is sufficiently large, deduce that $\pi$ is irrational.

**Remark 0.10: The Original Proof by Ivan Niven (1947)**

**A SIMPLE PROOF THAT $\pi$ IS IRRATIONAL**

Let $\pi = a/b$, the quotient of positive integers. We define the polynomials

$$f(x) = \frac{x^n(a - bx)^n}{n!},$$

$$F(x) = f(x) - f^{(2)}(x) + f^{(4)}(x) - \cdots + (-1)^n f^{(2n)}(x),$$

the positive integer $n$ being specified later. Since $n!f(x)$ has integral coefficients and terms in $x$ of degree not less than $n$, $f(x)$ and its derivatives $f^{(i)}(x)$ have integral values for $x = 0$; also for $x = \pi = a/b$, since $f(x) = f(a/b - x)$. By elementary calculus we have

$$\frac{d}{dx}\{F'(x)\sin x - F(x)\cos x\} = F''(x)\sin x + F(x)\sin x = f(x)\sin x$$

and

$$\int_0^\pi f(x)\sin x\,dx = \Big[F'(x)\sin x - F(x)\cos x\Big]_0^\pi = F(\pi) + F(0). \qquad (*)$$

Now $F(\pi) + F(0)$ is an integer, since $f^{(i)}(\pi)$ and $f^{(i)}(0)$ are integers. But for $0 < x < \pi$,

$$0 < f(x)\sin x < \frac{\pi^n a^n}{n!},$$

so that the integral in (*) is positive, but arbitrarily small for $n$ sufficiently large. Thus (*) is false, and so is our assumption that $\pi$ is rational.

*This elegant proof was published by Ivan Niven in the Bulletin of the American Mathematical Society in 1947. It demonstrates the power of combining calculus with number theory to establish fundamental results about irrational numbers.*

**Problem 4.40:** ramanujan summation & filters

**(a)** the indian mathematician srinivasa ramanujan discovered the series for $1/\pi$:

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801}\sum_{n=0}^{\infty} u_n(1103 + 26390n), \text{ where } u_n = \frac{(4n)!}{(n!)^4(396)^{4n}}. \text{ Do NOT prove this.}$$

(i) Show that for $n \geq 0$, the ratio of consecutive terms in the sequence $u_n$ is given by:

$$\frac{u_{n+1}}{u_n} = \frac{(4n+1)(4n+3)(2n+1)}{K(n+1)^3}, \text{ where } K = \frac{396^4}{8}.$$

(ii) Hence, prove that for all $n \geq 1$:

$$\frac{u_{n+1}}{u_n} < \frac{1}{9 \times 10^9}$$

(iii) Let $S_k$ be the sum of the first $k$ terms of the series. Using the result from part (ii), explain why the first term alone (i.e., $n = 0$) provides an approximation for $\pi$ that is accurate to at least 6 decimal places.

**(b)** ramanujan discovered distinct patterns in the partition function, $p(n)$; which counts the number of ways an integer $n$ can be written as a sum. for example, he proved that if $n$ is of the form $5k + 4$ (like 4, 9, 14...), then $p(n)$ is always divisible by 5. to isolate these specific terms, mathematicians often use filters.

Let $\omega = e^{i\frac{2\pi}{5}}$ be a complex fifth root of unity.

(i) Prove the identity:

$$\sum_{r=0}^{4} \omega^{kr} = \begin{cases} 5 & \text{if } k \text{ is a multiple of } 5 \\ 0 & \text{if } k \text{ is not a multiple of } 5 \end{cases}$$

(ii) Consider the polynomial $P(z) = \sum_{n=0}^{N} a_n z^n$. Using part (i), show that:

$$\sum_{r=0}^{4} \omega^{-4r} P(\omega^r z) = 5\sum_{k} a_{5k+4} z^{5k+4}$$

*Here, the symbol $\sum_k$ means we are summing over all integers $k$ so that $5k+4$ is an integer in the range from 0 to $N$.*

(iii) If $a_n$ represents the partition numbers $p(n)$, and we assume Ramanujan's congruence $p(5k + 4) \equiv 0 \pmod 5$ holds, determine the value of:

$$\sum_{r=0}^{4} \omega^{-4r} P(\omega^r)$$

*Here, the symbol $\equiv$ means "is congruent to" (acting like an equals sign), and* (mod 5) *means we are looking at the remainder when divided by 5; together,* $\equiv 0 \pmod 5$ *simply means "is divisible by 5."*

**Hint:**

- **(a)(i)** When substituting $n+1$, remember that $(4(n+1))! = (4n+4)!$, which expands to $(4n+4)(4n+3)(4n+2)(4n+1)(4n)!$. Write out the ratio carefully and cancel common factorial terms.
- **(a)(ii)** The expression for the ratio is strictly decreasing for $n \geq 1$. Evaluate the ratio at its maximum possible value (when $n = 1$) and compare it to the given bound.
- **(a)(iii)** If each subsequent term is less than $10^{-9}$ times the previous term, how fast is the error shrinking? Think about geometric progressions and decimal place accuracy.
- **(b)(i)** Recognize this as a finite geometric series with a common ratio of $\omega^k$. Apply the geometric sum formula for the case where $\omega^k \neq 1$.
- **(b)(ii)** Expand $P(\omega^r z)$ into its sum form. Swap the order of summation so you evaluate the sum over $r$ first. Notice that $\omega^{-4r}\omega^{rn} = \omega^{r(n-4)}$. Apply the result from (b)(i).
- **(b)(iii)** Substitute $z = 1$ into your result from (b)(ii). Relate $a_n$ to the partition function $p(n)$, and use the fact that $p(5k+4)$ is a multiple of 5.

## Solution 4.40

**(a)(i)** Starting from the definition of $u_n$,

$$\frac{u_{n+1}}{u_n} = \frac{(4n+4)!}{((n+1)!)^4 396^{4n+4}} \cdot \frac{(n!)^4 396^{4n}}{(4n)!} = \frac{(4n+4)(4n+3)(4n+2)(4n+1)}{(n+1)^4 396^4}.$$

Now $4n+4 = 4(n+1)$ and $4n+2 = 2(2n+1)$, so

$$\frac{u_{n+1}}{u_n} = \frac{8(n+1)(4n+3)(2n+1)(4n+1)}{(n+1)^4 396^4} = \frac{(4n+1)(4n+3)(2n+1)}{K(n+1)^3}, \qquad K = \frac{396^4}{8}.$$

**(a)(ii)** The ratio decreases for $n \geq 1$, so its largest value occurs at $n = 1$. Then

$$\frac{u_2}{u_1} = \frac{5 \cdot 7 \cdot 3}{K \cdot 2^3} = \frac{105}{396^4}.$$

Since $396^4 \approx 2.46 \times 10^{10}$, this is about $4.3 \times 10^{-9}$, so certainly

$$\frac{u_{n+1}}{u_n} < \frac{1}{9 \times 10^9} \qquad (n \geq 1).$$

**(a)(iii)** By part (ii), each term is less than $10^{-9}$ times the previous one, so the tail is bounded by an infinite geometric series with $a = u_1$ and $r = 10^{-9}$. This series converges to a value far smaller than the $6^{\text{th}}$ decimal place, making the tail completely negligible. In particular, the $n = 1$ term affects only about the 9th or 10th decimal place. Hence $S_0$ alone already gives at least 6 correct decimal places for $1/\pi$.

**(b)(i)** Let $S = \sum_{r=0}^{4} (\omega^k)^r$. If $5 \mid k$, then $\omega^k = e^{i2\pi m} = 1$, so $S = 5$. Otherwise $\omega^k \neq 1$, and

$$S = \frac{1 - (\omega^k)^5}{1 - \omega^k} = \frac{1 - (\omega^5)^k}{1 - \omega^k} = \frac{1 - 1}{1 - \omega^k} = 0.$$

**(b)(ii)** Since $P(z) = \sum_{n=0}^{N} a_n z^n$,

$$\sum_{r=0}^{4} \omega^{-4r} P(\omega^r z) = \sum_{n=0}^{N} a_n z^n \sum_{r=0}^{4} \omega^{r(n-4)}.$$

By part (i), the inner sum is 5 when $n = 5k+4$ and 0 otherwise, so

$$5 \sum_k a_{5k+4} z^{5k+4}.$$

**(b)(iii)** Setting $z = 1$ in part (ii) gives

$$\sum_{r=0}^{4} \omega^{-4r} P(\omega^r) = 5 \sum_k a_{5k+4}.$$

If $a_n = p(n)$, this becomes $5 \sum_k p(5k+4)$. Since $p(5k+4) = 5m$,

$$5 \sum_k p(5k+4) = 25 \sum_k m,$$

so the value is a multiple of 25.

**Takeaways 4.40**

1. **Hyper-Fast Convergence:** Ramanujan's series for $1/\pi$ demonstrates that analyzing the ratio of consecutive terms is key to proving how fast an infinite series converges to its target value.

2. **Roots of Unity Filter:** This is an elegant and immensely powerful algebraic tool. By summing over roots of unity, you cause unwanted terms to perfectly cancel each other out (summing to 0), cleanly isolating periodic sequences within a polynomial or power series.

3. **Connecting Universes:** Factorials, complex numbers, modular arithmetic, and $\pi$ shouldn't logically sit in the same problem. This is a masterclass in how seemingly isolated topics weave together to solve otherwise impossible problems.

4. **Roots of unity and filters (standard identity):** For a primitive $n$th root of unity $\omega = e^{2\pi i/n}$, the sum

$$\sum_{r=0}^{n-1} \omega^{kr} = \begin{cases} n & \text{if } n \mid k, \\ 0 & \text{otherwise,} \end{cases}$$

   acts like an on/off switch: it is $n$ exactly when the exponent index $k$ is a multiple of $n$, and 0 otherwise.

5. **Mini-example (polynomial filter):** If $P(z) = \sum_{m=0}^{N} a_m z^m$ and $\omega$ is a primitive $n$th root of unity, then

$$\sum_{r=0}^{n-1} P(\omega^r z) = \sum_{m=0}^{N} a_m z^m \sum_{r=0}^{n-1} \omega^{rm} = n \sum_{k} a_{kn} z^{kn},$$

   because the inner sum is $n$ only when $m$ is a multiple of $n$ and 0 otherwise. In other words, this sum "filters out" and keeps only the coefficients whose indices are multiples of $n$—exactly the mechanism used in part (b) to isolate the $5k+4$ terms.

## Problem 4.41: Uniform Convergence and Integration

This problem explores the relationship between sequences of functions, their limits, and integration, focusing on the critical concept of **Uniform Convergence**.

### Introduction to Uniform Convergence

A sequence of functions $\{f_n\}$ defined on a set $B$ is said to **converge uniformly** to a function $f$ if for every $\epsilon > 0$, there exists an integer $N$ such that $|f(x) - f_n(x)| < \epsilon$ for all $n \geq N$ and for all $x \in B$.
The key feature of uniform convergence is that the integer $N$ depends only on the tolerance $\epsilon$ and must work *simultaneously* for every $x$ in the domain $B$. This is a stronger requirement than merely having each individual $f_n(x)$ approach $f(x)$ for each fixed $x$, where different values of $x$ might require different values of $N$. The word "uniform" emphasizes that a single $N$ works uniformly across the entire domain.
Two fundamental theorems regarding uniform convergence are:

- **Theorem 1:** If $\{f_n\}$ converges uniformly to $f$ on $B$, then the limit and integral can be swapped:
$$\int_B f(x)\,dx = \lim_{n\to\infty} \int_B f_n(x)\,dx$$
- **Theorem 2:** If there exists a positive convergent series $\sum_{n=1}^{\infty} a_n$ such that $|f_n(x) - f_{n-1}(x)| \leq a_n$ for all $n$ and all $x \in B$, then $\{f_n\}$ converges uniformly to some function $f$.

(Do NOT prove these theorems; they are given as tools for solving the problem.)

### Problem Statement

Consider the interval $B = [0, 1]$.
**(i)** Let $f_n(x) = n^2 x(1-x)^n$ for $n = 1, 2, 3, \ldots$
Show that for any fixed $x \in [0, 1]$, the pointwise limit $f(x) = \lim_{n\to\infty} f_n(x)$ is 0.
**(ii)** For the sequence in part **(i)**, show that:

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx \neq \int_0^1 \left(\lim_{n\to\infty} f_n(x)\right) dx$$

Briefly explain, with reference to the definition of uniform convergence, why this result does not contradict Theorem 1.
**(iii)** Now consider a different sequence $\{g_n\}$ on $[0, 1]$ defined by the partial sums:

$$g_n(x) = \sum_{k=1}^{n} \frac{x^k}{k^2}$$

By considering the difference $|g_n(x) - g_{n-1}(x)|$, use Theorem 2 and the fact that $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges to prove that $\{g_n\}$ converges uniformly to some function $g(x)$ on $[0, 1]$.
**(iv)** Deduce that:

$$\int_0^1 g(x)\,dx = \sum_{k=1}^{\infty} \frac{1}{k^2(k+1)}$$

**Hint:**

- **For (i):** For $0 < x < 1$, consider the growth rate of $n^2$ versus the decay rate of $(1-x)^n$. Recall that for $r > 1$, $\lim_{n\to\infty} \frac{n^k}{r^n} = 0$.
- **For (ii):** Use the substitution $u = 1 - x$ to evaluate the integral $\int_0^1 x(1-x)^n \, dx$. Compare the resulting limit as $n \to \infty$ with the integral of the pointwise limit found in part **(i)**.
- **For (iii):** Apply Theorem 2 directly by finding a convergent series $\sum a_n$ that bounds the difference between successive terms for all $x \in [0, 1]$.
- **For (iv):** Apply Theorem 1. Term-by-term integration is permitted if the sequence of partial sums converges uniformly.

## Solution 4.41

**(i)**

- If $x = 0$, $f_n(0) = 0$ for all $n$, so $\lim_{n\to\infty} f_n(0) = 0$.
- If $x = 1$, $f_n(1) = 0$ for all $n$, so $\lim_{n\to\infty} f_n(1) = 0$.
- For $0 < x < 1$, let $1 - x = \frac{1}{r}$ where $r > 1$. Then $f_n(x) = \frac{n^2 x}{r^n}$. Since exponential growth dominates polynomial growth, $\lim_{n\to\infty} \frac{n^2 x}{r^n} = 0$.
- Thus, the pointwise limit is $f(x) = 0$.

**(ii)**

- Integral of the limit: $\int_0^1 0\,dx = 0$.
- Limit of the integral: $\int_0^1 n^2 x(1-x)^n\,dx$. Let $u = 1 - x$, $du = -dx$.

$$= n^2 \int_0^1 (1-u)u^n\,du = n^2 \left( \int_0^1 u^n\,du - \int_0^1 u^{n+1}\,du \right)$$
$$= n^2 \left( \frac{1}{n+1} - \frac{1}{n+2} \right) = \frac{n^2}{(n+1)(n+2)}$$

- $\lim_{n\to\infty} \frac{n^2}{n^2+3n+2} = 1$.
- Since $1 \neq 0$, the swap failed. This is because $\{f_n\}$ does not converge uniformly. The maximum value of $f_n(x)$ occurs at $x = \frac{1}{n+1}$ (found by setting $f_n'(x) = n^2(1-x)^{n-1}(1-(n+1)x) = 0$), where $f_n(\frac{1}{n+1}) \approx \frac{n}{e}$, which tends to $\infty$. Thus, no $N$ can be found that works for all $x$ simultaneously.

**(iii)**

- $|g_n(x) - g_{n-1}(x)| = \left|\frac{x^n}{n^2}\right| = \frac{x^n}{n^2}$.
- On the interval $[0, 1]$, $x^n \leq 1$, so $\frac{x^n}{n^2} \leq \frac{1}{n^2}$ for all $x \in B$.
- Let $a_n = \frac{1}{n^2}$. Since $\sum \frac{1}{n^2}$ is a positive convergent series, by Theorem 2, $\{g_n\}$ converges uniformly to $g(x)$.

**(iv)**

- By Theorem 1, since $\{g_n\}$ converges uniformly to $g$, the integral of the limit is the limit of the integrals: $\int_0^1 g(x)\,dx = \lim_{n\to\infty} \int_0^1 g_n(x)\,dx$.
- $\int_0^1 \sum_{k=1}^n \frac{x^k}{k^2}\,dx = \sum_{k=1}^n \int_0^1 \frac{x^k}{k^2}\,dx = \sum_{k=1}^n \left[\frac{x^{k+1}}{k^2(k+1)}\right]_0^1 = \sum_{k=1}^n \frac{1}{k^2(k+1)}$.
- As $n \to \infty$, $\int_0^1 g(x)\,dx = \sum_{k=1}^\infty \frac{1}{k^2(k+1)}$.

**Takeaways 4.41**

1. **The Global Requirement:** Uniform convergence requires a "global" $N$ that forces the entire function sequence into an $\epsilon$-tube around the limit function $f(x)$.
2. **Pointwise is Insufficient:** Part **(ii)** proves that simple pointwise convergence is not strong enough to preserve properties like the value of an integral through a limit process.
3. **The M-Test Logic:** Theorem 2 (often called the Weierstrass M-test) is the standard way to prove uniform convergence for series of functions by bounding them with a convergent numerical series.
4. **Integration and Limits:** Understanding when limits and integrals can be interchanged is crucial in advanced calculus and analysis.
5. **Exponential Dominance:** Exponential decay $(1-x)^n$ always dominates polynomial growth $n^2$ for $0 < x < 1$, a key observation in analyzing sequences.

## Problem 4.42: Dirichlet's Decay Factor: Taming $\sin^2 x$

**Historical Context:**
In the 19th century, Peter Gustav Lejeune Dirichlet faced a stubborn puzzle: the integral

$$\int_0^\infty \frac{\sin x}{x}\,dx$$

involves oscillations that never settle, yet the total area is finite. His key move was to introduce an exponential *decay factor* $e^{-tx}$—multiplying a wild wave by something that fades with time forces the tail to die away, so a limit can exist.

The same idea appears far beyond the classroom. A car hitting a speed bump sets the suspension oscillating; without **damping**, those vibrations would persist indefinitely. A shock absorber dissipates energy, so each bounce is smaller than the last—exactly the role of $e^{-x}$ here. The wave $\sin^2 x$ keeps producing positive "humps" forever, but $e^{-x}$ acts as a mathematical shock absorber, bleeding away their contribution until only a finite total area remains.

Having seen how exponential decay tames an endless wave, apply the same idea to

$$\int_0^\infty \frac{\sin^2 x}{e^x}\,dx$$

**Problem Statement:**
**(i)** Show that $f(x) = \frac{\sin^2 x}{e^x}$ can be written as:

$$f(x) = \frac{1}{2}e^{-x} - \frac{1}{2}e^{-x}\cos 2x$$

**(ii)** Use integration by parts to find

$$\int e^{-x}\cos 2x\,dx$$

**(iii)** Hence, or otherwise, evaluate the improper integral:

$$\int_0^\infty \frac{\sin^2 x}{e^x}\,dx$$

**Hint:**

- **Part (a):** Recall the double angle formula for $\cos 2x$. Make $\sin^2 x$ the subject and distribute $e^{-x}$.
- **Part (b):** Apply double integration by parts. Let $I = \int e^{-x}\cos 2x\,dx$. After two iterations, $I$ will reappear on the right-hand side. Solve algebraically for $I$.
- **Part (c):** Integrate the expression from (a) using your result from (b). As $x \to \infty$, remember that $e^{-x} \to 0$ while trigonometric functions remain strictly bounded.

## Solution 4.42

**Part (i)**
Using the double angle identity $\cos 2x = 1 - 2\sin^2 x$, rearrange for $\sin^2 x$:

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

Substitute this into $f(x)$:

$$f(x) = e^{-x}\left(\frac{1-\cos 2x}{2}\right) = \frac{1}{2}e^{-x} - \frac{1}{2}e^{-x}\cos 2x$$

**Part (ii)**
Let $I = \int e^{-x}\cos 2x\,dx$.
Apply integration by parts ($u = \cos 2x, dv = e^{-x}dx$):

$$I = -e^{-x}\cos 2x - 2\int e^{-x}\sin 2x\,dx$$

Apply integration by parts again to the new integral ($u = \sin 2x, dv = e^{-x}dx$):

$$\int e^{-x}\sin 2x\,dx = -e^{-x}\sin 2x + 2\int e^{-x}\cos 2x\,dx = -e^{-x}\sin 2x + 2I$$

Substitute this back into the equation for $I$:

$$I = -e^{-x}\cos 2x - 2(-e^{-x}\sin 2x + 2I)$$

$$I = -e^{-x}\cos 2x + 2e^{-x}\sin 2x - 4I$$

$$5I = e^{-x}(2\sin 2x - \cos 2x) \implies I = \frac{1}{5}e^{-x}(2\sin 2x - \cos 2x) + C$$

**Part (iii)**
Using the identity from (i) and the antiderivative from (ii):

$$\int_0^\infty \frac{\sin^2 x}{e^x}\,dx = \left[-\frac{1}{2}e^{-x} - \frac{1}{10}e^{-x}(2\sin 2x - \cos 2x)\right]_0^\infty$$

Evaluate at the bounds 0 and $\infty$:
**As $x \to \infty$:** $-\frac{1}{2}e^{-x} - \frac{1}{10}e^{-x}(2\sin 2x - \cos 2x) \to 0$ (by the Squeeze Theorem, since $(2\sin 2x - \cos 2x)$ is bounded while $e^{-x} \to 0$).
**At $x = 0$:** $-\frac{1}{2}e^0 - \frac{1}{10}e^0(0 - 1) = -\frac{1}{2} + \frac{1}{10} = -\frac{2}{5}$
Subtracting the lower bound from the upper bound gives:

$$= 0 - \left(-\frac{2}{5}\right) = \frac{2}{5}$$

**Takeaways 4.42**

- **The Power of Linearisation:** By converting $\sin^2 x$ into a linear $\cos 2x$ term, we sidestep an impossible integration and unlock the looping integration-by-parts method.
- **Generalisation ($e^{-ax}\sin bx$):** This strategy scales. Any integral in the form $\int_0^\infty e^{-ax}\sin^2(bx)\,dx$ or $\int_0^\infty e^{-ax}\cos(bx)\,dx$ can be solved using these exact same principles without altering the core logic.
- **Dirichlet's Taming Principle and the Laplace Transform:** Tame an endlessly repeating positive wave with exponential decay, then read off a finite area. This miniature version of Dirichlet's 19th-century idea reappears in damped harmonic motion, shock-absorber and vibration-control design, and the Laplace transform $\mathcal{L}\{f(x)\} = \int_0^\infty e^{-sx}f(x)\,dx$—solving this problem is exactly $\mathcal{L}\{\sin^2 x\}$ evaluated at $s = 1$.
- **Complex Analysis Shortcut:** While HSC students must use integration by parts, uni math often evaluate these oscillating real-valued integrals elegantly in a single step using **Euler's Formula** ($e^{i\theta} = \cos\theta + i\sin\theta$) or **Cauchy's Residue Theorem**.

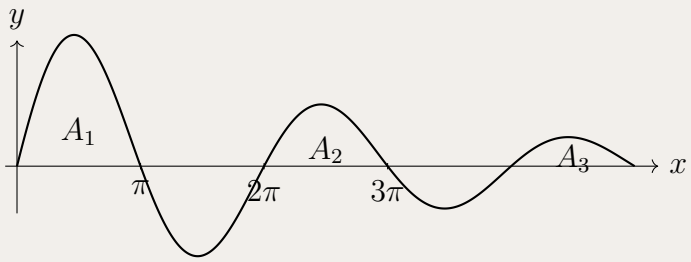
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**Remark 0.11: Normanhurst Boys High School Trial 2024, Ext 2 Q 16(b)**

If you like this style of problem, try Normanhurst Boys High School Trial 2024 Extension 2 Question 16(b) below:

The diagram shows a sketch of part of the curve with equation

$$y = e^{-x}\sin x \quad x \geq 0$$




(i) Show that

$$\int e^{-x}\sin x\,dx = -\frac{1}{2}e^{-x}(\sin x + \cos x) + C$$

(ii) The terms $A_1, A_2, \cdots, A_n$ represent successive areas above the $x$-axis and bounded by the curve $y = e^{-x}\sin x$.

The area $A_n$ is bounded by the curve $y = e^{-x}\sin x$ between $x = (2n-2)\pi$ and $x = (2n-1)\pi$.

The areas represented by $A_1$ and $A_2$ are shown in the diagram.

Show that

$$A_n = \frac{1}{2}\left(e^{(1-2n)\pi} + e^{(2-2n)\pi}\right)$$

(iii) Show that

$$A_1 + A_2 + A_3 + \cdots$$

is a geometric series, and has $\dfrac{e^{\pi}}{2(e^{\pi}-1)}$ as its limiting sum.

(iv) Given that $\lim\limits_{n\to\infty}\displaystyle\int_0^n e^{-x}\sin x\,dx = \frac{1}{2}$, find the exact value of

$$\lim_{n\to\infty}\int_0^n \left|e^{-x}\sin x\right|dx$$

**Remark 0.12: A cleaner route to part (iv)**

To avoid the pain of $|\sin x|$ and geometric series, there is a clean shortcut. Let

$$I = \int_0^\infty e^{-x} \, |\sin x| \, dx.$$

Split at $x = \pi$:

$$I = \int_0^\pi e^{-x} \, |\sin x| \, dx + \int_\pi^\infty e^{-x} \, |\sin x| \, dx.$$

Substitute $u = x - \pi$ in the second integral:

$$\int_\pi^\infty e^{-x} \, |\sin x| \, dx = \int_0^\infty e^{-(u+\pi)} \, |\sin(u+\pi)| \, du = e^{-\pi} \int_0^\infty e^{-u} \, |\sin u| \, du = e^{-\pi} I.$$

On $[0, \pi]$, $\sin x \geq 0$, so $|\sin x| = \sin x$. Hence

$$I = \int_0^\pi e^{-x} \sin x \, dx + e^{-\pi} I.$$

Apply double-integration by parts to $\int e^{-x} \sin x \, dx$ to find its value, then rearrange to express $I$ in terms of this integral, which can be refferenced from part (i).

# 5 Conclusion

Mastering Problem 16 represents the culmination of your HSC Mathematics Extension 2 journey. These "Last Resort" problems demand not just computational skills, but mathematical maturity, strategic thinking, and the ability to synthesize diverse topics under examination pressure.

The problems in this collection span the full range of techniques and topics that appear in Problem 16: from elegant inequality applications to sophisticated complex number theory, from vector optimization to advanced function analysis. Each problem has been chosen to develop specific aspects of mathematical reasoning while building your confidence in tackling unfamiliar scenarios.

Remember that Problem 16 success comes from:

- **Pattern Recognition:** Learning to identify familiar structures in new contexts
- **Technique Integration:** Combining multiple approaches within single problems
- **Proof Communication:** Expressing mathematical arguments clearly and completely
- **Strategic Thinking:** Choosing effective approaches when multiple paths exist
- **Persistence:** Working through complex multi-step problems systematically

Use this collection as a comprehensive training ground. Work through problems multiple times, focus on understanding the reasoning behind each step, and practice communicating your solutions clearly. With dedicated preparation, you can approach Problem 16 with confidence and skill.

Best of luck with your studies and HSC examinations!

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